Statistics for Artificial Intelligence
Appunti

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Chapter 1

Basics of Linear Algebra

This module collects the linear algebra tools used throughout the course, and in particular in the PCA module (it follows *Mathematics for Machine Learning*). The main themes are:

- **matrices and matrix operations**: dimensions, multiplication, rank, determinants, inverses, traces, block matrices, Kronecker products;
- **norms and distances**: measures of size and separation of vectors;
- **matrix decompositions**: Cholesky, eigendecomposition / spectral decomposition, SVD;
- **vector calculus**: gradients, quadratic forms, Jacobians, chain rule, Hessians;
- **R illustrations**.

Notation. Lowercase letters x are (column) vectors, uppercase letters A are matrices; 0_p is the zero vector of $\mathbb{R}^p$, I_p the $p \times p$ identity, $\mathbf{1}_n$ the vector of ones; $x^\top y$ is the Euclidean inner product and $\|x\|_2$ the Euclidean norm. Dimensions are written explicitly when useful, e.g. $A \in \mathbb{R}^{n \times p}$.

1.1 Matrices

Definition 1.1.1: Matrix

An $n \times m$ matrix is a rectangular array with n rows and m columns:

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix},$$

where a_{ij} is the entry in row i and column j.

1.1.1 Addition, scalar multiplication and transposition

Two matrices can be added only if they have the **same dimensions**: for $A, B \in \mathbb{R}^{n \times m}$, $A + B = [a_{ij} + b_{ij}]$, and for $\alpha \in \mathbb{R}$, $\alpha A = [\alpha a_{ij}]$. We have

$$A + B = B + A, \qquad (A + B) + C = A + (B + C), \qquad \alpha(A + B) = \alpha A + \alpha B.$$

Definition 1.1.2: Transpose

The transpose of $A \in \mathbb{R}^{n \times m}$ is the $m \times n$ matrix obtained by swapping rows and columns: $A^\top = [a_{ji}]$, i.e. $(A^\top)_{ij} = a_{ji}$.

Theorem 1.1.1 Properties of the transpose

$(A^\top)^\top = A$; for $B \in \mathbb{R}^{n\times m}$, $(A+B)^\top = A^\top + B^\top$; for $C \in \mathbb{R}^{m\times k}$,

$$(AC)^\top = C^\top A^\top.$$

Proof: The first two identities hold entry by entry: $((A+B)^\top)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji}$. For the product, with $i \le k$ and $j \le n$,

$$\big((AC)^\top\big)_{ij} = (AC)_{ji} = \sum_{\ell=1}^{m} a_{j\ell}\, c_{\ell i} = \sum_{\ell=1}^{m} (C^\top)_{i\ell}\,(A^\top)_{\ell j} = (C^\top A^\top)_{ij},$$

and both sides are $k \times n$.

Note:

Iterating: $(ABC)^\top = C^\top B^\top A^\top$. The order is reversed.

Exercise 1.1.1 Suggested exercise 1

Let $A, B \in \mathbb{R}^{2\times 3}$. Show that $(A+B)^\top = A^\top + B^\top$.

Solution: For $i = 1,2,3$ and $j = 1,2$: $((A+B)^\top)_{ij} = (A+B)_{ji} = a_{ji} + b_{ji} = (A^\top)_{ij} + (B^\top)_{ij}$. Explicitly,

$$(A+B)^\top = \begin{pmatrix} a_{11}+b_{11} & a_{21}+b_{21} \\ a_{12}+b_{12} & a_{22}+b_{22} \\ a_{13}+b_{13} & a_{23}+b_{23} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \\ b_{13} & b_{23} \end{pmatrix}.$$

Exercise 1.1.2 Suggested exercise 2

Let $A \in \mathbb{R}^{2\times 3}$ and $C \in \mathbb{R}^{3\times 2}$. Show that $(AC)^\top = C^\top A^\top$.

Solution: $(AC)_{ij} = \sum_{k=1}^{3} a_{ik} c_{kj}$, hence

$$\big((AC)^\top\big)_{ij} = (AC)_{ji} = \sum_{k=1}^{3} a_{jk} c_{ki} = \sum_{k=1}^{3} (C^\top)_{ik} (A^\top)_{kj} = (C^\top A^\top)_{ij}.$$

Both matrices are 2×2 and agree entry by entry. For instance, entry $(1,2)$ of $(AC)^\top$ is $(AC)_{21} = a_{21}c_{11} + a_{22}c_{21} + a_{23}c_{31}$, i.e. row 1 of $C^\top$ times column 2 of $A^\top$.

1.1.2 Multiplication

Definition 1.1.3: Matrix product

Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times k}$ (columns of A = rows of B). The product $C = AB \in \mathbb{R}^{n\times k}$ has entries

$$c_{ij} = \sum_{\ell=1}^{m} a_{i\ell}\, b_{\ell j},$$

i.e. the inner product of row i of A with column j of B.

For matrices of suitable dimensions:

$$(AB)C = A(BC), \qquad A(B+C) = AB + AC, \qquad (A+C)B = AB + CB,$$

but in general $AB \neq BA$.

Example 1.1.1

With $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$:

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \neq BA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Note:

Two useful ways of reading a product: the j-th column of AB is $Ab_j = \sum_\ell b_{\ell j} a_\ell$, a linear combination of the columns of A; and $AB = \sum_\ell a_\ell \tilde{b}_\ell^\top$ is a sum of *column* $\times$ *row* products (a_ℓ columns of A, $\tilde{b}_\ell^\top$ rows of B).

Exercise 1.1.3 Suggested exercise 3

Let $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{m\times k}$. Which dimension must C have for (1) $(AB)C = A(BC)$, (2) $A(B + C) = AB + AC$, (3) $(A + C)B = AB + CB$ to make sense?

Solution: 1. AB is $n \times k$, so C must have k rows: $C \in \mathbb{R}^{k\times \ell}$ for any ℓ. Then BC is $m \times \ell$ and $A(BC)$ is $n \times \ell$.

2. $B + C$ requires C of the same size as B: $C \in \mathbb{R}^{m\times k}$.

3. $A + C$ requires C of the same size as A: $C \in \mathbb{R}^{n\times m}$.

1.1.3 Identity and special matrices

Definition 1.1.4: Identity matrix

$I_n = [\delta_{ij}] \in \mathbb{R}^{n\times n}$ has ones on the diagonal and zeros elsewhere.

Proposition 1.1.1 Neutral element

For $A \in \mathbb{R}^{n\times m}$: $AI_m = I_n A = A$.

Proof: $(AI_m)_{ij} = \sum_{\ell=1}^m a_{i\ell}\delta_{\ell j} = a_{ij}$ and $(I_n A)_{ij} = \sum_{\ell=1}^n \delta_{i\ell} a_{\ell j} = a_{ij}$. Note that for rectangular A the identities on the two sides have *different sizes*.

Definition 1.1.5: Special types of matrices

- **zero matrix** 0; **vector** $= n \times 1$ matrix; $\mathbf{1}_n$ = vector of ones;
- **square**: $A \in \mathbb{R}^{n\times n}$; **symmetric**: $A = A^\top$;
- **orthonormal** (square): $A^\top A = AA^\top = I$; rows and columns are orthonormal vectors;
- $A \in \mathbb{R}^{n\times m}$, $m < n$, has **orthonormal columns** if $A^\top A = I_m$ (then $AA^\top \neq I_n$);
- **diagonal**: only diagonal entries may be nonzero; **triangular**: all entries above (lower triangular) or below (upper triangular) the diagonal are zero;
- **idempotent**: $A^2 = A$; **orthogonal projector**: $A^2 = A$ and $A = A^\top$;
- **positive definite** (symmetric A): $x^\top Ax > 0$ for all $x \neq 0$.

Proposition 1.1.2 Orthonormal matrices

Let $A \in \mathbb{R}^{n\times n}$ with $A^\top A = I$. Then $AA^\top = I$, $\det A = \pm 1$, and $\|Ax\|_2 = \|x\|_2$ for every x.

Proof: $\det(A^\top)\det(A) = \det(A)^2 = \det(I) = 1$, so $\det A = \pm 1 \neq 0$ and A is invertible. Multiplying $A^\top A = I$ on the right by A^{-1} gives $A^\top = A^{-1}$, hence $AA^\top = AA^{-1} = I$. Finally $\|Ax\|_2^2 = x^\top A^\top Ax = x^\top x = \|x\|_2^2$ (orthonormal matrices are rotations/reflections; similarly $(Ax)^\top(Ay) = x^\top y$). ☺

Proposition 1.1.3 Orthonormal columns

Let $A \in \mathbb{R}^{n\times m}$, $m < n$, with $A^\top A = I_m$. Then $AA^\top \neq I_n$, and $AA^\top$ is the orthogonal projector onto the column space of A.

Proof: $\operatorname{rank}(AA^\top) \leq \operatorname{rank}(A) \leq m < n = \operatorname{rank}(I_n)$, so $AA^\top \neq I_n$. $AA^\top$ is symmetric, and idempotent: $(AA^\top)(AA^\top) = A(A^\top A)A^\top = AI_mA^\top = AA^\top$. ☺

Note:

This is exactly the situation of $P_qP_q^\top$ in PCA.

Theorem 1.1.2 Eigenvalues of idempotent matrices

If $A^2 = A$, every eigenvalue of A is 0 or 1.

Proof: Let $Ap = \lambda p$, $p \neq 0$. Then $A^2p = \lambda Ap = \lambda^2 p$, but also $A^2p = Ap = \lambda p$. Hence $(\lambda^2 - \lambda)p = 0$ and, since $p \neq 0$, $\lambda \in \{0, 1\}$. ☺

Note:

If A is idempotent, so is $I - A$: $(I - A)^2 = I - 2A + A^2 = I - A$. If A is an orthogonal projector, $A = P\Lambda P^\top$ with Λ made of 0s and 1s, so $\operatorname{tr}(A) = \operatorname{rank}(A)$. For instance $\operatorname{tr}(P_qP_q^\top) = \operatorname{tr}(P_q^\top P_q) = q$.

Theorem 1.1.3 Positive definite matrices

If A is symmetric positive definite then (1) all its eigenvalues are > 0; (2) $a_{ii} > 0$ for every i; (3) A is invertible.

Proof: (1) If $Ap = \lambda p$ with $p \neq 0$: $0 < p^\top Ap = \lambda\, p^\top p = \lambda\|p\|^2$, so $\lambda > 0$. (2) Take $x = e_i$: $a_{ii} = e_i^\top Ae_i > 0$. (3) If $Ax = 0$ then $x^\top Ax = 0$, which forces $x = 0$: the kernel is trivial and A is invertible (equivalently $\det A = \prod_i \lambda_i > 0$). ☺

1.1.4 Rank

Definition 1.1.6: Rank

The *rank* of A, $\operatorname{rank}(A)$, is the number of linearly independent columns of A; equivalently the number of linearly independent rows, i.e. the dimension of the column space.

$$\operatorname{rank}(A) = \operatorname{rank}(A^\top) \leq \min(n, m).$$

A has **full rank** if $\operatorname{rank}(A) = \min(n, m)$.

For a square matrix, full rank is equivalent to invertibility (see below).

Theorem 1.1.4 Rank of products and sums

$$\operatorname{rank}(AB) \leq \min\{\operatorname{rank}(A), \operatorname{rank}(B)\}, \qquad \operatorname{rank}(A + B) \leq \operatorname{rank}(A) + \operatorname{rank}(B).$$

Proof: The j-th column of AB is Ab_j, a linear combination of the columns of A, so $\operatorname{col}(AB) \subseteq \operatorname{col}(A)$ and $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$. Applying this to $B^\top A^\top$: $\operatorname{rank}(AB) = \operatorname{rank}((AB)^\top) = \operatorname{rank}(B^\top A^\top) \leq \operatorname{rank}(B^\top) = \operatorname{rank}(B)$. For the sum, the j-th column $a_j + b_j$ of $A + B$ lies in $\operatorname{col}(A) + \operatorname{col}(B)$, so $\operatorname{rank}(A + B) \leq \dim(\operatorname{col} A + \operatorname{col} B) \leq \dim \operatorname{col} A + \dim \operatorname{col} B$. ☺

Corollary 1.1.1 Rank-one matrices

If $u, v \neq 0$, then $\operatorname{rank}(uv^\top) = 1$. Hence $\sum_{i=1}^{r} \sigma_i u_i v_i^\top$ has rank $\leq r$.

Proof: The columns of $uv^\top$ are $v_j u$: all multiples of u, and not all zero. The second claim follows from the rank of a sum. ✆

1.1.5 Determinant

The determinant associates a scalar $|A| = \det(A)$ to a square matrix. For small matrices,

$$|a_{11}| = a_{11}, \qquad \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21},$$

and for 3×3 matrices one uses the Laplace (cofactor) expansion along the first row:

$$|A| = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}.$$

Theorem 1.1.5 Properties of the determinant

For $A, B \in \mathbb{R}^{n \times n}$ and $\alpha \in \mathbb{R}$:

$$|AB| = |A||B|, \qquad |A^\top| = |A|, \qquad |\alpha A| = \alpha^n |A|, \qquad |I_n| = 1.$$

A square matrix is **singular** if and only if $|A| = 0$.

Note:

$|\alpha A| = \alpha^n |A|$ and not $\alpha|A|$: writing $\alpha A = (\alpha I_n)A$ and using $|\alpha I_n| = \alpha^n$ (product of the diagonal) and the product rule. Also, from $AA^{-1} = I$, $|A||A^{-1}| = 1$, i.e. $|A^{-1}| = 1/|A|$.

Proposition 1.1.4 Triangular matrices

The determinant of a triangular matrix is the product of its diagonal entries.

Proof: Induction on n; $n = 1$ is trivial. For L lower triangular, expanding along the first row, the only nonzero entry is l_{11}, so $|L| = l_{11}|L'|$ where L' (removing row and column 1) is lower triangular of size $n - 1$; by induction $|L'| = \prod_{i \geq 2} l_{ii}$. For upper triangular U use $|U| = |U^\top|$. ✆

Exercise 1.1.4 Suggested exercise 4

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$ show that $|AB| = |A||B|$.

Solution: $AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$, so

$$\begin{aligned} |AB| &= (ae + bg)(cf + dh) - (af + bh)(ce + dg) \\ &= acef + adeh + bcfg + bdgh - acef - adfg - bceh - bdgh \\ &= adeh + bcfg - adfg - bceh = ad(eh - fg) - bc(eh - fg) \\ &= (ad - bc)(eh - fg) = |A||B|. \end{aligned}$$

✆

1.2 Matrix Operations

1.2.1 Linear systems

A system of n linear equations in p unknowns

$$\begin{cases} a_{11}x_1 + \cdots + a_{1p}x_p = b_1, \\ \quad\vdots \\ a_{n1}x_1 + \cdots + a_{np}x_p = b_n \end{cases}$$

is written compactly as $Ax = b$, with $A \in \mathbb{R}^{n\times p}$ the coefficient matrix (each row is an equation, each column contains the coefficients of one unknown), $x \in \mathbb{R}^p$ the unknowns and $b \in \mathbb{R}^n$ the right-hand side. A system may have no solution, exactly one, or infinitely many.

Theorem 1.2.1 Rouché–Capelli

- $Ax = b$ has a solution $\iff$ $b \in \operatorname{col}(A)$ $\iff$ $\operatorname{rank}(A) = \operatorname{rank}([A \mid b])$;
- if x_0 is a solution, the set of all solutions is $x_0 + \ker(A)$;
- the solution is unique $\iff$ $\ker(A) = \{0\}$ $\iff$ $\operatorname{rank}(A) = p$; otherwise there are infinitely many.

Proof: $Ax = \sum_j x_j a_j$ ranges exactly over $\operatorname{col}(A)$, which gives the first point. If $Ax_0 = Ax_1 = b$ then $A(x_1 - x_0) = 0$, i.e. $x_1 \in x_0 + \ker A$; conversely $A(x_0 + k) = b$ for $k \in \ker A$. Finally $\dim\ker A = p - \operatorname{rank}(A)$ (rank–nullity). ☺

Exercise 1.2.1 Suggested exercise 5

Solve $x_1 + x_2 + x_3 = 3$, $x_1 - x_2 + 2x_3 = 2$, $x_2 + x_3 = 2$. Is the solution unique?

Solution: First minus third: $x_1 = 1$. Then the second gives $-x_2 + 2x_3 = 1$; adding the third, $3x_3 = 3$, so $x_3 = 1$ and $x_2 = 1$: $x = (1, 1, 1)^\top$. It is unique: every step was forced; equivalently

$$\det\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 0 & 1 & 1 \end{pmatrix} = 1(-1-2) - 1(1-0) + 1(1-0) = -3 \neq 0,$$

so the matrix is invertible and $x = M^{-1}b$ is the only solution. ☺

1.2.2 Inverse matrix

Definition 1.2.1: Inverse

For a full-rank square matrix $A \in \mathbb{R}^{n\times n}$, the inverse A^{-1} satisfies $A^{-1}A = AA^{-1} = I_n$. Then $Ax = b$ has the solution $x = A^{-1}b$.

Proposition 1.2.1 Uniqueness of the inverse

If B and C are both inverses of A, then $B = C$.

Proof: $B = BI = B(AC) = (BA)C = IC = C$. ☺

Theorem 1.2.2 Invertibility

For $A \in \mathbb{R}^{n\times n}$ the following are equivalent: A is invertible; $|A| \neq 0$; A has full rank; $Ax = 0 \Rightarrow x = 0$.

Proof (sketch): Invertible $\Rightarrow |A||A^{-1}| = 1 \Rightarrow |A| \neq 0$. Conversely, if $|A| \neq 0$ then $A^{-1} = \frac{1}{|A|}\operatorname{adj}(A)$ (transposed cofactor matrix). Full rank $\iff$ independent columns $\iff$ $(Ax = \sum_j x_j a_j = 0 \Rightarrow x = 0)$. A trivial kernel means A is injective, hence (square, rank–nullity) surjective, hence invertible. ☺

Note:

Consequently a homogeneous system $Mx = 0$ has a nontrivial solution $\iff |M| = 0$: this is the key fact behind the computation of eigenvalues.

For a 2×2 matrix,

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \Longrightarrow \quad A^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix},$$

provided $a_{11}a_{22} - a_{12}a_{21} \neq 0$ (swap the diagonal, change sign off the diagonal, divide by the determinant).

Theorem 1.2.3 Properties of the inverse

For $A, B \in \mathbb{R}^{n \times n}$ invertible:

$$(A^{-1})^{-1} = A, \qquad (AB)^{-1} = B^{-1}A^{-1}, \qquad (A^{\top})^{-1} = (A^{-1})^{\top}.$$

Proof: $A^{-1}A = AA^{-1} = I$ says exactly that A is the inverse of A^{-1}. $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$ and similarly $(B^{-1}A^{-1})(AB) = I$; conclude by uniqueness. Finally $A^{\top}(A^{-1})^{\top} = (A^{-1}A)^{\top} = I^{\top} = I$.

Exercise 1.2.2 Suggested exercise 6

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $ad - bc \neq 0$ show that $AA^{-1} = A^{-1}A = I_2$.

Solution: With $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$:

$$AA^{-1} = \frac{1}{ad - bc}\begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = I_2, \qquad A^{-1}A = \frac{1}{ad - bc}\begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = I_2.$$

1.2.3 Diagonal and trace

Definition 1.2.2: Trace

The *diagonal* of a square matrix is $a_{11}, \ldots, a_{nn}$; the *trace* is their sum: $\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}$.

$\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ and $\operatorname{tr}(\alpha A) = \alpha \operatorname{tr}(A)$.

Theorem 1.2.4 Cyclic property

For $A \in \mathbb{R}^{n \times m}$ and $B \in \mathbb{R}^{m \times n}$: $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Proof:

$$\operatorname{tr}(AB) = \sum_{i=1}^{n}\sum_{\ell=1}^{m} a_{i\ell}b_{\ell i} = \sum_{\ell=1}^{m}\sum_{i=1}^{n} b_{\ell i}a_{i\ell} = \operatorname{tr}(BA).$$

Note that AB is $n \times n$ while BA is $m \times m$.

Note:

Consequences: $\operatorname{tr}(ABC) = \operatorname{tr}(CAB) = \operatorname{tr}(BCA)$ (cyclic permutations only); $x^{\top}Ax = \operatorname{tr}(Axx^{\top})$; $\operatorname{tr}(A^{\top}A) = \sum_{i,j} a_{ij}^2$.

Corollary 1.2.1 Trace and determinant from eigenvalues

If $A = P\Lambda P^{-1}$ with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, then $\operatorname{tr}(A) = \sum_i \lambda_i$ and $\det(A) = \prod_i \lambda_i$.

Proof: $\operatorname{tr}(P\Lambda P^{-1}) = \operatorname{tr}(\Lambda P^{-1}P) = \operatorname{tr}(\Lambda)$ and $\det(P\Lambda P^{-1}) = \det P \det\Lambda \det P^{-1} = \det\Lambda$. In PCA: the total variance is $\operatorname{tr}(S) = \sum_j \lambda_j$. ☺

> **Exercise 1.2.3** Suggested exercise 7
>
> For generic $A, B \in \mathbb{R}^{2\times 2}$ show that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Solution: With $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$: $\operatorname{tr}(AB) = (ae + bg) + (cf + dh)$ and $BA = \begin{pmatrix} ea + fc & eb + fd \\ ga + hc & gb + hd \end{pmatrix}$, so $\operatorname{tr}(BA) = ea + fc + gb + hd$: the same four terms. ☺

1.2.4 Partitioned matrices

Many computations are easier when a matrix is split into blocks, $A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$. For **conformably** partitioned matrices one computes as with scalars, *keeping the order* of the factors:

$$A + B = \begin{bmatrix} A_{11} + B_{11} & A_{12} + B_{12} \\ A_{21} + B_{21} & A_{22} + B_{22} \end{bmatrix}, \qquad AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

For instance $\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{bmatrix}$. A **block-diagonal** matrix has square blocks A_{ii} (not necessarily of the same size) on the diagonal and zeros elsewhere.

> **Theorem 1.2.5** Block determinant (Schur complement)
>
> If A_{11} is invertible,
>
> $$|A| = |A_{11}|\left|A_{22} - A_{21}A_{11}^{-1}A_{12}\right|;$$
>
> if A_{22} is invertible, $|A| = |A_{22}|\left|A_{11} - A_{12}A_{22}^{-1}A_{21}\right|$.

Proof: By block multiplication one checks

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I & 0 \\ A_{21}A_{11}^{-1} & I \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} - A_{21}A_{11}^{-1}A_{12} \end{bmatrix};$$

the second block row gives $A_{21}A_{11}^{-1}A_{11} = A_{21}$ and $A_{21}A_{11}^{-1}A_{12} + A_{22} - A_{21}A_{11}^{-1}A_{12} = A_{22}$. The first factor is triangular with unit diagonal (determinant 1); the second is block triangular, whose determinant is the product of the determinants of the diagonal blocks. Conclude by the product rule. The second formula is symmetric. ☺

> **Corollary 1.2.2** Block-diagonal matrices
>
> $$\begin{vmatrix} A_{11} & 0 \\ 0 & A_{22} \end{vmatrix} = |A_{11}|\,|A_{22}|, \qquad \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}^{-1} = \begin{bmatrix} A_{11}^{-1} & 0 \\ 0 & A_{22}^{-1} \end{bmatrix}.$$

Proof: The determinant is the Schur formula with $A_{12} = A_{21} = 0$. For the inverse, block multiplication gives $\operatorname{diag}(A_{11}, A_{22})\operatorname{diag}(A_{11}^{-1}, A_{22}^{-1}) = \operatorname{diag}(I, I) = I$ (the mixed terms vanish). ☺

1.2.5 Kronecker product

Definition 1.2.3: Kronecker product

For $A \in \mathbb{R}^{n\times m}$ and $B \in \mathbb{R}^{k\times l}$,

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1m}B \\ \vdots & \ddots & \vdots \\ a_{n1}B & \cdots & a_{nm}B \end{bmatrix} \in \mathbb{R}^{nk\times ml}.$$

Example 1.2.1

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \otimes \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 3 & 0 & 4 & 0 \end{bmatrix}.$$

It is useful for structured covariance matrices (e.g. $I_n \otimes \Sigma$ for n independent stacked vectors) and vectorization identities. Some properties (beyond the slides):

Proposition 1.2.2 Mixed product

$(A \otimes B)(C \otimes D) = AC \otimes BD$ (when the products are defined); hence $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$. Also $(A \otimes B)^\top = A^\top \otimes B^\top$, but in general $A \otimes B \neq B \otimes A$.

Proof: The block (i, j) of $(A \otimes B)(C \otimes D)$ is $\sum_\ell (a_{i\ell} B)(c_{\ell j} D) = (AC)_{ij}\, BD$, which is the block (i, j) of $AC \otimes BD$. With $C = A^{-1}$, $D = B^{-1}$ we get $I \otimes I = I$. ☺

1.3 Norms and Distances

Definition 1.3.1: Norm

Let V be a vector space. A function $\|\cdot\| : V \to \mathbb{R}$ is a *norm* if for all $x, y \in V$ and $\lambda \in \mathbb{R}$:

1. **positive definiteness**: $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$;
2. **absolute homogeneity**: $\|\lambda x\| = |\lambda|\,\|x\|$;
3. **triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$.

These properties guarantee that a norm behaves like a sensible measure of length.

1.3.1 Common norms on $\mathbb{R}^n$

$$\|x\|_1 = \sum_{i=1}^n |x_i| \text{ (Manhattan)}, \qquad \|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2} \text{ (Euclidean)},$$

$$\|x\|_p = \Big(\sum_{i=1}^n |x_i|^p\Big)^{1/p} \ (p \geq 1), \qquad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i| \text{ (maximum)}.$$

ℓ_1 and ℓ_2 are special cases of ℓ_p; $\|x\|_\infty$ measures the largest absolute component.

Proposition 1.3.1 Relations between norms

$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n\|x\|_\infty$, and $\|x\|_p \to \|x\|_\infty$ as $p \to \infty$.

Proof: $\max_i x_i^2 \leq \sum_i x_i^2 \leq \big(\sum_i |x_i|\big)^2$ (the cross terms are nonnegative), and $\sum_i |x_i| \leq n \max_i |x_i|$. For the limit, with $M = \max_i |x_i|$, $M \leq \|x\|_p \leq (nM^p)^{1/p} = n^{1/p} M \to M$. ☺

Theorem 1.3.1 Cauchy–Schwarz and triangle inequality

For $x, y \in \mathbb{R}^n$: $|x^\top y| \leq \|x\|_2 \|y\|_2$, and consequently $\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$.

Proof: For every $t \in \mathbb{R}$, $0 \leq \|x + ty\|_2^2 = \|x\|^2 + 2t\, x^\top y + t^2 \|y\|^2$. A degree-two polynomial in t that is always nonnegative has nonpositive discriminant: $4(x^\top y)^2 - 4\|x\|^2\|y\|^2 \leq 0$. Then

$$\|x + y\|_2^2 = \|x\|^2 + 2x^\top y + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\|_2 + \|y\|_2)^2.$$

For ℓ_1 and ℓ_∞ the triangle inequality follows from $|x_i + y_i| \le |x_i| + |y_i|$ (sum, resp. max over i).

Note:

In $\mathbb{R}^2$ the unit balls $\{\|x\| \le 1\}$ are: a **diamond** for ℓ_1, the **disk** for ℓ_2, the **square** $[-1,1]^2$ for ℓ_∞. They are nested, consistently with $\|x\|_\infty \le \|x\|_2 \le \|x\|_1$.

Exercise 1.3.1 Suggested exercise 8

Compute the ℓ_1, ℓ_2 and maximum norms of $x = (1,-2,2)^\top$.

Solution: $\|x\|_1 = 1+2+2 = 5$, $\|x\|_2 = \sqrt{1+4+4} = 3$, $\|x\|_\infty = \max(1,2,2) = 2$.

Exercise 1.3.2 Suggested exercise 9

Show that $\|\lambda x\|_2 = |\lambda|\,\|x\|_2$ for $x \in \mathbb{R}^n$, $\lambda \in \mathbb{R}$.

Solution: $\|\lambda x\|_2 = \sqrt{\sum_i (\lambda x_i)^2} = \sqrt{\lambda^2 \sum_i x_i^2} = \sqrt{\lambda^2}\sqrt{\sum_i x_i^2} = |\lambda|\,\|x\|_2$.

1.3.2 Distances

Definition 1.3.2: Distance

Given a norm, the distance between x and y is $d(x,y) = \|x - y\|$.

Proposition 1.3.2 Properties of the induced distance

$d(x,y) \ge 0$; $d(x,y) = 0 \iff x = y$; $d(x,y) = d(y,x)$; $d(x,z) \le d(x,y) + d(y,z)$.

Proof: The first two come from positive definiteness applied to $x - y$. Symmetry: $\|x-y\| = \|-(y-x)\| = |-1|\,\|y-x\|$. Triangle: $\|x-z\| = \|(x-y)+(y-z)\| \le \|x-y\| + \|y-z\|$.

In particular $d(x,0) = \|x\|$: the length of a vector is its distance from the origin.

Exercise 1.3.3 Suggested exercise 10

Compute the ℓ_1, ℓ_2 and maximum distances between $x = (1,0,-1)^\top$ and $y = (2,-2,3)^\top$.

Solution: $x - y = (-1,2,-4)^\top$, so $d_1 = 1+2+4 = 7$, $d_2 = \sqrt{1+4+16} = \sqrt{21} \approx 4.583$, $d_\infty = 4$.

1.4 Matrix Decompositions

1.4.1 Cholesky decomposition

Theorem 1.4.1 Cholesky decomposition

If $A \in \mathbb{R}^{n\times n}$ is **square, symmetric and positive definite**, it admits a unique factorization

$$A = LL^\top,$$

with L lower triangular with positive diagonal entries (the *Cholesky factor*).

It is the matrix analogue of a square root. Conversely, the conditions are necessary:

Proposition 1.4.1

If $A = LL^\top$ with L lower triangular with positive diagonal, then A is symmetric positive definite.

Proof: $(LL^\top)^\top = LL^\top$. Since $\det L = \prod_i l_{ii} > 0$, $L^\top$ is invertible, so $L^\top x \neq 0$ for $x \neq 0$ and $x^\top A x = x^\top L L^\top x = \|L^\top x\|_2^2 > 0$. ☺

Computing L. From $a_{ij} = \sum_{k \le j} l_{ik} l_{jk}$ $(i \ge j)$, column by column:

$$l_{jj} = \sqrt{a_{jj} - \sum_{k<j} l_{jk}^2}, \qquad l_{ij} = \frac{a_{ij} - \sum_{k<j} l_{ik} l_{jk}}{l_{jj}} \quad (i > j).$$

If a square root of a nonpositive number appears, A is not positive definite.

Example 1.4.1

For $A = \begin{bmatrix} 4 & 2 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$: $l_{11} = 2$, $l_{21} = l_{31} = 1$; $l_{22} = \sqrt{2-1} = 1$, $l_{32} = (1-1)/1 = 0$; $l_{33} = \sqrt{3-1-0} = \sqrt{2}$:

$$L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & \sqrt{2} \end{bmatrix}.$$

For a diagonal matrix $A = \operatorname{diag}(a_{11}, a_{22}, a_{33})$ a valid Cholesky factor is simply $L = \operatorname{diag}(\sqrt{a_{11}}, \sqrt{a_{22}}, \sqrt{a_{33}})$.

Exercise 1.4.1 Suggested exercise 11

Find the Cholesky factor of $A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$.

Solution: $L = \begin{pmatrix} l_{11} & 0 \\ l_{21} & l_{22} \end{pmatrix}$ gives $LL^\top = \begin{pmatrix} l_{11}^2 & l_{11} l_{21} \\ l_{11} l_{21} & l_{21}^2 + l_{22}^2 \end{pmatrix}$. Then $l_{11}^2 = 4 \Rightarrow l_{11} = 2$; $l_{11} l_{21} = 2 \Rightarrow l_{21} = 1$; $1 + l_{22}^2 = 3 \Rightarrow l_{22} = \sqrt{2}$:

$$L = \begin{pmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{pmatrix}, \qquad LL^\top = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix} = A.$$

☺

Exercise 1.4.2 Suggested exercise 12

Show that for $A = \operatorname{diag}(a_{11}, a_{22}, a_{33})$ a valid Cholesky factor is $L = \operatorname{diag}(\sqrt{a_{11}}, \sqrt{a_{22}}, \sqrt{a_{33}})$.

Solution: A positive definite means $a_{ii} > 0$, so the square roots are real and positive. L is diagonal, hence lower triangular with positive diagonal, and $L^\top = L$: $LL^\top = L^2 = \operatorname{diag}(a_{11}, a_{22}, a_{33}) = A$. By uniqueness it is *the* Cholesky factor. ☺

Solving systems using Cholesky factors

The Cholesky decomposition is computationally important because it allows us to

- solve linear systems efficiently when A is symmetric positive definite;
- compute determinants efficiently: $\det(A) = \det(L)^2 = \left(\prod_{i=1}^n l_{ii}\right)^2$ (L triangular, $|L^\top| = |L|$, product rule);
- avoid computing a full inverse.

Indeed, if $A = LL^\top$, then $Ax = b$ is solved in two triangular steps:

$$\text{(1) solve } Lz = b, \qquad \text{(2) solve } L^\top x = z.$$

Triangular systems are easy: *forward substitution* gives $z_1 = b_1/l_{11}$, $z_i = \left(b_i - \sum_{k<i} l_{ik} z_k\right)/l_{ii}$, at cost $O(n^2)$, while the factorization costs $O(n^3)$ once and can be reused for many right-hand sides.

Exercise 1.4.3 Suggested exercise 13

A symmetric positive definite with Cholesky factor L. Show that $Ax = b$ can be solved in two triangular steps: $Lz = b$, then $L^\top x = z$.

Solution: $Ax = b \iff L(L^\top x) = b$. Set $z := L^\top x$. (1) L is lower triangular with positive diagonal, hence invertible: $Lz = b$ is solved by forward substitution. (2) $L^\top$ is upper triangular and invertible: $L^\top x = z$ is solved by backward substitution. Check: $Ax = LL^\top x = Lz = b$. Since A is invertible, this is the unique solution.

1.4.2 Spectral decomposition

Definition 1.4.1: Eigenvalues and eigenvectors

A nonzero $p \in \mathbb{R}^n$ is an *eigenvector* of $A \in \mathbb{R}^{n\times n}$ with *eigenvalue* λ if

$$Ap = \lambda p :$$

applying A to p only changes its scale, not its direction.

Definition 1.4.2: Diagonalization

A is *diagonalizable* if there is an invertible P such that $\Lambda = P^{-1}AP$ is diagonal. If the columns of P are eigenvectors and Λ contains the corresponding eigenvalues, $A = P\Lambda P^{-1}$ is the *eigendecomposition* of A. For a real symmetric matrix P can be chosen orthonormal, $P^\top P = I_n$, and

$$\Lambda = P^\top AP, \qquad A = P\Lambda P^\top \qquad (\textbf{spectral decomposition}).$$

Exercise 1.4.4 Suggested exercise 14

Let $P = [p_1 \cdots p_n]$ be invertible with $Ap_i = \lambda_i p_i$. Show that $A = P\Lambda P^{-1}$ with $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

Solution: Column by column, $AP = [Ap_1 \mid \cdots \mid Ap_n] = [\lambda_1 p_1 \mid \cdots \mid \lambda_n p_n]$. The i-th column of $P\Lambda$ is $P(\lambda_i e_i) = \lambda_i p_i$, so $AP = P\Lambda$. Multiply on the right by P^{-1}: $A = P\Lambda P^{-1}$.

How to compute eigenvalues and eigenvectors

$Ax = \lambda x$ with $x \neq 0$ is equivalent to $(A - \lambda I)x = 0$: for a given λ, an eigenvector is a nonzero solution of a homogeneous system with matrix $A - \lambda I$. Such a solution exists precisely when $A - \lambda I$ is not invertible, i.e.

$$\det(A - \lambda I) = 0.$$

The roots of this polynomial of degree n (the *characteristic polynomial*) are the eigenvalues. Procedure:

1. form $A - \lambda I$ and compute $\det(A - \lambda I)$;
2. solve $\det(A - \lambda I) = 0$;
3. for each eigenvalue, solve $(A - \lambda I)x = 0$; any nonzero solution is an eigenvector; normalize it, $\|x\|_2 = 1$, to obtain an orthonormal P.

Example 1.4.2

$S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$: $(2-\lambda)^2 - 1 = 0 \Rightarrow \lambda = 3, 1$. For $\lambda = 3$: $-x_1 + x_2 = 0$, $p_1 = \frac{1}{\sqrt{2}}(1,1)^\top$; for $\lambda = 1$: $x_1 + x_2 = 0$, $p_2 = \frac{1}{\sqrt{2}}(-1,1)^\top$ (as returned by `eigen(S)` in R).

Exercise 1.4.5 Suggested exercise 15

Diagonalize $A = \frac{1}{2}\begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$ as $A = P\Lambda P^\top$.

Solution: For $M = \begin{bmatrix} 5 & -2 \\ -2 & 5 \end{bmatrix}$: $(5-\mu)^2 - 4 = 0 \Rightarrow \mu = 7, 3$, so the eigenvalues of $A = M/2$ are $\frac{7}{2}$ and $\frac{3}{2}$. $\lambda = \frac{7}{2}$: $A - \frac{7}{2}I = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$, $p_1 = \frac{1}{\sqrt{2}}(1,-1)^\top$. $\lambda = \frac{3}{2}$: $A - \frac{3}{2}I = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, $p_2 = \frac{1}{\sqrt{2}}(1,1)^\top$.

$$\Lambda = \begin{bmatrix} 7/2 & 0 \\ 0 & 3/2 \end{bmatrix}, \qquad P = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \qquad P^{-1} = P^\top.$$

Check: $A(1,-1)^\top = \frac{1}{2}(7,-7)^\top = \frac{7}{2}(1,-1)^\top$.

Symmetric matrices

Theorem 1.4.2 Real eigenvalues

The eigenvalues of a real symmetric matrix are real.

Proof: Let $Ap = \lambda p$ with $\lambda \in \mathbb{C}$, $p \in \mathbb{C}^n \setminus \{0\}$, and $p^* = \bar{p}^\top$. Then $p^*Ap = \lambda\, p^*p = \lambda\|p\|^2$ with $\|p\|^2 > 0$ real. The scalar p^*Ap is real: its conjugate is $p^\top A\bar{p}$ (A real), and transposing this scalar gives $\bar{p}^\top A^\top p = p^*Ap$ ($A = A^\top$). Hence $\lambda = p^*Ap/\|p\|^2 \in \mathbb{R}$.

Theorem 1.4.3 Orthogonal eigenvectors

Eigenvectors of a symmetric matrix associated with distinct eigenvalues are orthogonal.

Proof: Let $Ap_1 = \lambda_1 p_1$, $Ap_2 = \lambda_2 p_2$, $\lambda_1 \neq \lambda_2$. Then

$$\lambda_1\, p_1^\top p_2 = (Ap_1)^\top p_2 = p_1^\top A^\top p_2 = p_1^\top A p_2 = \lambda_2\, p_1^\top p_2,$$

so $(\lambda_1 - \lambda_2)p_1^\top p_2 = 0$ and $p_1^\top p_2 = 0$. For repeated eigenvalues one chooses an orthonormal basis of the eigenspace.

Theorem 1.4.4 Spectral theorem

Every real symmetric $A \in \mathbb{R}^{n\times n}$ can be written as $A = P\Lambda P^\top$ with P orthonormal and Λ real diagonal. Equivalently, as a sum of rank-one matrices,

$$A = \sum_{j=1}^{n} \lambda_j\, p_j p_j^\top.$$

Proof (idea): Induction on n. There is a real eigenvalue λ_1 with unit eigenvector p_1. Complete it to an orthonormal basis $Q = [p_1\ Q_2]$. Then $Q_2^\top Ap_1 = \lambda_1 Q_2^\top p_1 = 0$ and, by symmetry, $p_1^\top AQ_2 = 0$, so

$$Q^\top AQ = \begin{bmatrix} \lambda_1 & 0 \\ 0 & Q_2^\top AQ_2 \end{bmatrix},$$

with $Q_2^\top AQ_2$ symmetric of size $n-1$: apply the induction hypothesis. The sum form is $P\Lambda P^\top$ written as column $\times$ row products.

Additional properties

For a diagonal matrix the eigendecomposition is immediate: $\Lambda = A$, $P = I$, with the standard basis vectors as eigenvectors.

Proposition 1.4.2 Functions of a diagonalizable matrix

If $A = P\Lambda P^{-1}$ then $A^k = P\Lambda^k P^{-1}$ with $\Lambda^k = \mathrm{diag}(\lambda_1^k, \dots, \lambda_n^k)$. If moreover A is symmetric ($P^{-1} = P^\top$): $A^{-1} = P\Lambda^{-1}P^\top$ when all $\lambda_i \neq 0$, and $A^{1/2} = P\Lambda^{1/2}P^\top$ when all $\lambda_i \geq 0$.

Proof: Powers: see Exercise 16. Inverse: $P\Lambda P^\top P\Lambda^{-1}P^\top = P\Lambda\Lambda^{-1}P^\top = PP^\top = I$. Square root: $P\Lambda^{1/2}P^\top$ is symmetric PSD and $P\Lambda^{1/2}P^\top P\Lambda^{1/2}P^\top = P\Lambda P^\top = A$. ☺

Exercise 1.4.6 Suggested exercise 16

If $A = P\Lambda P^{-1}$, show that $A^k = P\Lambda^k P^{-1}$ with $\Lambda^k = \mathrm{diag}(\lambda_1^k, \dots, \lambda_n^k)$.

Solution: Induction on k; $k = 1$ is the hypothesis. Then

$$A^{k+1} = A^k A = P\Lambda^k P^{-1} P\Lambda P^{-1} = P\Lambda^k (P^{-1}P)\Lambda P^{-1} = P\Lambda^{k+1}P^{-1}.$$

The product of diagonal matrices is diagonal with the products of the entries, so $\Lambda^k = \mathrm{diag}(\lambda_i^k)$. ☺

1.4.3 Singular value decomposition

The SVD extends eigendecomposition to **any** real matrix: square or rectangular, full rank or rank deficient, symmetric or not.

Theorem 1.4.5 Singular value decomposition

Every $A \in \mathbb{R}^{m\times n}$ can be written as

$$A = U\Sigma V^\top,$$

with $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ orthonormal, and $\Sigma \in \mathbb{R}^{m\times n}$ diagonal (in the rectangular sense) with nonnegative entries, the *singular values* $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$, where $r = \mathrm{rank}(A)$.

If $\mathrm{rank}(A) = r$, the **reduced SVD** $A = U_r\Sigma_r V_r^\top$, with $U_r \in \mathbb{R}^{m\times r}$, $\Sigma_r \in \mathbb{R}^{r\times r}$, $V_r \in \mathbb{R}^{n\times r}$, keeps only the singular vectors associated with nonzero singular values.

Proposition 1.4.3 Singular vectors and rank-one expansion

For each $i \leq r$: $Av_i = \sigma_i u_i$ and $A^\top u_i = \sigma_i v_i$. Moreover

$$A = \sum_{i=1}^{r} \sigma_i\, u_i v_i^\top,$$

a sum of rank-one matrices ordered from most to least important.

Proof: Multiplying $A = U\Sigma V^\top$ on the right by V: $AV = U\Sigma$, whose i-th column reads $Av_i = \sigma_i u_i$. Similarly $A^\top U = V\Sigma^\top$ gives $A^\top u_i = \sigma_i v_i$. Writing the reduced SVD as a sum of column × row products: $U_r\Sigma_r V_r^\top = \sum_{i=1}^{r}(\sigma_i u_i)v_i^\top$. ☺

Note:

Truncating the sum to the first q terms gives the best rank-q approximation of A (Eckart–Young–Mirsky, see the PCA chapter).

Relation with eigenvalues

Theorem 1.4.6 SVD and eigendecomposition

$$A^\top A = V(\Sigma^\top\Sigma)V^\top, \qquad AA^\top = U(\Sigma\Sigma^\top)U^\top.$$

Hence the right singular vectors V are eigenvectors of $A^\top A$, the left singular vectors U are eigenvectors of

$AA^\top$, and the nonzero eigenvalues of both are $\sigma_1^2, \dots, \sigma_r^2$.

Proof: $A^\top A = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^\top\Sigma V^\top$, with $\Sigma^\top\Sigma = \mathrm{diag}(\sigma_1^2, \dots, \sigma_r^2, 0, \dots, 0)$: this is a spectral decomposition. The same computation works for $AA^\top$. ☺

Note:

So $\sigma_i = \sqrt{\lambda_i(A^\top A)}$; for a symmetric PSD matrix SVD and spectral decomposition coincide; for a symmetric matrix $\sigma_i = |\lambda_i|$. In PCA, $S = \frac{1}{n}X^\top X$ has eigenvalues σ_i^2/n.

Example 1.4.3

$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$: $AA^\top = \begin{bmatrix} 17 & 8 \\ 8 & 17 \end{bmatrix}$ has eigenvalues $25, 9$, so $\sigma_1 = 5$, $\sigma_2 = 3$. With $u_1 = \frac{1}{\sqrt{2}}(1,1)^\top$, $u_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$ and $v_i = A^\top u_i/\sigma_i$:

$$v_1 = \tfrac{1}{\sqrt{2}}(1,1,0)^\top, \qquad v_2 = \tfrac{1}{3\sqrt{2}}(1,-1,4)^\top \approx (0.236, -0.236, 0.943)^\top.$$

R returns the same vectors up to sign (u_i and v_i can be flipped together).

1.5 Vector Calculus

Many estimation and learning problems are **optimization problems**: we choose parameters to minimize a loss or maximize a likelihood. Gradients tell us how the objective changes with the parameters and give the directions used by optimization algorithms.

1.5.1 Gradient

For $f : \mathbb{R}^p \to \mathbb{R}$, a partial derivative $\frac{\partial f}{\partial x_j}$ measures the rate of change with respect to x_j keeping the other variables fixed. For instance, if $f(x_1, x_2) = x_1^2 x_2 + 3x_2^2$ then $\frac{\partial f}{\partial x_1} = 2x_1x_2$ and $\frac{\partial f}{\partial x_2} = x_1^2 + 6x_2$.

Definition 1.5.1: Gradient

The gradient of $f : \mathbb{R}^p \to \mathbb{R}$ is the **column** vector of partial derivatives

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_p} \end{bmatrix}.$$

Theorem 1.5.1 Steepest ascent

The gradient points in the direction of steepest increase of f.

Proof: For a unit vector u, the directional derivative is $D_u f(x) = \frac{d}{dt} f(x + tu)\big|_{t=0} = \nabla f(x)^\top u$. By Cauchy–Schwarz, $\nabla f^\top u \le \|\nabla f\|_2 \|u\|_2 = \|\nabla f\|_2$, with equality iff $u = \nabla f/\|\nabla f\|$. ☺

Note:

Hence gradient descent $x_{t+1} = x_t - \eta\nabla f(x_t)$ moves along the steepest decrease; stationary points satisfy $\nabla f = 0$.

Exercise 1.5.1 Suggested exercise 17

Compute the gradient of $f(x_1, x_2) = x_1^2 + 3x_1x_2 + x_2^2$.

Solution: $\frac{\partial f}{\partial x_1} = 2x_1 + 3x_2$, $\frac{\partial f}{\partial x_2} = 3x_1 + 2x_2$, so $\nabla f(x) = (2x_1 + 3x_2,\ 3x_1 + 2x_2)^\top$. Matrix check: $f = x^\top Ax$ with $A = \begin{bmatrix} 1 & 3/2 \\ 3/2 & 1 \end{bmatrix}$ symmetric, and $\nabla f = 2Ax$.

Useful results

Theorem 1.5.2 Linear and quadratic forms

$$\nabla_x(a^\top x) = \nabla_x(x^\top a) = a, \qquad \nabla_x(x^\top x) = 2x, \qquad \nabla_x(x^\top Ax) = (A + A^\top)x,$$

and $\nabla_x(x^\top Ax) = 2Ax$ if A is symmetric.

Proof: See Supplementary exercises 1, 4 and 5 below ($x^\top a = a^\top x$ is a scalar, and $x^\top x = x^\top Ix$).

Quadratic forms

Definition 1.5.2: Quadratic form

A quadratic form in $x \in \mathbb{R}^n$ with $A \in \mathbb{R}^{n\times n}$ is $f(x) = x^\top Ax$. For A symmetric:

- **positive definite** if $x^\top Ax > 0$ for all $x \neq 0$;
- **positive semidefinite** if $x^\top Ax \geq 0$;
- **negative definite** if $x^\top Ax < 0$ for all $x \neq 0$;
- **indefinite** if the sign depends on x.

These notions appear everywhere in statistics, especially for covariance matrices and Hessians.

Note:

One can always assume A symmetric: $x^\top Ax$ is a scalar, so $x^\top Ax = x^\top A^\top x = x^\top \frac{A+A^\top}{2} x$; the antisymmetric part does not contribute.

Theorem 1.5.3 Definiteness through eigenvalues

Let A be symmetric. A is positive definite $\iff$ all $\lambda_j > 0$; positive semidefinite $\iff$ all $\lambda_j \geq 0$; negative definite $\iff$ all $\lambda_j < 0$; indefinite $\iff$ it has eigenvalues of both signs.

Proof: Write $A = P\Lambda P^\top$ and $y = P^\top x$ (a bijection, with $x \neq 0 \iff y \neq 0$). Then $x^\top Ax = y^\top \Lambda y = \sum_j \lambda_j y_j^2$, which is positive for all $y \neq 0$ iff every $\lambda_j > 0$ (take $y = e_j$ for necessity). The other cases are analogous.

1.5.2 Jacobian and Hessian matrices

Definition 1.5.3: Jacobian

For $f : \mathbb{R}^p \to \mathbb{R}^n$ with components $f_1, \ldots, f_n$,

$$J(x) = \frac{\partial f}{\partial x^\top} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_p} \end{bmatrix} \in \mathbb{R}^{n\times p}.$$

The Jacobian gives the best local linear approximation $f(x + h) \approx f(x) + J(x)h$. For scalar f, $J = \nabla f^\top$.

Proposition 1.5.1 Jacobian of a linear map

If $f(x) = Ax$ then $J(x) = A$.

Proof: $f_i(x) = \sum_k a_{ik} x_k$, so $\frac{\partial f_i}{\partial x_j} = a_{ij}$.

Chain rule

Theorem 1.5.4 Chain rule

Let $g : \mathbb{R}^p \to \mathbb{R}^k$, $f : \mathbb{R}^k \to \mathbb{R}^n$ and $h = f \circ g$. Then $J_h(x) = J_f(g(x))\, J_g(x)$. In the scalar-output case ($n = 1$):

$$\nabla_x h(x) = J_g(x)^\top \nabla_u f(u)\big|_{u=g(x)}.$$

Proof: By the multivariable chain rule,

$$\frac{\partial h}{\partial x_j} = \sum_{\ell=1}^{k} \frac{\partial f}{\partial u_\ell} \frac{\partial g_\ell}{\partial x_j} = \sum_{\ell} (J_g)_{\ell j} (\nabla f)_\ell = (J_g^\top \nabla f)_j.$$

Dimension check: $(p \times k)(k \times 1) = p \times 1$.

Exercise 1.5.2 Suggested exercise 18

Let $g(x) = x_1 + 2x_2$ and $f(u) = u^2$. Use the chain rule to compute ∇h for $h(x) = f(g(x))$.

Solution: $\nabla h(x) = f'(g(x))\, \nabla g(x) = 2(x_1 + 2x_2) \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2x_1 + 4x_2 \\ 4x_1 + 8x_2 \end{bmatrix}$. Direct check: $h = x_1^2 + 4x_1 x_2 + 4x_2^2$.

Exercise 1.5.3 Suggested exercise 19

Least squares: $L(\beta) = \|y - X\beta\|^2$ with $X \in \mathbb{R}^{n \times p}$ of full rank. Compute ∇L and the minimizer.

Solution: $L(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X \beta$ ($y^\top X \beta = \beta^\top X^\top y$ is a scalar). Hence

$$\nabla L(\beta) = -2X^\top y + 2X^\top X \beta = -2X^\top (y - X\beta).$$

$\nabla L = 0 \iff X^\top X \beta = X^\top y$ (*normal equations*). Full rank makes $X^\top X$ positive definite ($v^\top X^\top X v = \|Xv\|^2 > 0$ for $v \neq 0$), hence invertible:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

It is the global minimum since the Hessian $2X^\top X$ is positive definite.

Hessian matrix

Definition 1.5.4: Hessian

For $f : \mathbb{R}^p \to \mathbb{R}$,

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_p \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_p^2} \end{bmatrix},$$

i.e. the Jacobian of the gradient. If $f \in C^2$ it is symmetric (Schwarz).

Note:

By Taylor, $f(x+h) \approx f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h$. At a stationary point the sign of $h^\top H h$ decides: H positive definite $\Rightarrow$ local minimum, negative definite $\Rightarrow$ local maximum, indefinite $\Rightarrow$ saddle point. If H is PSD everywhere, f is convex.

Example 1.5.1

$\nabla(x^\top A x) = (A + A^\top)x$ is linear, so the Hessian is the constant $A + A^\top$. For least squares, $\nabla L = -2X^\top y + 2X^\top X\beta$ and $H = 2X^\top X$, which is PSD since $h^\top X^\top X h = \|Xh\|^2 \geq 0$: L is convex.

Exercise 1.5.4 Suggested exercise 20

Compute the Hessian of $f = x_1^2 + 3x_1x_2 + x_2^2$, knowing $\nabla f = (2x_1 + 3x_2,\ 3x_1 + 2x_2)^\top$.

Solution:

$$\nabla^2 f = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix},$$

constant and symmetric. Its eigenvalues are 5 and -1: it is indefinite, so the origin is a saddle point.

1.6 Supplementary Exercises

Throughout these exercises, gradients are column vectors, and for $g : \mathbb{R}^p \to \mathbb{R}^m$ the Jacobian is

$$J_g(x) = \frac{\partial g}{\partial x^\top} = \left[\frac{\partial g_i}{\partial x_j}\right]_{i=1..m,\ j=1..p}$$

(i row index, j column index). The chain rule for gradients reads $\nabla_x h(g(x)) = J_g(x)^\top \nabla h(g(x))$.

Exercise 1.6.1 Supplementary 1

Let $a \in \mathbb{R}^p$ be fixed and $f(x) = a^\top x$. Using componentwise differentiation, show that $\nabla_x(a^\top x) = a$.

Solution: $f(x) = \sum_{k=1}^p a_k x_k$, so $\frac{\partial f}{\partial x_j} = \sum_k a_k \frac{\partial x_k}{\partial x_j} = \sum_k a_k \delta_{kj} = a_j$. Stacking the components, $\nabla_x(a^\top x) = a$.

Exercise 1.6.2 Supplementary 2

Let $g(b) = b$, $b \in \mathbb{R}^p$. Compute $J_g(b) = \frac{\partial g}{\partial b^\top}$.

Solution: $g_i(b) = b_i$, so $(J_g)_{ij} = \frac{\partial b_i}{\partial b_j} = \delta_{ij}$, i.e. $J_g(b) = I_p$.

Exercise 1.6.3 Supplementary 3

Let $y \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$ and $g(\beta) = y - X\beta$. Show that $J_g(\beta) = \frac{\partial g(\beta)}{\partial \beta^\top} = -X$.

Solution: $g_i(\beta) = y_i - \sum_{k=1}^p X_{ik}\beta_k$, hence $(J_g)_{ij} = \frac{\partial g_i}{\partial \beta_j} = -\sum_k X_{ik}\delta_{kj} = -X_{ij}$ (y does not depend on β).

Exercise 1.6.4 Supplementary 4

Let $A \in \mathbb{R}^{p \times p}$ be fixed and $f(x) = x^\top A x$. Using componentwise differentiation, show that $\nabla_x(x^\top A x) = (A + A^\top)x$.

Solution: $f(x) = \sum_{k=1}^{p}\sum_{l=1}^{p} x_k A_{kl} x_l$. By the product rule,

$$\frac{\partial f}{\partial x_j} = \sum_{k,l}\left(\delta_{kj}A_{kl}x_l + x_k A_{kl}\delta_{lj}\right) = \sum_{l} A_{jl}x_l + \sum_{k} A_{kj}x_k = (Ax)_j + (A^\top x)_j.$$

Hence $\nabla_x(x^\top Ax) = (A + A^\top)x$, and $2Ax$ if A is symmetric.

Exercise 1.6.5 Supplementary 5

Let $f(x) = \|x\|_2^2$. Using the quadratic-form result, show that $\nabla_x \|x\|_2^2 = 2x$.

Solution: $\|x\|_2^2 = x^\top I_p x$; by S4 with $A = I_p$: $\nabla_x\|x\|_2^2 = (I_p + I_p^\top)x = 2x$.

Exercise 1.6.6 Supplementary 6

Let $f(x, y) = x^\top Ay$ with $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$. Compute $\nabla_x f(x, y)$ and $\nabla_y f(x, y)$.

Solution: $f = \sum_{i,j} x_i A_{ij} y_j$.

1. $\frac{\partial f}{\partial x_k} = \sum_j A_{kj}y_j = (Ay)_k$, so $\nabla_x f = Ay$ (equivalently $f = (Ay)^\top x$ and S1).
2. $\frac{\partial f}{\partial y_k} = \sum_i x_i A_{ik} = (A^\top x)_k$, so $\nabla_y f = A^\top x$ (equivalently $f = (A^\top x)^\top y$).

Exercise 1.6.7 Supplementary 7

Let $f(x) = a^\top(Bx + c)$ with $x \in \mathbb{R}^p$, $a, c \in \mathbb{R}^m$, $B \in \mathbb{R}^{m\times p}$. Show that $\nabla_x f(x) = B^\top a$.

Solution: $f(x) = a^\top Bx + a^\top c = (B^\top a)^\top x + \text{const}$, so by S1 $\nabla_x f = B^\top a$. Componentwise: $f = \sum_i a_i\big(\sum_j B_{ij}x_j + c_i\big)$ and $\partial_{x_k} f = \sum_i a_i B_{ik} = (B^\top a)_k$.

Exercise 1.6.8 Supplementary 8

Let $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$ with $A \in \mathbb{R}^{p\times p}$, $b \in \mathbb{R}^p$, $c \in \mathbb{R}$. (1) Compute $\nabla_x f(x)$. (2) Compute the Hessian $\nabla_x^2 f(x)$. (3) Simplify both when A is symmetric.

Solution:

1. By S4 and S1: $\nabla_x f = \frac{1}{2}(A + A^\top)x - b$.
2. The Hessian is the Jacobian of the gradient: $\nabla_x^2 f = \frac{\partial}{\partial x^\top}\left(\frac{1}{2}(A + A^\top)x - b\right) = \frac{1}{2}(A + A^\top)$, constant.
3. If $A = A^\top$: $\nabla f = Ax - b$ and $\nabla^2 f = A$. The stationary point solves $Ax = b$; if A is positive definite it is the global minimum.

Exercise 1.6.9 Supplementary 9

Let $f(x) = \varphi(a^\top x + b)$ with $x, a \in \mathbb{R}^p$, $b \in \mathbb{R}$ and $\varphi : \mathbb{R} \to \mathbb{R}$ differentiable. Show that $\nabla_x f(x) = \varphi'(a^\top x + b)\, a$. Use this to compute the gradients of $f_1(x) = (a^\top x + b)^k$ and $f_2(x) = \exp(a^\top x + b)$.

Solution: Let $u = a^\top x + b$. By the scalar chain rule $\frac{\partial f}{\partial x_j} = \varphi'(u)\frac{\partial u}{\partial x_j} = \varphi'(u)\, a_j$, hence $\nabla_x f = \varphi'(a^\top x + b)\, a$.

- $\varphi(u) = u^k$, $\varphi'(u) = ku^{k-1}$: $\nabla f_1 = k(a^\top x + b)^{k-1}\, a$.
- $\varphi(u) = e^u$, $\varphi'(u) = e^u$: $\nabla f_2 = \exp(a^\top x + b)\, a$.

Exercise 1.6.10 Supplementary 10

Let $f(x) = \varphi(x^\top Ax)$ with $A \in \mathbb{R}^{p\times p}$ and φ differentiable. Show that $\nabla_x f(x) = \varphi'(x^\top Ax)(A + A^\top)x$. Then compute the gradient of $\exp(-x^\top Ax)$ for A symmetric.

Solution: Let $u = x^\top Ax$. Then $\frac{\partial f}{\partial x_j} = \varphi'(u)\frac{\partial u}{\partial x_j}$ and by S4 $\nabla_x u = (A + A^\top)x$, so $\nabla_x f = \varphi'(x^\top Ax)(A + A^\top)x$. With $\varphi(u) = e^{-u}$, $\varphi'(u) = -e^{-u}$ and $A + A^\top = 2A$:

$$\nabla_x \exp(-x^\top Ax) = -2\exp(-x^\top Ax)\, Ax$$

(the gradient of a Gaussian kernel).

Exercise 1.6.11 Supplementary 11

Let $g(x) = Ax - b$ with $A \in \mathbb{R}^{m\times p}$, $x \in \mathbb{R}^p$, $b \in \mathbb{R}^m$, and $f(x) = \|g(x)\|_2^2$. Use the chain rule to show that $\nabla_x f(x) = 2A^\top(Ax - b)$.

Solution: $f = h \circ g$ with $h(u) = \|u\|_2^2$, $\nabla_u h = 2u$ (S5), and $J_g(x) = A$ (as in S3). By the chain rule

$$\nabla_x f = J_g^\top \nabla h(g(x)) = A^\top \cdot 2(Ax - b) = 2A^\top(Ax - b).$$

With $A = X$, $b = y$, $x = \beta$ this is the least-squares gradient (Exercise 19).

Exercise 1.6.12 Supplementary 12

Let $S(\beta) = (y - X\beta)^\top W(y - X\beta)$ with $W \in \mathbb{R}^{n\times n}$ symmetric. Use the chain rule to show that $\nabla_\beta S(\beta) = -2X^\top W(y - X\beta)$.

Solution: $r(\beta) = y - X\beta$ has $J_r = -X$ (S3); $h(r) = r^\top Wr$ has $\nabla_r h = (W + W^\top)r = 2Wr$ (S4, W symmetric). Hence

$$\nabla_\beta S = J_r^\top \nabla_r h = (-X)^\top 2W(y - X\beta) = -2X^\top W(y - X\beta).$$

(Weighted least squares: $\nabla S = 0 \Rightarrow \hat\beta = (X^\top WX)^{-1}X^\top Wy$.)

Exercise 1.6.13 Supplementary 13

Let $\sigma(z) = \frac{1}{1+\exp(-z)}$ and $f(\beta) = \sigma(x^\top\beta)$ with $x, \beta \in \mathbb{R}^p$. (1) Show that $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. (2) Use the chain rule to compute $\nabla_\beta f(\beta)$.

Solution:

1. $\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}}$, and $1 - \sigma(z) = \frac{1+e^{-z}-1}{1+e^{-z}} = \frac{e^{-z}}{1+e^{-z}}$. Hence $\sigma' = \sigma(1 - \sigma)$.
2. By S9 with $\varphi = \sigma$ and $a = x$: $\nabla_\beta\, \sigma(x^\top\beta) = \sigma(x^\top\beta)\big(1 - \sigma(x^\top\beta)\big)\, x$.

Exercise 1.6.14 Supplementary 14

Let $y \in \{-1, 1\}$ and $\ell(\beta) = \log\big(1 + \exp(-yx^\top\beta)\big)$. Use the chain rule to show that

$$\nabla_\beta \ell(\beta) = -\frac{yx}{1 + \exp(yx^\top\beta)}.$$

Solution: Let $u = -yx^\top\beta$, so $\nabla_\beta u = -yx$ (S1). $\ell = \log(1 + e^u)$ and $\frac{d\ell}{du} = \frac{e^u}{1+e^u}$. Then

$$\nabla_\beta \ell = \frac{e^u}{1 + e^u}(-yx) = -\frac{yx}{e^{-u} + 1} = -\frac{yx}{1 + \exp(yx^\top\beta)},$$

dividing numerator and denominator by e^u and using $-u = yx^\top\beta$. Equivalently $\nabla\ell = -\sigma(-yx^\top\beta)\, yx$.

Exercise 1.6.15 Supplementary 15

Let $r(\theta) = (r_1(\theta), \dots, r_n(\theta))^\top \in \mathbb{R}^n$ be a differentiable vector of residuals, $\theta \in \mathbb{R}^p$, and $S(\theta) = \frac{1}{2}\|r(\theta)\|_2^2$. Show that $\nabla_\theta S(\theta) = J_r(\theta)^\top r(\theta)$.

Solution: $S = \frac{1}{2}\sum_{i=1}^n r_i(\theta)^2$, so

$$\frac{\partial S}{\partial \theta_j} = \sum_{i=1}^n r_i \frac{\partial r_i}{\partial \theta_j} = \sum_i (J_r)_{ij}\, r_i = (J_r^\top r)_j,$$

i.e. $\nabla_\theta S = J_r(\theta)^\top r(\theta)$ (or: chain rule with $h(u) = \frac{1}{2}\|u\|^2$, $\nabla h = u$). This is the basis of Gauss–Newton; for $r = y - X\theta$, $J_r = -X$ and we recover $-X^\top(y - X\theta)$.

1.7 R Illustrations

In practice one does **not** compute inverses explicitly unless necessary: to solve a system use `solve(A, b)` rather than `solve(A) %*% b`, which is numerically better and more efficient.

1.7.1 Basic operations, products, systems

```
A <- matrix(c(1, 2, 3,
              4, 5, 6), nrow = 2, byrow = TRUE)
B <- matrix(c(1, 0, 1,
              0, 1, 1), nrow = 2, byrow = TRUE)
A + B # elementwise sum
t(A) # transpose
A %*% t(B) # matrix product (A * B is elementwise!)

A <- matrix(c(2, 3, 5,
              4, -2, -7,
              9, 5, -3), nrow = 3, byrow = TRUE)
b <- c(1, 8, 2)
solve(A, b) # 2.445378 -3.285714 1.193277
```

1.7.2 Rank, determinant, inverse, Kronecker

```
A <- matrix(c(1, 2,
              3, 5), nrow = 2, byrow = TRUE)
det(A) # -1
solve(A) # inverse: [-5 2; 3 -1]
qr(A)$rank # 2

A <- matrix(c(1, 2, 3, 4), nrow = 2, byrow = TRUE)
B <- matrix(c(0, 1, 1, 0), nrow = 2, byrow = TRUE)
kronecker(A, B) # 4 x 4 matrix
```

1.7.3 Norms and distances

```
x <- c(3, -4)
c(l1 = sum(abs(x)), l2 = sqrt(sum(x^2)), linf = max(abs(x))) # 7 5 4

distance_p <- function(x, y, p = 2) {
  z <- x - y
  if (is.infinite(p)) {
    return(max(abs(z)))
  }
```

```
  (sum(abs(z)^p))^(1/p)
}
distance_p(c(1, 0, -1), c(2, -2, 3), p = 2) # 4.582576
distance_p(c(1, 0, -1), c(2, -2, 3), p = 1) # 7
distance_p(c(1, 0, -1), c(2, -2, 3), p = Inf) # 4
```

1.7.4 Decompositions

```
# Cholesky: chol() returns the UPPER factor R with A = t(R) %*% R
A <- matrix(c(4, 2, 2,
              2, 2, 1,
              2, 1, 3), nrow = 3, byrow = TRUE)
L <- t(chol(A))
L %*% t(L) # gives back A

# Eigendecomposition: values in decreasing order, unit eigenvectors (sign arbitrary)
S <- matrix(c(2, 1, 1, 2), nrow = 2, byrow = TRUE)
eigen(S) # values 3 1

# SVD
A <- matrix(c(3, 2, 2,
              2, 3, -2), nrow = 2, byrow = TRUE)
s <- svd(A) # s$d = 5 3, s$u, s$v
A_rec <- s$u %*% diag(s$d, nrow = length(s$d)) %*% t(s$v)
round(A_rec, 10) # gives back A
```

Note:

`nrow = length(s$d)` avoids that `diag(3)`, with a single value, builds I_3 instead of the 1×1 matrix [3].

Chapter 2

Principal Component Analysis (PCA)

2.1 Empirical Covariance Matrices

Before deriving PCA, it is useful to make explicit the role of the *empirical covariance matrix*: it is the object PCA works on directly.

Setting. We have n observations $x_1, \dots, x_n \in \mathbb{R}^p$, i.e. n individuals on which we measure p variables. We write x_{ij} for the j-th variable of the i-th observation.

Definition 2.1.1: Empirical mean

The *empirical mean* of the dataset is the vector

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{bmatrix} \in \mathbb{R}^p, \qquad \bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}.$$

Each component $\bar{x}_j$ is simply the mean of the j-th variable.

Definition 2.1.2: Empirical variance

The *empirical variance* of the j-th variable is

$$s_j^2 = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2 \geqslant 0,$$

i.e. the mean squared deviation of the values $x_{1j}, \dots, x_{nj}$ from their mean $\bar{x}_j$. It measures how spread out the variable is; $s_j = \sqrt{s_j^2}$ is the *standard deviation*.

Note:

Some texts divide by $n-1$ instead of n (unbiased estimator). In these notes we always use $\frac{1}{n}$; for PCA the choice is irrelevant, since it only rescales everything by a constant.

Definition 2.1.3: Empirical covariance matrix

The *empirical covariance matrix* is

$$S = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^\top \in \mathbb{R}^{p\times p}.$$

Entry by entry:

$$S_{jk} = \frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

Note:

Each term $(x_i - \bar{x})(x_i - \bar{x})^\top$ is a *column* $\times$ *row* product ($p \times 1$ times $1 \times p$), hence a $p \times p$ matrix (of rank 1). S is the average of these n matrices.

Note:

n or $n-1$? Here we divide by n: this is the empirical-distribution convention (it coincides with the maximum-likelihood estimator in the Gaussian case). The usual *unbiased* estimator divides by $n-1$ instead:

$$S_{\text{unb}} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^\top = \frac{n}{n-1}\,S.$$

For PCA nothing changes: the two matrices differ by a positive constant, so they have the same eigenvectors (the principal directions) and the same explained-variance ratios; only the numerical eigenvalues change, all multiplied by $\frac{n}{n-1}$.

2.1.1 Matrix form

Definition 2.1.4: Data matrix

The *data matrix* $X \in \mathbb{R}^{n\times p}$ contains the observations **by row**:

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix} \qquad \text{(rows = observations, columns = variables).}$$

The data are *centered* if $\bar{x} = 0_p$, i.e. every column of X has zero mean.

Theorem 2.1.1

If the data are centered ($\bar{x} = 0_p$), then

$$S = \frac{1}{n}X^\top X.$$

In general, letting $X_c = X - \mathbf{1}_n\bar{x}^\top$ be the centered matrix ($\bar{x}^\top$ is subtracted from every row), we always have $S = \frac{1}{n}X_c^\top X_c$.

Proof: The product $X^\top X$ can be written as a sum of column $\times$ row products: the i-th column of $X^\top$ is x_i and the i-th row of X is $x_i^\top$, so

$$X^\top X = \sum_{i=1}^{n} x_i x_i^\top.$$

If $\bar{x} = 0_p$, then $x_i - \bar{x} = x_i$ and the definition of S becomes $S = \frac{1}{n}\sum_i x_i x_i^\top = \frac{1}{n}X^\top X$. In the general case, apply the same computation to X_c, whose rows are $(x_i - \bar{x})^\top$.

Note:

This is why in practice **data are always centered** before PCA: from then on the covariance is simply $\frac{1}{n}X^\top X$.

2.1.2 How to read S

The covariance matrix summarizes how the variables vary, individually and jointly:

- **diagonal**: $S_{jj} = \frac{1}{n}\sum_i (x_{ij} - \bar{x}_j)^2$ is the variance s_j^2 of variable j;
- **off-diagonal**: S_{jk} is the *covariance* between variables j and k;
  - $S_{jk} > 0$: when j is above its mean, k tends to be above its mean too;
  - $S_{jk} < 0$: when one goes up, the other tends to go down;
  - $S_{jk} \approx 0$: no *linear* association.

Example 2.1.1

If $p = 3$ (variables: height, weight, age):

$$S = \begin{bmatrix} \text{Var(height)} & \text{Cov(height, weight)} & \text{Cov(height, age)} \\ \text{Cov(weight, height)} & \text{Var(weight)} & \text{Cov(weight, age)} \\ \text{Cov(age, height)} & \text{Cov(age, weight)} & \text{Var(age)} \end{bmatrix}.$$

Note:

The value of S_{jk} depends on the units of measurement: a "large" covariance must be judged *relative to the scales* of the variables. To compare, one uses the correlation

$$r_{jk} = \frac{S_{jk}}{\sqrt{S_{jj}\, S_{kk}}} \in [-1, 1].$$

2.1.3 Basic properties

Proposition 2.1.1 Symmetry

$S^\top = S$.

Proof: Each term is symmetric: $\big((x_i - \bar{x})(x_i - \bar{x})^\top\big)^\top = (x_i - \bar{x})(x_i - \bar{x})^\top$, and a sum of symmetric matrices is symmetric.

Proposition 2.1.2 Positive semidefiniteness

For every vector $a \in \mathbb{R}^p$,

$$a^\top S a \geqslant 0.$$

More precisely, $a^\top S a$ is the empirical variance of the data projected along a, i.e. of the numbers $a^\top x_1, \ldots, a^\top x_n$.

Proof:

$$a^\top S a = \frac{1}{n}\sum_{i=1}^n a^\top (x_i - \bar{x})(x_i - \bar{x})^\top a = \frac{1}{n}\sum_{i=1}^n \big(a^\top x_i - a^\top \bar{x}\big)^2 \;\geqslant\; 0,$$

since $a^\top(x_i - \bar{x})$ is a scalar, so it equals its transpose $(x_i - \bar{x})^\top a$. The last expression is exactly the empirical variance of the values $a^\top x_i$, whose mean is $a^\top \bar{x}$.

Corollary 2.1.1 Nonnegative eigenvalues

S has p real eigenvalues $\lambda_1 \geqslant \ldots \geqslant \lambda_p \geqslant 0$ and an orthonormal basis of eigenvectors.

Proof: Since S is symmetric, the spectral theorem gives real eigenvalues and an orthonormal basis of eigenvectors. If $Sv = \lambda v$ with $\|v\| = 1$, then by positive semidefiniteness

$$\lambda = \lambda\, v^\top v = v^\top S v \geqslant 0.$$

Note:

This is why positive semidefiniteness is crucial: eigenvalues of S are *variances* (along the eigenvector directions), so they cannot be negative. It is also the bridge to PCA: finding the direction a (with $\|a\| = 1$) along which the data vary the most means maximizing $a^\top S a$.

Proposition 2.1.3 Rank bound

If $X \in \mathbb{R}^{n\times p}$, then

$$\operatorname{rank}(X^\top X) = \operatorname{rank}(X) \leqslant \min(p, n).$$

For centered data, $S = \frac{1}{n} X_c^\top X_c$ satisfies the sharper bound

$$\operatorname{rank}(S) \leqslant \min(p,\, n-1).$$

Proof: $\operatorname{rank}(X^\top X) = \operatorname{rank}(X)$**.** We show $\ker(X^\top X) = \ker(X)$. If $Xv = 0$ then clearly $X^\top X v = 0$. Conversely, if $X^\top X v = 0$ then

$$0 = v^\top X^\top X v = \|Xv\|^2 \quad \Longrightarrow \quad Xv = 0.$$

Both matrices have p columns, so by rank–nullity they have the same rank. And $\operatorname{rank}(X) \leqslant \min(p, n)$ because X has n rows and p columns.

Centered case. Centering imposes one linear constraint on the rows: they sum to zero,

$$\sum_{i=1}^{n} (x_i - \bar{x}) = n\bar{x} - n\bar{x} = 0, \qquad \text{i.e.} \qquad \mathbf{1}_n^\top X_c = 0.$$

So the n rows of X_c are linearly dependent and span a space of dimension at most $n-1$. Hence $\operatorname{rank}(S) = \operatorname{rank}(X_c) \leqslant \min(p, n-1)$ (the factor $\frac{1}{n}$ does not change the rank).

Note:

High-dimensional setting. When $n \ll p$ (few observations, many variables), the rank bound gives $\operatorname{rank}(S) \leqslant n - 1 < p$: S is low rank, hence singular, with at least $p - n + 1$ zero eigenvalues. This situation is called a *high-dimensional setting* and requires special treatment.

Example 2.1.2

With $n = 2$ observations in $\mathbb{R}^3$, after centering $x_2 - \bar{x} = -(x_1 - \bar{x})$: all the data lie on a single line through $\bar{x}$, so $\operatorname{rank}(S) \leqslant 1$ even though S is 3×3.

Exercise 2.1.1 Suggested exercise 21

Let S be the empirical covariance matrix of the n p-dimensional vectors x_i. Show that for every vector $a \in \mathbb{R}^p$, $a^\top S a \geqslant 0$.

Solution:

$$a^\top S a = \frac{1}{n}\sum_{i=1}^{n} a^\top (x_i - \bar{x})(x_i - \bar{x})^\top a = \frac{1}{n}\sum_{i=1}^{n} \left(a^\top (x_i - \bar{x})\right)^2 \geqslant 0,$$

because $a^\top (x_i - \bar{x})$ is a scalar, hence equal to its transpose $(x_i - \bar{x})^\top a$: every term is a square. It is the empirical variance of the projected data $a^\top x_i$.

2.2 Link With Linear Algebra

PCA is one of the main applications of the linear algebra of the previous chapter. The following facts are fundamental:

- the covariance matrix S is **symmetric positive semidefinite**;
- its eigenvectors form an **orthonormal basis** of principal directions (spectral theorem);
- with $P_q = [\,\boldsymbol{p}_1 \mid \cdots \mid \boldsymbol{p}_q\,]$, the matrix $P_q P_q^\top$ is the **orthogonal projector** onto the principal subspace;
- hence the reconstruction
$$\hat{X} = X P_q P_q^\top$$
is the orthogonal projection of the data (each row $x_i^\top$) onto that subspace.

So PCA is not only a statistical method: it is a concrete application of spectral decomposition, orthogonal projection and low-rank approximation. These are also the three equivalent points of view on PCA used in this module: *variance maximization*, *projection and reconstruction*, *best rank-q approximation* of the data matrix.
Notation. Throughout the chapter: $X \in \mathbb{R}^{n\times p}$ centered data matrix, $S = \frac{1}{n}X^\top X = P\Lambda P^\top$, P_q the first q principal directions, $Z = XP_q$ the latent scores, $\hat{X} = XP_qP_q^\top$ the rank-q reconstruction.

> **Exercise 2.2.1** Suggested exercise 22
> Let $X \in \mathbb{R}^{n\times p}$ be centered, $S = \frac{1}{n}X^\top X$, and $P = [\,\boldsymbol{p}_1 \cdots \boldsymbol{p}_p\,]$ the orthonormal matrix of eigenvectors of S. The reconstructed centered data matrix is $\hat{X}_q = XP_qP_q^\top$. Show that with $q = p$, $\hat{X}_p = X$.

Solution: For $q = p$, $P_p = P$ is *square* with $P^\top P = I_p$. A square matrix with a left inverse $P^\top$ has the same matrix as right inverse, so also $PP^\top = I_p$. (Alternatively: $PP^\top = \sum_{j=1}^{p} \boldsymbol{p}_j \boldsymbol{p}_j^\top$ is the projector onto the whole $\mathbb{R}^p$, i.e. the identity.) Hence
$$\hat{X}_p = XPP^\top = XI_p = X.$$
Keeping all components the reconstruction is exact (lost variance $J_0 = 0$). ☺

2.3 Maximum Variance Principle

The main objective of PCA is to find a **low-dimensional subspace** such that the **projected data have maximum variance**. The idea:

- if the projected points still *spread out* substantially, the projection retains information;
- if the projected points *collapse* into a small region, much structure has been removed.

From now on the data are **centered** ($\bar{x} = 0_p$) and $S = \frac{1}{n}X^\top X$.

> **Note:**
> **Notation clash.** The course uses p both for the number of variables and for a direction vector. To keep them apart, here the direction is written in bold: $\boldsymbol{p} \in \mathbb{R}^p$.

2.3.1 One direction

> **Definition 2.3.1: Latent score**
> Given a direction $\boldsymbol{p} \in \mathbb{R}^p$, the *latent score* of observation i is
> $$z_i = \boldsymbol{p}^\top x_i \in \mathbb{R},$$
> i.e. the coordinate of x_i along $\boldsymbol{p}$ (its projection, when $\|\boldsymbol{p}\| = 1$).

By the positive semidefiniteness proposition (the data are centered, so the scores have mean 0), the variance of the latent scores is

$$\operatorname{var}(z) = \frac{1}{n}\sum_{i=1}^{n} z_i^2 = \boldsymbol{p}^\top S \boldsymbol{p}.$$

Note:

Why the constraint $\boldsymbol{p}^\top \boldsymbol{p} = 1$. Without it the problem has no finite maximum: for any scalar $c \in \mathbb{R}$,

$$\operatorname{var}(cz) = (c\boldsymbol{p})^\top S (c\boldsymbol{p}) = c^2\, \boldsymbol{p}^\top S \boldsymbol{p},$$

which, whenever $\boldsymbol{p}^\top S \boldsymbol{p} > 0$, becomes arbitrarily large by taking $|c|$ large. The constraint fixes the scale, so that we compare only *directions*, not lengths.

Exercise 2.3.1 Suggested exercise 23

Let $X \in \mathbb{R}^{n\times p}$ be centered, $S = \frac{1}{n} X^\top X = P \Lambda P^\top$, let $P_2 = [\boldsymbol{p}_1, \boldsymbol{p}_2] \in \mathbb{R}^{p\times 2}$ contain the first two eigenvectors and $Z_2 = XP_2$ the matrix of scores on the first two principal components. Show that the score vector $\boldsymbol{z}_i$ (the i-th row of Z_2) can be written as

$$\boldsymbol{z}_i = \sum_{j=1}^{p} x_{ij} \begin{bmatrix} p_{j1} & p_{j2} \end{bmatrix}.$$

Solution: The i-th row of a product XP_2 is the i-th row of X times P_2: $\boldsymbol{z}_i = x_i^\top P_2$ ($1\times p$ times $p \times 2$). Its k-th entry, $k \in \{1, 2\}$, is

$$(\boldsymbol{z}_i)_k = (XP_2)_{ik} = \sum_{j=1}^{p} x_{ij} (P_2)_{jk} = \sum_{j=1}^{p} x_{ij}\, p_{jk},$$

where p_{jk} is the j-th component of $\boldsymbol{p}_k$. Putting the two entries side by side,

$$\boldsymbol{z}_i = \Big(\sum_j x_{ij} p_{j1},\ \sum_j x_{ij} p_{j2} \Big) = \sum_{j=1}^{p} x_{ij} \begin{bmatrix} p_{j1} & p_{j2} \end{bmatrix}.$$

So $\boldsymbol{z}_i$ is a linear combination of the *rows* of P_2 (the loadings of the variables, see the section on the latent space), weighted by the values x_{ij} of observation i.

Exercise 2.3.2 Suggested exercise 24

Let $\tilde{x}_i = x_i - \bar{x}$, $i = 1, \dots, n$, be the centered data and $S = \frac{1}{n}\sum_{i=1}^{n} \tilde{x}_i \tilde{x}_i^\top$. Show that the variance of the projected data $z_i = \boldsymbol{p}^\top \tilde{x}_i$ is $\operatorname{var}(z) = \boldsymbol{p}^\top S \boldsymbol{p}$, where $\boldsymbol{p}$ is an eigenvector.

Solution: The scores have mean zero:

$$\bar{z} = \frac{1}{n}\sum_i \boldsymbol{p}^\top \tilde{x}_i = \boldsymbol{p}^\top \Big(\frac{1}{n} \sum_i \tilde{x}_i \Big) = \boldsymbol{p}^\top 0_p = 0.$$

Using $z_i^2 = (\boldsymbol{p}^\top \tilde{x}_i)(\tilde{x}_i^\top \boldsymbol{p})$,

$$\operatorname{var}(z) = \frac{1}{n}\sum_i (z_i - \bar{z})^2 = \frac{1}{n}\sum_i \boldsymbol{p}^\top \tilde{x}_i \tilde{x}_i^\top \boldsymbol{p} = \boldsymbol{p}^\top \Big(\frac{1}{n} \sum_i \tilde{x}_i \tilde{x}_i^\top \Big) \boldsymbol{p} = \boldsymbol{p}^\top S \boldsymbol{p}.$$

The computation holds for any $\boldsymbol{p}$; if $\boldsymbol{p}$ is a unit eigenvector with eigenvalue λ, then $\operatorname{var}(z) = \lambda\, \boldsymbol{p}^\top \boldsymbol{p} = \lambda$.

Definition 2.3.2: PCA optimization problem (first direction)

$$\max_{\boldsymbol{p} \in \mathbb{R}^p} \boldsymbol{p}^\top S \boldsymbol{p} \qquad \text{subject to} \qquad \boldsymbol{p}^\top \boldsymbol{p} = 1.$$

Theorem 2.3.1

Every stationary point of the problem satisfies the eigenvalue equation

$$S\boldsymbol{p} = \lambda \boldsymbol{p},$$

and at such a point the objective equals $\boldsymbol{p}^\top S\boldsymbol{p} = \lambda$. Therefore:

- the **principal directions** are (unit) **eigenvectors** of S;
- the **variance along each principal direction** is the corresponding **eigenvalue**;
- the maximum is λ_1, the largest eigenvalue, attained at its eigenvector $\boldsymbol{p}_1$.

Proof: **Lagrange multiplier.** Consider

$$\mathcal{L}(\boldsymbol{p}, \lambda) = \boldsymbol{p}^\top S\boldsymbol{p} - \lambda\,(\boldsymbol{p}^\top \boldsymbol{p} - 1).$$

Since S is symmetric, $\nabla_{\boldsymbol{p}}\, \boldsymbol{p}^\top S\boldsymbol{p} = 2S\boldsymbol{p}$ and $\nabla_{\boldsymbol{p}}\, \boldsymbol{p}^\top \boldsymbol{p} = 2\boldsymbol{p}$, so

$$\nabla_{\boldsymbol{p}} \mathcal{L} = 2S\boldsymbol{p} - 2\lambda \boldsymbol{p} = 0 \quad \Longleftrightarrow \quad S\boldsymbol{p} = \lambda \boldsymbol{p}.$$

Multiplying on the left by $\boldsymbol{p}^\top$ and using $\boldsymbol{p}^\top \boldsymbol{p} = 1$: $\boldsymbol{p}^\top S\boldsymbol{p} = \lambda$. Among the stationary points, the best one is the eigenvector with the largest eigenvalue.

Direct check that λ_1 is the maximum. Write $S = P\Lambda P^\top$ (see below) and $\boldsymbol{y} = P^\top \boldsymbol{p}$; since P is orthogonal, $\|\boldsymbol{y}\| = \|\boldsymbol{p}\| = 1$. Then

$$\boldsymbol{p}^\top S\boldsymbol{p} = \boldsymbol{y}^\top \Lambda \boldsymbol{y} = \sum_{j=1}^{p} \lambda_j y_j^2 \;\leqslant\; \lambda_1 \sum_{j=1}^{p} y_j^2 = \lambda_1,$$

with equality for $\boldsymbol{y} = e_1$, i.e. $\boldsymbol{p} = \boldsymbol{p}_1$.

Exercise 2.3.3 Suggested exercise 25

Show that the optimization problem $\max_{\boldsymbol{p}} \boldsymbol{p}^\top S\boldsymbol{p}$ subject to $\boldsymbol{p}^\top \boldsymbol{p} = 1$, using a Lagrange multiplier, leads to $S\boldsymbol{p} = \lambda \boldsymbol{p}$.

Solution: Lagrangian: $\mathcal{L}(\boldsymbol{p}, \lambda) = \boldsymbol{p}^\top S\boldsymbol{p} - \lambda(\boldsymbol{p}^\top \boldsymbol{p} - 1)$. From the vector-calculus rules of the previous chapter, $\nabla_{\boldsymbol{p}}(\boldsymbol{p}^\top S\boldsymbol{p}) = (S + S^\top)\boldsymbol{p} = 2S\boldsymbol{p}$ (S symmetric) and $\nabla_{\boldsymbol{p}}(\boldsymbol{p}^\top \boldsymbol{p}) = 2\boldsymbol{p}$. First-order conditions:

$$\nabla_{\boldsymbol{p}} \mathcal{L} = 2S\boldsymbol{p} - 2\lambda \boldsymbol{p} = 0 \iff S\boldsymbol{p} = \lambda \boldsymbol{p}, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = 0 \iff \boldsymbol{p}^\top \boldsymbol{p} = 1.$$

So $\boldsymbol{p}$ is a unit eigenvector, the multiplier λ is its eigenvalue, and the objective equals $\boldsymbol{p}^\top S\boldsymbol{p} = \lambda$: the maximum is reached for $\lambda = \lambda_1$.

2.3.2 Spectral decomposition and q directions

Theorem 2.3.2

(Spectral decomposition.) Since S is symmetric,

$$S = P\Lambda P^\top,$$

where

- $P = [\,\boldsymbol{p}_1 \mid \cdots \mid \boldsymbol{p}_p\,] \in \mathbb{R}^{p \times p}$ is orthogonal ($P^\top P = I$), its columns are the orthonormal eigenvectors $\boldsymbol{p}_j$;
- $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_p)$ contains the eigenvalues, ordered as

  $$\lambda_1 \geqslant \lambda_2 \geqslant \ldots \geqslant \lambda_p \geqslant 0$$

  (nonnegative by the corollary on eigenvalues).

To project onto a q-dimensional subspace ($q < p$) we take an orthonormal basis $W = [\, \boldsymbol{w}_1 \mid \cdots \mid \boldsymbol{w}_q \,] \in \mathbb{R}^{p\times q}$, $W^\top W = I_q$. The variance retained is the sum of the variances of the q scores:

$$\sum_{k=1}^{q} \boldsymbol{w}_k^\top S \boldsymbol{w}_k = \operatorname{tr}(W^\top S W).$$

Note:

Why the trace. The k-th score is $\boldsymbol{z}_k = X_c \boldsymbol{w}_k$, whose variance is $\boldsymbol{w}_k^\top S \boldsymbol{w}_k$, i.e. the (k,k) entry of $W^\top S W$. Summing over k gives the trace. Moreover the trace depends only on the *subspace*, not on the chosen basis: if $W' = WQ$ with $Q \in \mathbb{R}^{q\times q}$ orthogonal, then

$$\operatorname{tr}(Q^\top W^\top S W Q) = \operatorname{tr}(W^\top S W\, Q Q^\top) = \operatorname{tr}(W^\top S W).$$

Theorem 2.3.3

(Best q-dimensional subspace.) Let $S = P\Lambda P^\top$ with $\lambda_1 \geqslant \ldots \geqslant \lambda_p \geqslant 0$. For every $W \in \mathbb{R}^{p\times q}$ with $W^\top W = I_q$,

$$\operatorname{tr}(W^\top S W) \leqslant \lambda_1 + \cdots + \lambda_q,$$

with equality for $W = [\, \boldsymbol{p}_1 \mid \cdots \mid \boldsymbol{p}_q \,]$. In other words

$$\max_{W^\top W = I_q} \operatorname{tr}(W^\top S W) = \lambda_1 + \cdots + \lambda_q,$$

and the best q-dimensional PCA subspace is span($\boldsymbol{p}_1, \ldots, \boldsymbol{p}_q$), spanned by the eigenvectors of the q largest eigenvalues.

Proof: **Step 1: change of basis.** Substituting $S = P\Lambda P^\top$,

$$\operatorname{tr}(W^\top S W) = \operatorname{tr}(W^\top P\, \Lambda\, P^\top W).$$

Let $Y = P^\top W \in \mathbb{R}^{p\times q}$, so that $W^\top P = (P^\top W)^\top = Y^\top$ and

$$\operatorname{tr}(W^\top S W) = \operatorname{tr}(Y^\top \Lambda Y).$$

Y still has orthonormal columns, since P is orthogonal ($PP^\top = I_p$):

$$Y^\top Y = W^\top P P^\top W = W^\top W = I_q.$$

Meaning: Y is the same basis W written in the coordinates of the eigenvectors, where S is diagonal.
Step 2: writing out the trace. Let $Y = (y_{ij})$, $i = 1, \ldots, p$, $j = 1, \ldots, q$. Since Λ is diagonal, $(\Lambda Y)_{ij} = \lambda_i y_{ij}$, hence

$$(Y^\top \Lambda Y)_{jj} = \sum_{i=1}^{p} y_{ij}\, \lambda_i\, y_{ij} = \sum_{i=1}^{p} \lambda_i y_{ij}^2.$$

Summing the diagonal and swapping the sums:

$$\operatorname{tr}(Y^\top \Lambda Y) = \sum_{j=1}^{q} \sum_{i=1}^{p} \lambda_i y_{ij}^2 = \sum_{i=1}^{p} \lambda_i \underbrace{\Big(\sum_{j=1}^{q} y_{ij}^2 \Big)}_{=:\,\alpha_i},$$

so

$$\boxed{\operatorname{tr}(W^\top S W) = \sum_{i=1}^{p} \lambda_i \alpha_i}$$

where α_i is the squared norm of the i-th row of Y. The retained variance is a *weighted sum of the eigenvalues*, with weights α_i.

Step 3: constraints on the weights.
(a) The weights sum to q. Each column $\boldsymbol{y}_j$ of Y has norm 1, so

$$\sum_{i=1}^{p} \alpha_i = \sum_{i=1}^{p}\sum_{j=1}^{q} y_{ij}^2 = \sum_{j=1}^{q} \|\boldsymbol{y}_j\|^2 = q \qquad \big(= \operatorname{tr}(Y^\top Y) = \operatorname{tr}(I_q)\big).$$

(b) Each weight lies in $[0,1]$. $\alpha_i \geqslant 0$ is obvious (sum of squares). For $\alpha_i \leqslant 1$: complete the q orthonormal columns of Y to an orthonormal basis of $\mathbb{R}^p$, obtaining an orthogonal matrix $\tilde{Y} = [\, Y \mid Y' \,] \in \mathbb{R}^{p\times p}$. An orthogonal matrix also satisfies $\tilde{Y}\tilde{Y}^\top = I_p$, so every *row* of $\tilde{Y}$ has norm 1:

$$\underbrace{\sum_{j=1}^{q} y_{ij}^2}_{\alpha_i} + \underbrace{\sum_{j=q+1}^{p} \tilde{y}_{ij}^2}_{\geqslant 0} = 1 \quad\Longrightarrow\quad \alpha_i \leqslant 1.$$

(Equivalently: $\alpha_i = \|Y^\top e_i\|^2$ is the squared norm of the orthogonal projection of e_i onto the column space of Y, and a projection does not lengthen vectors.)
Step 4: the reduced problem. We are left with

$$\max \sum_{i=1}^{p} \lambda_i \alpha_i \qquad \text{subject to} \qquad 0 \leqslant \alpha_i \leqslant 1, \quad \sum_{i=1}^{p} \alpha_i = q :$$

q units of weight to distribute, each eigenvalue receiving at most one. Intuitively we fill up the q largest eigenvalues. Formally:

$$\sum_{i=1}^{p} \lambda_i \alpha_i - \sum_{i=1}^{q} \lambda_i = \sum_{i=1}^{q} \lambda_i (\alpha_i - 1) + \sum_{i=q+1}^{p} \lambda_i \alpha_i.$$

- for $i \leqslant q$: $\alpha_i - 1 \leqslant 0$ and $\lambda_i \geqslant \lambda_q$, so $\lambda_i(\alpha_i - 1) \leqslant \lambda_q(\alpha_i - 1)$ (multiplying by a nonpositive number reverses the inequality);
- for $i > q$: $\alpha_i \geqslant 0$ and $\lambda_i \leqslant \lambda_q$, so $\lambda_i \alpha_i \leqslant \lambda_q \alpha_i$.

Therefore

$$\sum_{i=1}^{p} \lambda_i \alpha_i - \sum_{i=1}^{q} \lambda_i \leqslant \lambda_q \Big(\sum_{i=1}^{q} (\alpha_i - 1) + \sum_{i=q+1}^{p} \alpha_i \Big) = \lambda_q \Big(\sum_{i=1}^{p} \alpha_i - q \Big) = \lambda_q (q - q) = 0,$$

i.e. $\operatorname{tr}(W^\top S W) \leqslant \lambda_1 + \cdots + \lambda_q$ for every W with $W^\top W = I_q$.
Step 5: the bound is attained. For $W = [\, \boldsymbol{p}_1 \mid \cdots \mid \boldsymbol{p}_q \,]$, since $\boldsymbol{p}_i^\top \boldsymbol{p}_j = \delta_{ij}$,

$$Y = P^\top W = [\, e_1 \mid \cdots \mid e_q \,],$$

so rows $1, \ldots, q$ of Y contain a single 1 and the other rows are zero:

$$\alpha_1 = \cdots = \alpha_q = 1, \qquad \alpha_{q+1} = \cdots = \alpha_p = 0,$$

and $\operatorname{tr}(W^\top S W) = \lambda_1 + \cdots + \lambda_q$. The upper bound is reached, so it is the maximum. ☺

Note:

- For $q = 1$ this is exactly the theorem on the first principal component, with $\alpha_i = y_i^2$.
- If $\lambda_q = \lambda_{q+1}$ the optimal subspace is *not unique*: weight can be moved between λ_q and λ_{q+1} without changing the sum.

Example 2.3.1

Eigenvalues $10, 6, 2, 1$ and $q = 2$. The first two principal directions retain $10+6 = 16$ out of 19 ($\approx 84\%$). Any

other plane corresponds to weights $\alpha_i \in [0,1]$ summing to 2, e.g. $\alpha = (1, \frac{1}{2}, \frac{1}{2}, 0)$ gives $10+3+1 = 14 < 16$.

Definition 2.3.3: Captured and lost variance

The variance *captured* by the first q principal components is

$$V_q = \sum_{j=1}^{q} \lambda_j,$$

and the variance *lost* is

$$J_{p-q} = \sum_{j=q+1}^{p} \lambda_j.$$

Note:

$V_q + J_{p-q} = \sum_{j=1}^{p} \lambda_j = \operatorname{tr}(S) = \sum_j S_{jj}$, the *total variance* (sum of the variances of all variables). So the fraction of **explained variance** is

$$\frac{V_q}{\operatorname{tr}(S)} = \frac{\lambda_1 + \cdots + \lambda_q}{\lambda_1 + \cdots + \lambda_p}.$$

This ratio is unchanged if S is rescaled (e.g. n vs $n-1$).

Example 2.3.2

Suppose $p = 3$ and S has eigenvalues $\lambda_1 = 5$, $\lambda_2 = 2$, $\lambda_3 = 1$, so $\operatorname{tr}(S) = 8$.

- $q = 1$: $V_1 = 5$, $J_2 = 3$, explained variance $5/8 = 62.5\%$;
- $q = 2$: $V_2 = 7$, $J_1 = 1$, explained variance $7/8 = 87.5\%$.

Projecting onto the plane spanned by $\boldsymbol{p}_1, \boldsymbol{p}_2$ loses only 12.5% of the variance.

Exercise 2.3.4 Suggested exercise 26

Let $x_1, \ldots, x_n \in \mathbb{R}^p$ be centered and $S = \frac{1}{n}\sum_i x_i x_i^\top = P\Lambda P^\top$, with eigenvalues $\lambda_1 \geqslant \ldots \geqslant \lambda_p \geqslant 0$ and orthonormal eigenvectors $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_p$. Show that $\operatorname{tr}(S) = \sum_{j=1}^{p} \lambda_j$.

Solution: By the cyclic property $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ and $P^\top P = I_p$:

$$\operatorname{tr}(S) = \operatorname{tr}(P\Lambda P^\top) = \operatorname{tr}(\Lambda P^\top P) = \operatorname{tr}(\Lambda) = \sum_{j=1}^{p} \lambda_j.$$

Interpretation: the total variance $\sum_j S_{jj}$ equals the sum of the variances along the principal directions.

Exercise 2.3.5 Suggested exercise 27

Let $S = P\Lambda P^\top \in \mathbb{R}^{p\times p}$ be the spectral decomposition of a covariance matrix, with orthonormal eigenvectors and $\lambda_1 \geqslant \ldots \geqslant \lambda_p \geqslant 0$. For $q \leqslant p$ let $P_q = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_q]$ and $\tilde{S}_q = P_q \Lambda_q P_q^\top$, with Λ_q the diagonal matrix of the first q eigenvalues. Show that

$$\Lambda_q = P_q^\top \tilde{S}_q P_q = \operatorname{diag}(\lambda_1, \ldots, \lambda_q) \qquad \text{and} \qquad \operatorname{tr} \tilde{S}_q = \sum_{j=1}^{q} \lambda_j.$$

Solution: By orthonormality of the columns, $P_q^\top P_q = I_q$ (entry (j,k) is $\boldsymbol{p}_j^\top \boldsymbol{p}_k = \delta_{jk}$). Hence

$$P_q^\top \tilde{S}_q P_q = (P_q^\top P_q)\,\Lambda_q\,(P_q^\top P_q) = I_q \Lambda_q I_q = \Lambda_q.$$

For the trace, again by cyclicity,

$$\operatorname{tr} \tilde{S}_q = \operatorname{tr}(P_q \Lambda_q P_q^\top) = \operatorname{tr}(\Lambda_q P_q^\top P_q) = \operatorname{tr}(\Lambda_q) = \sum_{j=1}^{q} \lambda_j.$$

Careful: $P_q P_q^\top \neq I_p$ for $q < p$, but inside the trace only $P_q^\top P_q = I_q$ is needed.

2.4 A Worked Example in R

We apply everything to a simulated dataset with $n = 150$ observations and $p = 2$ variables.

```
set.seed(123)
n <- 150
x1 <- rnorm(n, sd = 2)
x2 <- 0.6 * x1 + rnorm(n, sd = 0.5) # x2 depends linearly on x1, plus noise
X <- cbind(x1, x2)
```

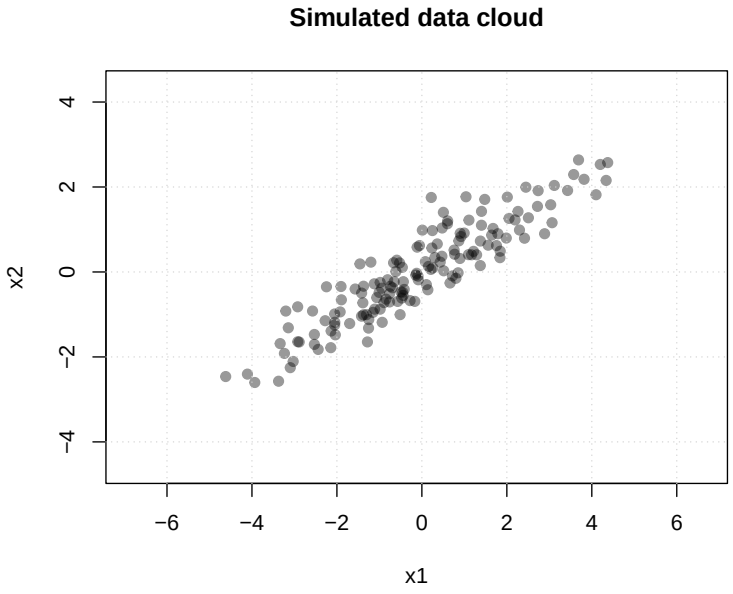


Figure 2.1: The simulated data cloud.

2.4.1 Interpretation of the scatter plot

This graph is the starting point. PCA is a **geometric** method, so we first look at the *shape* of the data cloud: if the cloud is **elongated**, most of the variation may lie close to one direction. PCA formalizes that intuition.

2.4.2 Covariance matrix and spectral decomposition

We center the data and compute $S = \frac{1}{n} X_c^\top X_c$, then its eigen-decomposition:

```
X_centered <- scale(X, center = TRUE, scale = FALSE) # X_c: subtract column means
S <- crossprod(X_centered) / nrow(X_centered) # crossprod(A) = t(A) %*% A
```

```r
S
## x1 x2
## x1 3.584051 2.004268
## x2 2.004268 1.334890

eig <- eigen(S)
eig$values
## [1] 4.757681 0.161260
eig$vectors
## [,1] [,2]
## [1,] -0.8629393 0.5053075
## [2,] -0.5053075 -0.8629393
```

Example 2.4.1

Reading the output with the theory:

- $S_{11} \approx 3.58$ and $S_{22} \approx 1.33$ are the variances; $S_{12} \approx 2.00 > 0$: x_1 and x_2 grow together (correlation $2.00/\sqrt{3.58 \cdot 1.33} \approx 0.92$);
- $\lambda_1 \approx 4.76$, $\lambda_2 \approx 0.16$: almost all variance is along $\boldsymbol{p}_1$;
- the columns of `eig$vectors` are $\boldsymbol{p}_1, \boldsymbol{p}_2$: unit length and orthogonal;
- check: $\lambda_1 + \lambda_2 = 4.918941 = S_{11} + S_{22} = \operatorname{tr}(S)$.

Note:

Sign of eigenvectors. If $\boldsymbol{p}$ is a unit eigenvector, so is $-\boldsymbol{p}$. Here R returns $\boldsymbol{p}_1 = (-0.86, -0.51)^\top$, but $(0.86, 0.51)^\top$ is equally valid. So principal directions and scores are defined only *up to sign*.

Note:

n vs $n-1$ again. Here S uses denominator n. By contrast, `prcomp()` reports component variances with denominator $n-1$ (R's sample-covariance convention). This multiplies the eigenvalues by $n/(n-1)$ but does not change the principal directions, the latent scores (up to sign), the reconstructions or the explained-variance ratios.

2.4.3 Principal directions

We draw each principal direction as an arrow:

```r
mu <- colMeans(X)
P <- eig$vectors
lambda <- eig$values

arrows_df <- data.frame(
  x0 = mu[1], y0 = mu[2],
  x1 = mu[1] + 2 * sqrt(lambda[1]) * P[1, 1],
  y1 = mu[2] + 2 * sqrt(lambda[1]) * P[2, 1],
  x2 = mu[1] + 2 * sqrt(lambda[2]) * P[1, 2],
  y2 = mu[2] + 2 * sqrt(lambda[2]) * P[2, 2]
)
## x0 y0 x1 y1 x2 y2
## x1 -0.04872574 0.0173874 -3.813231 -2.186977 0.3571088 -0.6756769
```

(then plotted with `ggplot2`: `geom_point` for the data and one `geom_segment` with an arrow for each direction.)

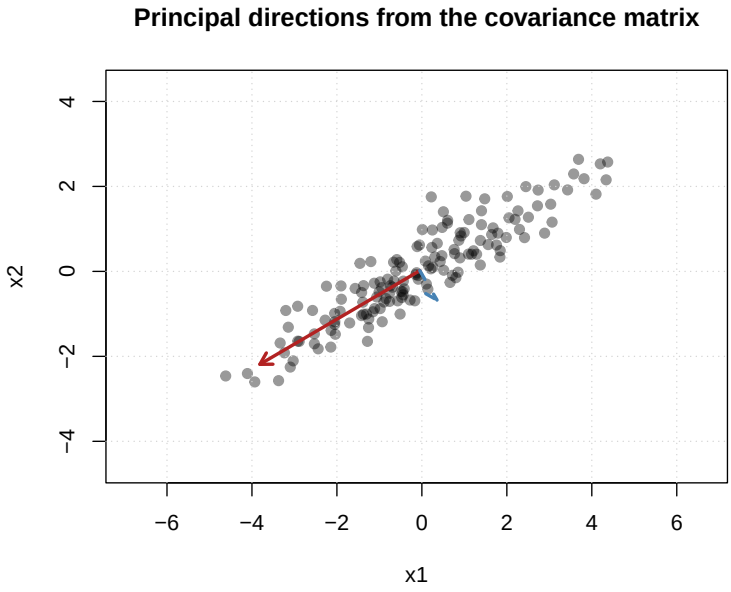


Figure 2.2: First principal direction (solid, red) and second (dashed, blue).

Interpretation

Each arrow is

$$\underbrace{\mu}_{\text{start}} + \underbrace{2\sqrt{\lambda_j}}_{\text{length}} \underbrace{\boldsymbol{p}_j}_{\text{direction}}, \qquad \text{i.e. in R: } \texttt{mu + 2 * sqrt(lambda[j]) * P[, j]}.$$

- **start**: the empirical mean $\mu = \frac{1}{n}\sum_i x_i$, the center of the cloud; this is why both arrows start at `(mu[1], mu[2])`;
- **direction**: the eigenvector, `P[, 1]` for the first principal direction and `P[, 2]` for the second;
- **length**: λ_j is the *variance* along $\boldsymbol{p}_j$, so $\sqrt{\lambda_j}$ is a *standard deviation*, which has the same units as the data. The factor 2 is only a visualization choice to make the arrows clearly visible.

2.4.4 Explained variance ratio

```
prop_var <- eig$values / sum(eig$values)
```

Component	Eigenvalue	Explained variance ratio
1	4.7577	0.9672
2	0.1613	0.0328

In this example **one component already explains** 96.7% **of the variability**: with $q = 1$, $V_1 = \lambda_1$ and the lost variance is only $J_1 = \lambda_2$.

2.4.5 Projection and reconstruction

With one retained component, each observation is projected onto the line through μ spanned by the first principal direction.

Definition 2.4.1: Score and reconstruction ($q = 1$)

The *score* of observation i on the first component and its *reconstruction* are

$$z_i = \boldsymbol{p}_1^\top (x_i - \mu), \qquad \hat{x}_i = \mu + z_i \boldsymbol{p}_1 = \mu + \boldsymbol{p}_1 \boldsymbol{p}_1^\top (x_i - \mu).$$

The *reconstruction error* is the vector $x_i - \hat{x}_i$.

We compute this with `prcomp()`, where `$x` are the scores z_i, `$rotation` the loadings (the matrix P) and `$center` the mean μ:

```
pca_fit <- prcomp(X, center = TRUE, scale. = FALSE)

scores1 <- pca_fit$x[, 1, drop = FALSE] # z_i (n x 1)
loadings1 <- pca_fit$rotation[, 1, drop = FALSE] # p_1 (2 x 1)

Xhat1 <- scores1 %*% t(loadings1) # z_i p_1^T (n x 2)
Xhat1 <- sweep(Xhat1, 2, pca_fit$center, "+") # add mu back to each row

head(data.frame(x1 = X[, 1], x2 = X[, 2],
                x1_hat = Xhat1[, 1], x2_hat = Xhat1[, 2]))
## x1 x2 x1_hat x2_hat
## 1 -1.1209513 -0.27870135 -0.97628325 -0.52575832
## 2 -0.4603550 0.10830813 -0.31560537 -0.13888812
## 3 3.1174166 2.03655127 3.18944333 1.91354762
## 4 0.1410168 -0.41957823 -0.09797001 -0.01144835
## 5 0.2585755 0.09541898 0.21413615 0.17131026
## 6 3.4301300 1.91788032 3.37056335 2.01960526
```

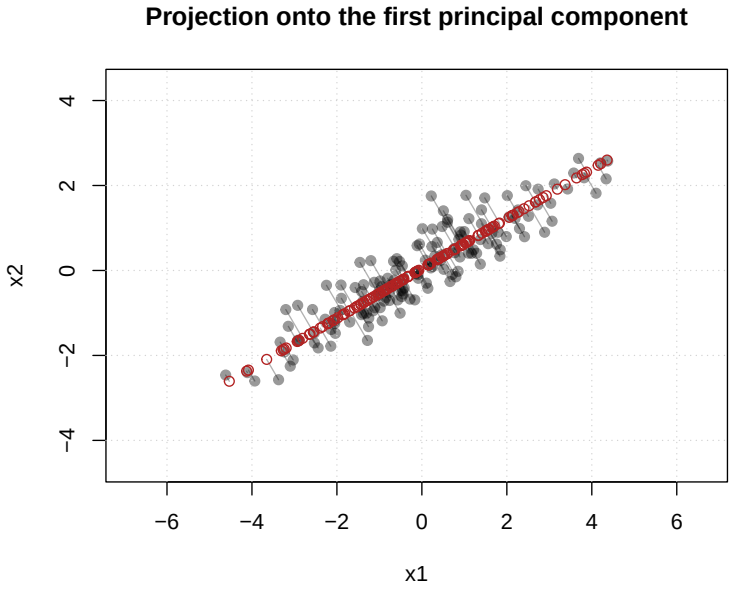


Figure 2.3: Original points (black), reconstructions $\hat{x}_i$ (red circles) and reconstruction errors (segments).

Interpretation

There are three geometric objects in this figure:

- the **original data points** x_i (`x1`, `x2`);

- the **reconstructed points** $\hat{x}_i$ (`x1_hat, x2_hat`): the projections of the original observations onto the first principal component, so they all lie on one line;
- the **segments** between them, each representing the reconstruction error $x_i - \hat{x}_i$.

This is why the graph is useful: it shows visually what PCA does when only one component is kept.

Note:

The segments are **perpendicular** to the line: $x_i - \hat{x}_i$ is the component of $x_i - \mu$ along $\boldsymbol{p}_2$. Hence the average squared length of the segments is exactly the lost variance:

$$\frac{1}{n}\sum_{i=1}^{n}\|x_i - \hat{x}_i\|^2 = \lambda_2 = J_1 \approx 0.16.$$

Maximizing the variance kept is the same as minimizing the reconstruction error.

2.5 Choosing the Dimension of the Projection Subspace

As before, the data are **centered**, so projections pass through the origin.

2.5.1 Projection and reconstruction error

Definition 2.5.1: Projection onto a q-dimensional subspace

Let $P_q = [\,\boldsymbol{p}_1 \mid \cdots \mid \boldsymbol{p}_q\,] \in \mathbb{R}^{p\times q}$ have **orthonormal columns**, i.e. an orthonormal basis of a q-dimensional subspace. The projection of x_i onto that subspace is

$$\tilde{x}_i = P_q P_q^\top x_i = \sum_{j=1}^{q} (\boldsymbol{p}_j^\top x_i)\,\boldsymbol{p}_j.$$

Read right to left: $P_q^\top x_i \in \mathbb{R}^q$ are the q scores (coordinates in the basis), and multiplying by P_q maps them back to $\mathbb{R}^p$.

Proposition 2.5.1 Properties of P_q

1. $P_q^\top P_q = I_q$;
2. $P_q P_q^\top$ is an **orthogonal projector**:

$$(P_q P_q^\top)^\top = P_q P_q^\top \quad \text{(symmetric)}, \qquad (P_q P_q^\top)^2 = P_q P_q^\top \quad \text{(idempotent)}.$$

Proof: 1. The (j,k) entry of $P_q^\top P_q$ is $\boldsymbol{p}_j^\top \boldsymbol{p}_k$, which is 1 if $j = k$ and 0 otherwise, by orthonormality.

2. $(P_q P_q^\top)^\top = (P_q^\top)^\top P_q^\top = P_q P_q^\top$, and $(P_q P_q^\top)(P_q P_q^\top) = P_q \underbrace{(P_q^\top P_q)}_{I_q} P_q^\top = P_q P_q^\top$.

Note:

Idempotent means "projecting twice = projecting once": a point already in the subspace does not move. Note that $P_q P_q^\top \neq I_p$ when $q < p$ (it is a $p \times p$ matrix of rank q), while $P_q^\top P_q = I_q$ always.

Theorem 2.5.1

For every P_q with orthonormal columns, the average squared reconstruction error is

$$\frac{1}{n}\sum_{i=1}^{n}\|x_i - \tilde{x}_i\|^2 = \operatorname{tr}(S) - \operatorname{tr}(P_q^\top S P_q).$$

Hence **minimizing the reconstruction error** and **maximizing the explained variance** $\operatorname{tr}(P_q^\top S P_q)$ are the same problem. The minimum is attained by the first q eigenvectors of S, and equals the lost variance:

$$J_{p-q} = \sum_{j=q+1}^{p}\lambda_j = \frac{1}{n}\sum_{i=1}^{n}\|x_i - \tilde{x}_i\|^2.$$

Proof: Let $\Pi = P_q P_q^\top$. By the previous proposition, $x_i - \tilde{x}_i = (I - \Pi)x_i$ is orthogonal to $\tilde{x}_i = \Pi x_i$ (since $\Pi^\top(I - \Pi) = \Pi - \Pi^2 = 0$), so by Pythagoras

$$\|x_i - \tilde{x}_i\|^2 = \|x_i\|^2 - \|\tilde{x}_i\|^2 = \|x_i\|^2 - \|P_q^\top x_i\|^2,$$

where $\|\tilde{x}_i\|^2 = x_i^\top P_q P_q^\top P_q P_q^\top x_i = \|P_q^\top x_i\|^2$. Averaging over i (data centered, $S = \frac{1}{n}\sum_i x_i x_i^\top$):

$$\frac{1}{n}\sum_i \|x_i\|^2 = \operatorname{tr}(S), \qquad \frac{1}{n}\sum_i \|P_q^\top x_i\|^2 = \operatorname{tr}(P_q^\top S P_q).$$

By the "best q-dimensional subspace" theorem, $\operatorname{tr}(P_q^\top S P_q) \leqslant \lambda_1 + \cdots + \lambda_q$, with equality for the first q eigenvectors. So the minimal error is $\operatorname{tr}(S) - (\lambda_1 + \cdots + \lambda_q) = \lambda_{q+1} + \cdots + \lambda_p = J_{p-q}$.

Note:

This is exactly what we saw in the R example: with $q = 1$ the average squared length of the segments was $\lambda_2 \approx 0.16 = J_1$.

Exercise 2.5.1 Suggested exercise 28

Let $S = P\Lambda P^\top$ with orthonormal eigenvectors $P = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_p]$ and $\lambda_1 \geqslant \ldots \geqslant \lambda_p \geqslant 0$. For $q \leqslant p$ let $P_q = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_q]$. Show that $\Pi_q := P_q P_q^\top$ is an orthogonal projector.

Solution: An orthogonal projector is a matrix that is **symmetric** and **idempotent**.

- Symmetry: $\Pi_q^\top = (P_q P_q^\top)^\top = (P_q^\top)^\top P_q^\top = P_q P_q^\top = \Pi_q$.
- Idempotence: $\Pi_q^2 = P_q(P_q^\top P_q)P_q^\top = P_q I_q P_q^\top = \Pi_q$.

Moreover it projects onto $\operatorname{span}(\boldsymbol{p}_1, \ldots, \boldsymbol{p}_q)$: $\Pi_q \boldsymbol{p}_k = \sum_{j \leqslant q} \boldsymbol{p}_j(\boldsymbol{p}_j^\top \boldsymbol{p}_k)$ equals $\boldsymbol{p}_k$ if $k \leqslant q$ and 0 if $k > q$. The residual is orthogonal to the image: $\Pi_q^\top(I - \Pi_q) = \Pi_q - \Pi_q^2 = 0$.

Exercise 2.5.2 Suggested exercise 29

Let $x_1, \ldots, x_n \in \mathbb{R}^p$ be centered, $S = \frac{1}{n}\sum_i x_i x_i^\top$ with eigenvalues $\lambda_1 \geqslant \ldots \geqslant \lambda_p \geqslant 0$ and orthonormal eigenvectors $\boldsymbol{p}_1, \ldots, \boldsymbol{p}_p$. Let $P_q = [\boldsymbol{p}_1, \ldots, \boldsymbol{p}_q]$ and $\tilde{x}_i = P_q P_q^\top x_i$. Show that

$$J_{p-q} = \sum_{j=q+1}^{p}\lambda_j = \frac{1}{n}\sum_{i=1}^{n}\|x_i - \tilde{x}_i\|^2.$$

Solution: (A direct proof, alternative to the one of the theorem above.) Expand x_i in the orthonormal basis of eigenvectors: $x_i = \sum_{j=1}^{p}(\boldsymbol{p}_j^\top x_i)\,\boldsymbol{p}_j$, while $\tilde{x}_i = \sum_{j \leqslant q}(\boldsymbol{p}_j^\top x_i)\,\boldsymbol{p}_j$. Then

$$x_i - \tilde{x}_i = \sum_{j=q+1}^{p}(\boldsymbol{p}_j^\top x_i)\,\boldsymbol{p}_j \quad \Longrightarrow \quad \|x_i - \tilde{x}_i\|^2 = \sum_{j=q+1}^{p}(\boldsymbol{p}_j^\top x_i)^2$$

(Pythagoras in an orthonormal basis). Averaging over i,

$$\frac{1}{n}\sum_i \|x_i - \tilde{x}_i\|^2 = \sum_{j=q+1}^{p} \boldsymbol{p}_j^\top \Big(\frac{1}{n}\sum_i x_i x_i^\top\Big)\boldsymbol{p}_j = \sum_{j=q+1}^{p} \boldsymbol{p}_j^\top S \boldsymbol{p}_j = \sum_{j=q+1}^{p} \lambda_j,$$

since $\boldsymbol{p}_j^\top S \boldsymbol{p}_j = \lambda_j \boldsymbol{p}_j^\top \boldsymbol{p}_j = \lambda_j$.

2.5.2 How many components should we keep?

A practical question in PCA: **how large should q be?**

- **too few** components: we may lose important structure in the data;
- **too many** components: we keep noise and lose the benefit of dimensionality reduction.

Recall that the eigenvalues of S measure how much variance each principal direction explains, and the **total variance** of the data is

$$\operatorname{tr}(S) = \sum_{j=1}^{p} \lambda_j.$$

Definition 2.5.2: Rank-q covariance and proportion of explained variance

Let $\Lambda_q = \operatorname{diag}(\lambda_1, \ldots, \lambda_q)$ and

$$\tilde{S}_q = P_q \Lambda_q P_q^\top \in \mathbb{R}^{p \times p},$$

i.e. the spectral decomposition $S = P\Lambda P^\top$ truncated to the first q terms: $\tilde{S}_q = \sum_{j=1}^{q} \lambda_j \boldsymbol{p}_j \boldsymbol{p}_j^\top$ (it is the covariance of the projected data $\tilde{x}_i$). The *cumulative proportion of explained variance* is

$$\mathrm{PEV}(q) = \frac{\operatorname{tr}(\tilde{S}_q)}{\operatorname{tr}(S)} = \frac{\sum_{j=1}^{q} \lambda_j}{\sum_{j=1}^{p} \lambda_j} \in [0, 1].$$

Note:

Why $\operatorname{tr}(\tilde{S}_q) = \sum_{j \leqslant q} \lambda_j$: by the cyclic property of the trace, $\operatorname{tr}(P_q \Lambda_q P_q^\top) = \operatorname{tr}(\Lambda_q P_q^\top P_q) = \operatorname{tr}(\Lambda_q)$. $\mathrm{PEV}(q)$ is increasing in q, with $\mathrm{PEV}(p) = 1$.

Definition 2.5.3: Threshold criterion

Fix a threshold τ (typically $\tau = 0.80$, 0.90 or 0.95) and choose

$$q^* = \min\{\, q : \mathrm{PEV}(q) \geqslant \tau \,\},$$

the **smallest** number of components explaining at least a fraction τ of the total variance.

Definition 2.5.4: Scree plot

The *scree plot* displays the individual eigenvalues λ_j (or the explained-variance ratios $\lambda_j / \operatorname{tr}(S)$) against the component index j. One looks for an **elbow**:

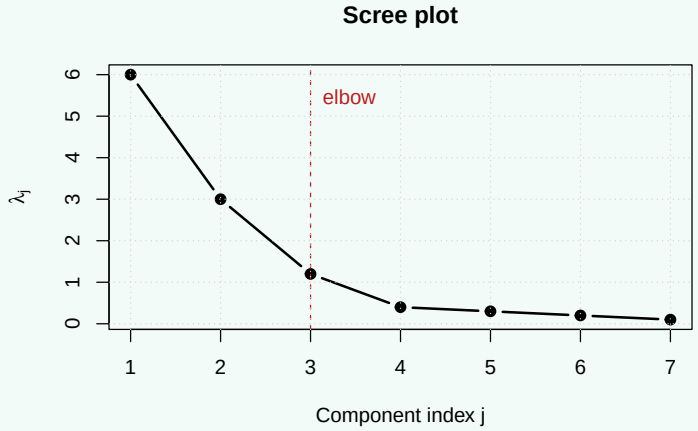
- **before** the elbow, eigenvalues decrease quickly: components usually considered *important*;
- **after** the elbow, they flatten out: components often interpreted as mostly *noise* or weak structure.

Example 2.5.1

Suppose $p = 7$ with eigenvalues $6,\ 3,\ 1.2,\ 0.4,\ 0.3,\ 0.2,\ 0.1$, so $\operatorname{tr}(S) = 11.2$.

q	1	2	3	4	5	6	7
PEV(q)	0.536	0.804	0.911	0.946	0.973	0.991	1

- $\tau = 0.80$: $q^* = 2$;
- $\tau = 0.90$: $q^* = 3$;
- $\tau = 0.95$: $q^* = 5$ (since PEV(4) = 0.946 < 0.95).



The scree plot drops fast up to $j = 3$ and is almost flat afterwards: the elbow suggests $q = 3$. Different criteria can give different answers, so the choice of q is partly a judgment call.
In the R example of the previous section, PEV(1) = 0.967: $q = 1$ already satisfies all three thresholds.

2.5.3 Practical workflow for choosing q

In practice, a reasonable workflow is:

1. compute the PCA;
2. inspect the scree plot;
3. inspect the cumulative explained variance curve;
4. choose a first candidate value of q;
5. check whether that value is appropriate for the intended purpose.

The best value of q **depends on the goal**:

- for **visualization**, one often takes $q = 2$ or $q = 3$;
- for **compression**, one takes the smallest q giving an acceptable reconstruction quality;
- for **preprocessing in machine learning**, one may choose q based on the downstream predictive performance.

So the choice of q combines statistical criteria, interpretability and the final objective of the analysis.

2.5.4 Illustration in R

We simulate $n = 200$ observations of $p = 5$ variables driven by three latent factors z_1, z_2, z_3 with standard deviations 3, 1.5, 0.5: x_1, x_2 depend on z_1, x_3, x_4 on z_2, x_5 on z_3, each plus a small noise (sd = 0.2).

```r
set.seed(123)
n <- 200
z1 <- rnorm(n, sd = 3); z2 <- rnorm(n, sd = 1.5); z3 <- rnorm(n, sd = 0.5)
x1 <- z1 + rnorm(n, sd = 0.2)
x2 <- 0.8 * z1 + rnorm(n, sd = 0.2)
x3 <- z2 + rnorm(n, sd = 0.2)
x4 <- 0.7 * z2 + rnorm(n, sd = 0.2)
x5 <- z3 + rnorm(n, sd = 0.2)
X <- scale(cbind(x1, x2, x3, x4, x5), center = TRUE, scale = FALSE)

pca <- prcomp(X, center = FALSE, scale. = FALSE)
eigenvalues <- pca$sdev^2
eigenvalues
## [1] 13.06986281 3.37695148 0.28067228 0.04648912 0.03056518
screeplot(pca, type = "lines", main = "Scree Plot from prcomp")

pev <- eigenvalues / sum(eigenvalues)
cum_pev <- cumsum(pev)
```

Since `prcomp()` uses the $n-1$ convention, these eigenvalues are on a slightly different scale from those computed with denominator n; the explained-variance ratios are unaffected.

Component	Eigenvalue	Explained variance ratio	Cumulative
1	13.0699	0.7778	0.7778
2	3.3770	0.2010	0.9787
3	0.2807	0.0167	0.9954
4	0.0465	0.0028	0.9982
5	0.0306	0.0018	1.0000

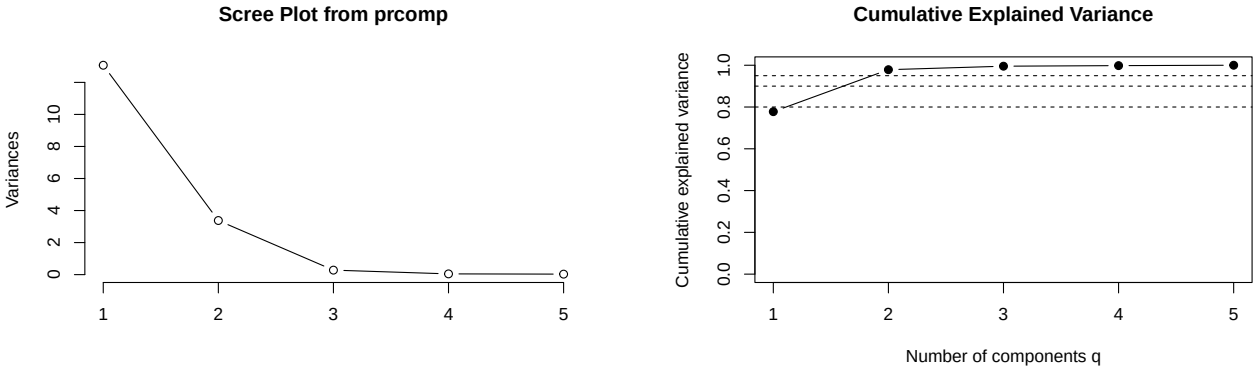


Figure 2.4: Scree plot and cumulative explained variance (dashed lines: $\tau = 0.80, 0.90, 0.95$).

Example 2.5.2

Reading the output:

- threshold criterion: PEV(1) = 0.778 < 0.80, while PEV(2) = 0.979 is above all three thresholds, so $q^* = 2$ for $\tau = 0.80, 0.90, 0.95$;
- scree plot: big drop from λ_2 to λ_3, then flat: the elbow also gives $q = 2$;
- this matches the simulation: two *strong* latent factors. The variance carried by z_1 through (x_1, x_2) is about $9\,(1 + 0.8^2) = 14.8$ and by z_2 through (x_3, x_4) about $2.25\,(1 + 0.7^2) = 3.35$, close to λ_1, λ_2; $\lambda_3 \approx 0.28 \approx 0.5^2 + 0.2^2$ is the weak factor z_3 in x_5, and $\lambda_4, \lambda_5 \approx 0.04 = 0.2^2$ are pure noise.

2.6 Projection into the Latent Space

It is important to distinguish **three objects**:

- the **original observations** x_i, in the variable space $\mathbb{R}^p$;
- the **latent scores**, the coordinates in the PCA space $\mathbb{R}^q$:

$$Z = XP_q \in \mathbb{R}^{n\times q};$$

- the **reconstructed observations**, mapped back to the original variable space:

$$\hat{X} = ZP_q^\top = XP_qP_q^\top \in \mathbb{R}^{n\times p}.$$

Proposition 2.6.1 The latent scores are uncorrelated

The columns of $Z = XP_q$ have mean zero, variances $\lambda_1, \dots, \lambda_q$ and are pairwise uncorrelated:

$$\frac{1}{n}Z^\top Z = \Lambda_q = \operatorname{diag}(\lambda_1, \dots, \lambda_q).$$

Proof: Mean: $\frac{1}{n}\mathbf{1}_n^\top Z = \left(\frac{1}{n}\mathbf{1}_n^\top X\right)P_q = 0$ because X is centered. Covariance: $\frac{1}{n}Z^\top Z = P_q^\top\left(\frac{1}{n}X^\top X\right)P_q = P_q^\top SP_q$. Since $S\boldsymbol{p}_j = \lambda_j\boldsymbol{p}_j$, we have $SP_q = P_q\Lambda_q$, hence $P_q^\top SP_q = P_q^\top P_q\Lambda_q = \Lambda_q$.

Note:

So PCA is a **rotation** of the data (for $q = p$, $Z = XP$ with P orthogonal) to new coordinates that are uncorrelated and ordered by decreasing variance.

2.6.1 Loadings

Definition 2.6.1: Loading vectors

The eigenvectors $\boldsymbol{p}_1, \dots, \boldsymbol{p}_p$ of $S = P\Lambda P^\top$ are also called *loading vectors*: they give the principal directions, while the eigenvalues measure the variance explained along them.

Keeping the first two directions, $P_2 = [\boldsymbol{p}_1, \boldsymbol{p}_2] \in \mathbb{R}^{p\times 2}$ and $Z_2 = XP_2 \in \mathbb{R}^{n\times 2}$: the rows of Z_2 are the coordinates of the observations in the two-dimensional PCA space. By Suggested exercise 23,

$$\boldsymbol{z}_i = x_i^\top P_2 = \sum_{j=1}^{p} x_{ij}\begin{bmatrix} p_{j1} & p_{j2}\end{bmatrix},$$

so the **row** (p_{j1}, p_{j2}) of P_2 contains the two coefficients of variable j in the first two principal directions:

- if $|p_{j1}|$ is large, variable j receives a large weight in PC1 (similarly for $|p_{j2}|$ and PC2);
- the **signs** tell whether variables enter a component in the same or in opposite directions.

Coefficient magnitudes are directly comparable only when the variables have comparable units or have been **standardized**.

Proposition 2.6.2 Correlation coordinates

Let s_j be the standard deviation of variable j. The correlation between variable j and the k-th latent score is

$$\operatorname{cor}(x_{\cdot j}, z_{\cdot k}) = \frac{\sqrt{\lambda_k}\,p_{jk}}{s_j}.$$

The pair $\left(\sqrt{\lambda_1}\,p_{j1}/s_j,\ \sqrt{\lambda_2}\,p_{j2}/s_j\right)$ is called the *correlation coordinates* of variable j.

Proof: Data are centered, so (with the same denominator n everywhere)

$$\operatorname{cov}(x_{\cdot j}, z_{\cdot k}) = \frac{1}{n}\sum_i x_{ij}\,(x_i^\top \boldsymbol{p}_k) = \Big(\frac{1}{n}\sum_i x_i x_i^\top \boldsymbol{p}_k\Big)_j = (S\boldsymbol{p}_k)_j = \lambda_k\, p_{jk}.$$

By the previous proposition $\operatorname{var}(z_{\cdot k}) = \lambda_k$, hence

$$\operatorname{cor}(x_{\cdot j}, z_{\cdot k}) = \frac{\lambda_k\, p_{jk}}{s_j\sqrt{\lambda_k}} = \frac{\sqrt{\lambda_k}\, p_{jk}}{s_j}.$$

Note:

Summing over all components, $\sum_{k=1}^{p} \operatorname{cor}(x_{\cdot j}, z_{\cdot k})^2 = \frac{1}{s_j^2}\sum_k \lambda_k p_{jk}^2 = \frac{S_{jj}}{s_j^2} = 1$ (because $S_{jj} = (P\Lambda P^\top)_{jj} = \sum_k \lambda_k p_{jk}^2$). So the squared length of the correlation-coordinate arrow in the PC1–PC2 plane is the fraction of the variance of variable j captured by the first two components: it is at most 1, and close to 1 when the variable is well represented in the plane.

2.6.2 Biplot

Definition 2.6.2: Biplot

A *biplot* is the joint representation of observations and variables in the plane of two principal components:

- **observation points**: each row of Z_2 is plotted as a point;
- **variable arrows**: each original variable is an arrow, based on its coordinates in the loading matrix.

Often several biplots are used (PC1–PC2, PC1–PC3, ...).

The exact interpretation depends on the scaling convention. In a latent score plot with correlation-coordinate arrows:

- observations close to each other have similar latent scores; observations far apart differ strongly along the displayed components; **clusters** may reveal structure;
- a **long** arrow means the variable is well represented in the PC1–PC2 plane; a **short** arrow means a substantial part of its variation lies in later components;
- arrows pointing in **similar** directions: positively associated variables; **opposite** directions: negatively associated; close to **orthogonal**: weak association in the displayed plane;
- the **angle** between arrows approximates the correlation between the variables, when the plane represents them well;
- an observation lying far in the direction of an arrow tends to have a relatively **large** value of that variable; in the opposite direction, a relatively **small** value.

Note:

In practice arrows are often rescaled for visibility: one plots $c\,[\,p_{j1}\ p_{j2}\,]$ for some $c > 0$. This changes the visual length but not the direction; after an arbitrary rescaling, the arrow length must not be read as a correlation or as a quality of representation. When using a biplot:

- always check the explained variance of PC1 and PC2;
- standardize variables if they are on very different scales;
- avoid over-interpreting tiny differences in arrow lengths;
- use it together with scree plots and explained-variance summaries.

2.7 PCA of Iris Data

The `iris` dataset (150 flowers, 3 species, 4 measurements in cm) is a classical PCA example.

```
data(iris)
X <- iris[, 1:4]
species <- iris$Species
pca_iris <- prcomp(X, center = TRUE, scale. = TRUE)
summary(pca_iris)
## Importance of components:
## PC1 PC2 PC3 PC4
## Standard deviation 1.7084 0.9560 0.38309 0.14393
## Proportion of Variance 0.7296 0.2285 0.03669 0.00518
## Cumulative Proportion 0.7296 0.9581 0.99482 1.00000
pca_iris$rotation[, 1:2]
## PC1 PC2
## Sepal.Length 0.5210659 -0.37741762
## Sepal.Width -0.2693474 -0.92329566
## Petal.Length 0.5804131 -0.02449161
## Petal.Width 0.5648565 -0.06694199
biplot(pca_iris, cex = 0.6)
```

Note:

Standardization. With `scale. = TRUE` every variable is divided by its standard deviation, so S becomes the **correlation matrix** R ($R_{jj} = 1$). Then the total variance is $\mathrm{tr}(R) = p = 4$, and indeed $1.7084^2/4 = 2.919/4 = 0.7296$. Standardizing prevents a variable from dominating only because of its unit of measurement.

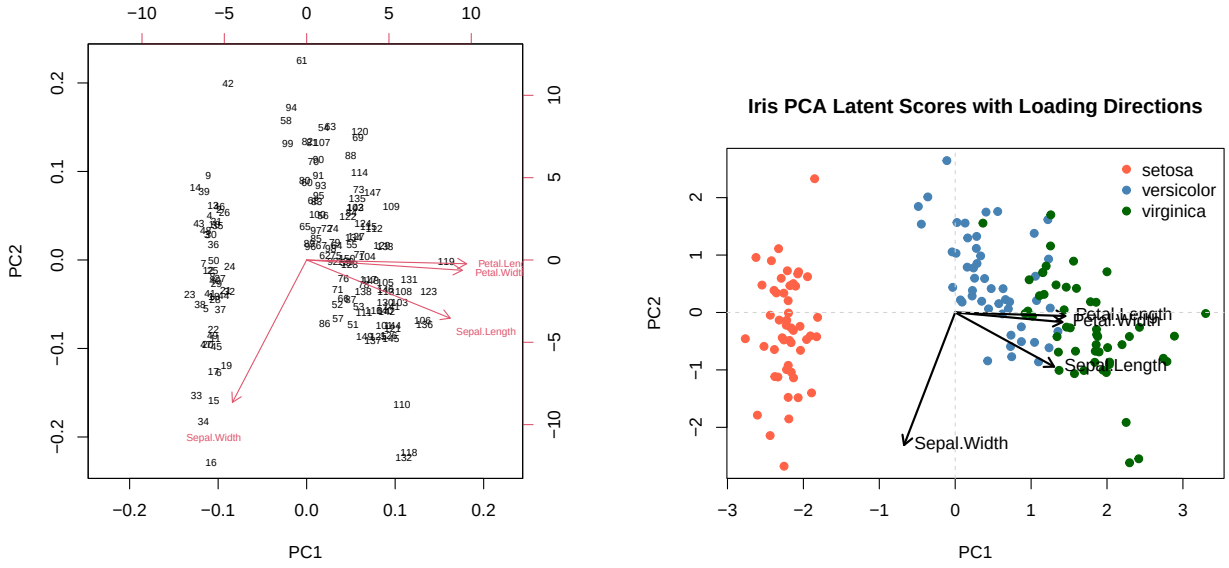


Figure 2.5: Standard `biplot()` and a colored manual version (scores coloured by species, loading arrows scaled by 2.5).

The colored version is built from `scores <- pca_iris$x[, 1:2]` and `loadings <- pca_iris$rotation[, 1:2]`, with `arrows(0, 0, 2.5 * loadings[, 1], 2.5 * loadings[, 2])`.

Example 2.7.1

- PC1 and PC2 explain 95.8% of the variance: the plane is a faithful summary.

- PC1 has loadings ≈ 0.58 and 0.56 on petal length and width and 0.52 on sepal length: it is roughly an average of them, i.e. a **flower (petal) size** component. Sepal width enters with the opposite sign.
- PC2 is dominated by sepal width (-0.92).
- The three species are well separated along PC1 alone: *setosa* (small petals) on one side, *versicolor* and *virginica* on the other.

2.8 PCA in High-Dimensional Settings

2.8.1 Decompositions

The centered data matrix is $X \in \mathbb{R}^{n\times p}$ (observations by row) and $S = \frac{1}{n}X^\top X \in \mathbb{R}^{p\times p}$. In the **high-dimensional setting** $n \ll p$, S may be very large and is **low rank** ($\operatorname{rank}(S) \leqslant n-1$). Computing $S = P\Lambda P^\top$ directly and keeping the $q \leqslant n$ largest eigenvalues is possible in principle, but it requires building and diagonalizing a $p \times p$ matrix: memory intensive and slow when p is in the thousands or millions (images, text). Two equivalent alternatives:

1. the **SVD** of X;
2. the spectral decomposition of the **small matrix**

$$\tilde{S} = \frac{1}{n}XX^\top \in \mathbb{R}^{n\times n}.$$

Theorem 2.8.1

(PCA via the SVD.) Let $X = U\Sigma V^\top$ with $U \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{p\times p}$ orthogonal and $\Sigma \in \mathbb{R}^{n\times p}$ with nonzero diagonal entries $\sigma_1 \geqslant \ldots \geqslant \sigma_r > 0$, where $r = \operatorname{rank}(X) \leqslant \min(n-1, p)$ for centered data. Then

- the nonzero eigenvalues of S are $\lambda_j = \sigma_j^2/n$, $j = 1, \ldots, r$;
- the columns of V are eigenvectors of S: P may be chosen equal to V (they agree up to sign changes).

The proof is Suggested exercise 30.

Exercise 2.8.1 Suggested exercise 30

Let $X \in \mathbb{R}^{n\times p}$ be a centered data matrix with rank $r \leqslant n < p$ and SVD $X = U\Sigma V^\top$, with nonzero singular values $\sigma_1, \ldots, \sigma_r$. Show that the spectral decomposition $S = \frac{1}{n}X^\top X = P\Lambda P^\top$ is such that $\lambda_j = \sigma_j^2/n$ for $j = 1, \ldots, r$. Show also that the columns of V are eigenvectors of S, so that P may be chosen equal to V.

Solution: Using $U^\top U = I_n$,

$$S = \frac{1}{n}X^\top X = \frac{1}{n}V\Sigma^\top U^\top U\Sigma V^\top = V\,\frac{\Sigma^\top\Sigma}{n}\,V^\top.$$

$\Sigma^\top\Sigma \in \mathbb{R}^{p\times p}$ is diagonal, $\Sigma^\top\Sigma = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_r^2, 0, \ldots, 0)$. Multiplying on the right by V ($V^\top V = I_p$):

$$SV = V\,\frac{\Sigma^\top\Sigma}{n} \quad\Longrightarrow\quad S\boldsymbol{v}_j = \frac{\sigma_j^2}{n}\,\boldsymbol{v}_j \ (j \leqslant r), \qquad S\boldsymbol{v}_j = 0 \ (j > r).$$

So $S = V\Lambda V^\top$ with V orthogonal and $\Lambda = \Sigma^\top\Sigma/n$ diagonal (ordered, since the σ_j are): this is a spectral decomposition, with $\lambda_j = \sigma_j^2/n$, and one may take $P = V$. The eigenvectors are unique only up to sign (or up to a rotation inside an eigenspace if eigenvalues are repeated), which is why P and V agree "up to possible sign changes".

Theorem 2.8.2

(PCA via the small matrix.) Let $\tilde{S} = \frac{1}{n}XX^\top = \tilde{P}\tilde{\Lambda}\tilde{P}^\top$, with $\tilde{P} \in \mathbb{R}^{n\times n}$ orthogonal and $\tilde{\Lambda} = \operatorname{diag}(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_n)$, $\tilde{\lambda}_j \geqslant 0$. Then

- S and $\tilde{S}$ have the **same nonzero eigenvalues**: $\lambda_j = \tilde{\lambda}_j$;
- for every $\tilde{\lambda}_j > 0$, the corresponding unit eigenvector of S is

$$\boldsymbol{p}_j = \frac{X^\top \tilde{\boldsymbol{p}}_j}{\sqrt{n\tilde{\lambda}_j}}.$$

The proof is Suggested exercise 31.

Exercise 2.8.2 Suggested exercise 31

Let $X \in \mathbb{R}^{n\times p}$ be centered, with rank $r \leqslant n < p$, and $\tilde{S} = \frac{1}{n}XX^\top = \tilde{P}\tilde{\Lambda}\tilde{P}^\top$. Show that the positive eigenvalues $\tilde{\lambda}_j$ of $\tilde{S}$ are the same as those of $S = \frac{1}{n}X^\top X = P\Lambda P^\top$, and that for every $\tilde{\lambda}_j > 0$ the corresponding eigenvector of S is $\boldsymbol{p}_j = X^\top \tilde{\boldsymbol{p}}_j / \sqrt{n\tilde{\lambda}_j}$.

Solution: **From $\tilde{S}$ to S.** Let $\tilde{S}\tilde{\boldsymbol{p}}_j = \tilde{\lambda}_j \tilde{\boldsymbol{p}}_j$ with $\tilde{\lambda}_j > 0$, $\|\tilde{\boldsymbol{p}}_j\| = 1$. Then

$$S\,(X^\top \tilde{\boldsymbol{p}}_j) = \frac{1}{n}X^\top XX^\top \tilde{\boldsymbol{p}}_j = X^\top\left(\frac{1}{n}XX^\top \tilde{\boldsymbol{p}}_j\right) = \tilde{\lambda}_j\,(X^\top \tilde{\boldsymbol{p}}_j).$$

The vector $X^\top \tilde{\boldsymbol{p}}_j$ is not zero, since

$$\|X^\top \tilde{\boldsymbol{p}}_j\|^2 = \tilde{\boldsymbol{p}}_j^\top XX^\top \tilde{\boldsymbol{p}}_j = n\,\tilde{\boldsymbol{p}}_j^\top \tilde{S}\tilde{\boldsymbol{p}}_j = n\tilde{\lambda}_j > 0.$$

So $\tilde{\lambda}_j$ is an eigenvalue of S, with unit eigenvector $\boldsymbol{p}_j = X^\top \tilde{\boldsymbol{p}}_j / \sqrt{n\tilde{\lambda}_j}$.

From S to $\tilde{S}$. Symmetrically, if $S\boldsymbol{p} = \lambda \boldsymbol{p}$ with $\lambda > 0$, then $\tilde{S}(X\boldsymbol{p}) = \frac{1}{n}XX^\top X\boldsymbol{p} = \lambda\, X\boldsymbol{p}$ and $\|X\boldsymbol{p}\|^2 = n\lambda > 0$. So the positive eigenvalues coincide.

Same multiplicities. The map preserves orthonormality:

$$\boldsymbol{p}_j^\top \boldsymbol{p}_k = \frac{\tilde{\boldsymbol{p}}_j^\top XX^\top \tilde{\boldsymbol{p}}_k}{n\sqrt{\tilde{\lambda}_j\tilde{\lambda}_k}} = \frac{n\tilde{\lambda}_k\,\tilde{\boldsymbol{p}}_j^\top \tilde{\boldsymbol{p}}_k}{n\sqrt{\tilde{\lambda}_j\tilde{\lambda}_k}} = \delta_{jk},$$

so orthonormal eigenvectors of $\tilde{S}$ give orthonormal eigenvectors of S, and each positive eigenvalue has the same multiplicity in both matrices.

Corollary 2.8.1 Latent scores from the small matrix

With $\boldsymbol{p}_j = X^\top \tilde{\boldsymbol{p}}_j / \sqrt{n\tilde{\lambda}_j}$, the j-th column of latent scores is

$$X\boldsymbol{p}_j = \sqrt{n\tilde{\lambda}_j}\,\tilde{\boldsymbol{p}}_j.$$

In SVD terms ($\tilde{\boldsymbol{p}}_j = \boldsymbol{u}_j$, $n\tilde{\lambda}_j = \sigma_j^2$): $Z = XV = U\Sigma$, i.e. $X\boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$.

Proof: $X\boldsymbol{p}_j = \dfrac{XX^\top \tilde{\boldsymbol{p}}_j}{\sqrt{n\tilde{\lambda}_j}} = \dfrac{n\tilde{\lambda}_j\,\tilde{\boldsymbol{p}}_j}{\sqrt{n\tilde{\lambda}_j}} = \sqrt{n\tilde{\lambda}_j}\,\tilde{\boldsymbol{p}}_j$. For the SVD form: $\tilde{S} = \frac{1}{n}U\Sigma\Sigma^\top U^\top$ is a spectral decomposition of $\tilde{S}$, so $\tilde{P} = U$ and $\tilde{\lambda}_j = \sigma_j^2/n$; and $XV = U\Sigma V^\top V = U\Sigma$.

Note:

Summary. $X = U\Sigma V^\top$ contains the whole PCA: V = principal directions, σ_j^2/n = variances, $U\Sigma$ = latent scores. The $n \times n$ route costs $O(n^2 p)$ to build $\tilde{S}$ plus $O(n^3)$ to diagonalize it, instead of $O(np^2) + O(p^3)$ for S.

2.8.2 Eckart–Young–Mirsky Theorem

Definition 2.8.1: Spectral norm

For $A \in \mathbb{R}^{n\times p}$, the *spectral norm* is

$$\|A\|_2 = \max_{\|\boldsymbol{w}\|_2=1} \|A\boldsymbol{w}\|_2 = \sigma_1(A),$$

the largest singular value of A.

Why $\max_{\|\boldsymbol{w}\|=1} \|A\boldsymbol{w}\| = \sigma_1$**:** $\|A\boldsymbol{w}\|^2 = \boldsymbol{w}^\top A^\top A\boldsymbol{w}$ and $A^\top A$ is symmetric PSD with largest eigenvalue σ_1^2. By the "direct check" in the maximum variance section, $\max_{\|\boldsymbol{w}\|=1} \boldsymbol{w}^\top A^\top A\boldsymbol{w} = \sigma_1^2$, attained at $\boldsymbol{w} = \boldsymbol{v}_1$.

Theorem 2.8.3

(Eckart–Young–Mirsky.) Let $X \in \mathbb{R}^{n\times p}$ have rank r and SVD $X = U\Sigma V^\top$, and for $q < r$ let

$$\hat{X}^{(q)} = \sum_{j=1}^{q} \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^\top$$

be the **truncated SVD**. Then

$$\hat{X}^{(q)} = \arg\min_{\operatorname{rank}(X^*)\leqslant q} \|X - X^*\|_2 \qquad \text{and} \qquad \|X - \hat{X}^{(q)}\|_2 = \sigma_{q+1}.$$

The truncated SVD is the best rank-q approximation of X in spectral norm: no other matrix of rank at most q produces a smaller error.

Proof: **Step 1: the error of the truncated SVD.** Since $X = \sum_{j=1}^{r} \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^\top$, the error matrix is

$$X - \hat{X}^{(q)} = \sum_{j=q+1}^{r} \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^\top = U \operatorname{diag}(0, \ldots, 0, \sigma_{q+1}, \ldots, \sigma_r) V^\top,$$

which is already an SVD: its singular values are exactly the discarded ones $\sigma_{q+1}, \ldots, \sigma_r$. The spectral norm is the largest of them, so $\|X - \hat{X}^{(q)}\|_2 = \sigma_{q+1}$. Moreover $\operatorname{rank}(\hat{X}^{(q)}) = q$.

Step 2: no rank-q matrix does better. Let X^* have rank $\leqslant q$. By rank–nullity, $\dim\ker X^* \geqslant p - q$. The subspace $\mathcal{V} = \operatorname{span}(\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{q+1})$ has dimension $q+1$. Since $(p-q) + (q+1) > p$, the two subspaces intersect nontrivially: there is $\boldsymbol{w} \in \ker X^* \cap \mathcal{V}$ with $\|\boldsymbol{w}\| = 1$, $\boldsymbol{w} = \sum_{j\leqslant q+1} c_j \boldsymbol{v}_j$, $\sum_j c_j^2 = 1$. Using $X\boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$ and the orthonormality of the $\boldsymbol{u}_j$,

$$\|X - X^*\|_2^2 \geqslant \|(X - X^*)\boldsymbol{w}\|^2 = \|X\boldsymbol{w}\|^2 = \Big\| \sum_{j\leqslant q+1} c_j \sigma_j \boldsymbol{u}_j \Big\|^2 = \sum_{j\leqslant q+1} c_j^2 \sigma_j^2 \geqslant \sigma_{q+1}^2 \sum_{j\leqslant q+1} c_j^2 = \sigma_{q+1}^2.$$

Hence $\|X - X^*\|_2 \geqslant \sigma_{q+1} = \|X - \hat{X}^{(q)}\|_2$.

Proposition 2.8.1 PCA reconstruction = truncated SVD

Choosing $P = V$, the rank-q PCA reconstruction coincides with the truncated SVD:

$$\hat{X} = XP_qP_q^\top = \sum_{j=1}^{q} \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^\top = \hat{X}^{(q)}.$$

Moreover, in Frobenius norm,

$$\frac{1}{n}\|X - \hat{X}\|_F^2 = \frac{1}{n}\sum_{i=1}^{n} \|x_i - \hat{x}_i\|^2 = \sum_{j=q+1}^{r} \frac{\sigma_j^2}{n} = J_{p-q}.$$

Proof: From $XV = U\Sigma$ we get $X\boldsymbol{v}_j = \sigma_j \boldsymbol{u}_j$, so $XP_qP_q^\top = \sum_{j=1}^{q}(X\boldsymbol{v}_j)\boldsymbol{v}_j^\top = \sum_{j=1}^{q} \sigma_j \boldsymbol{u}_j \boldsymbol{v}_j^\top$. The squared Frobenius norm is the sum of the squared norms of the rows, which gives the first equality; it is also the sum of the squared singular values, here $\sigma_{q+1}^2, \ldots, \sigma_r^2$ (Step 1 above); finally $\sigma_j^2/n = \lambda_j$.

Note:

This closes the circle of the **three points of view**: the subspace of *maximum variance*, the subspace of *minimum reconstruction error* and the *best rank-q approximation* of X are all given by the first q principal directions.

2.8.3 High-Dimensional Image Example

Data

A common application of PCA in high dimension is the analysis of images, such as handwritten digits. Each grayscale image is one observation: an image of height h and width w has $h \times w$ pixels and, after **vectorization**, becomes a point of $\mathbb{R}^{hw}$. For 28×28 images, $p = 784$. Each coordinate is a pixel intensity in $[0, 1]$ (0 = background, 1 = highest intensity).
The goals of PCA here are to:

- represent each image with a smaller number of coordinates;
- identify the dominant modes of variation in the dataset;
- reconstruct the images approximately from a few principal components;
- show that many images can be compressed without losing their main visual structure.

We use **synthetic** data: handwritten-style digit-8 images generated from a smooth reference shape plus random perturbations of position, size, thickness and pixel noise. Since the reference shape is known (it is not observed in practice), PCA's results can be compared with it. The motivation is the same as for MNIST, a collection of 70 000 labeled 28×28 grayscale images of handwritten digits widely used as a benchmark.

```
img_h <- 28; img_w <- 28
make_digit8 <- function(h = 28, w = 28, top_y = 10, bottom_y = 19, x0 = 14,
                        a1 = 5.0, b1 = 4.0, a2 = 5.5, b2 = 4.5,
                        thickness = 0.35, noise_sd = 0.05) {
  xx <- matrix(rep(1:w, each = h), nrow = h, ncol = w)
  yy <- matrix(rep(1:h, times = w), nrow = h, ncol = w)
  top_loop <- exp(-((sqrt(((xx - x0)/a1)^2 + ((yy - top_y)/b1)^2) - 1)^2) / thickness)
  bottom_loop <- exp(-((sqrt(((xx - x0)/a2)^2 + ((yy - bottom_y)/b2)^2) - 1)^2) / thickness)
  bridge <- exp(-((xx - x0)^2) / 7 - ((yy - (top_y + bottom_y)/2)^2) / 15)
  img <- pmax(top_loop, bottom_loop) + 0.35 * bridge
  img <- img + matrix(rnorm(h * w, sd = noise_sd), nrow = h, ncol = w)
  img <- pmax(img, 0)
  img / max(img)
}

n <- 60; p <- img_h * img_w # n = 60 images, p = 784 pixels
```

```
X <- matrix(0, nrow = n, ncol = p)
set.seed(5678)
for (i in 1:n) {
  img <- make_digit8(
    top_y = round(rnorm(1, 10, 0.7)), bottom_y = round(rnorm(1, 19, 0.8)),
    x0 = round(rnorm(1, 14, 0.7)),
    a1 = rnorm(1, 5.0, 0.4), b1 = rnorm(1, 4.0, 0.3),
    a2 = rnorm(1, 5.4, 0.5), b2 = rnorm(1, 4.5, 0.3),
    thickness = abs(rnorm(1, 0.34, 0.05)), noise_sd = 0.04)
  X[i, ] <- as.vector(img) # one image = one row
}
```

The two loops are Gaussian "rings" around two ellipses (centers $(x_0, \text{top_y})$ and $(x_0, \text{bottom_y})$, semi-axes a_1, b_1 and a_2, b_2), joined by a small bridge; each image draws its own positions, sizes and thickness.

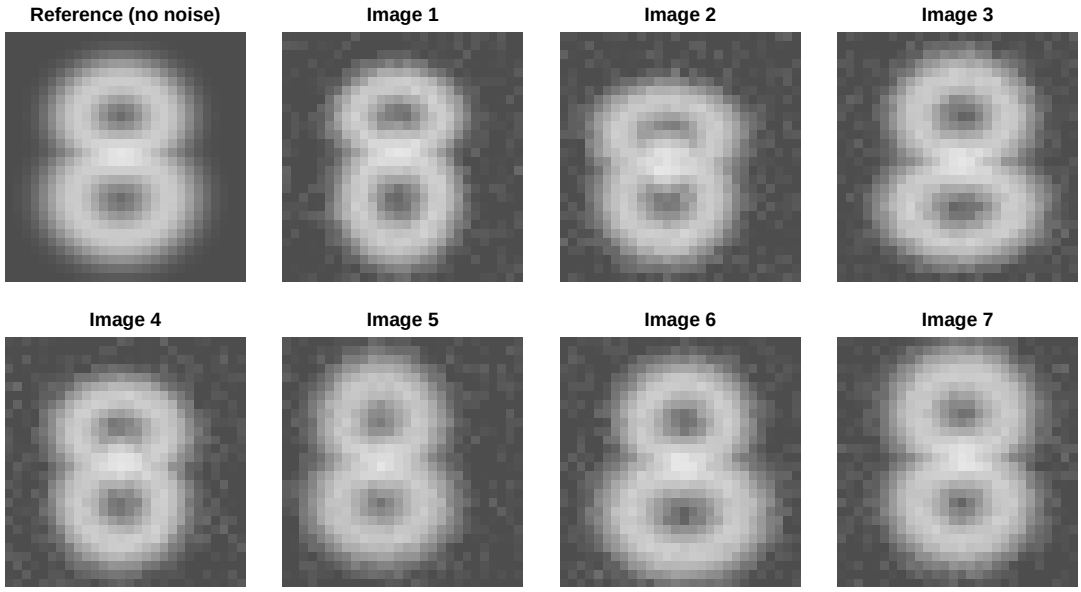


Figure 2.6: The noiseless reference digit and seven of the 60 observed images.

PCA

We center the data and, since $n = 60 \ll p = 784$, use the small matrix $\tilde{S} = \frac{1}{n} X X^\top \in \mathbb{R}^{60\times 60}$:

```
X_centered <- scale(X, center = TRUE, scale = FALSE)
mu <- attr(X_centered, "scaled:center") # mean image
S_small <- (1 / n) * X_centered %*% t(X_centered)
eig_small <- eigen(S_small)
lambda <- eig_small$values; P_tilde <- eig_small$vectors
head(lambda)
## [1] 2.6141122 1.9787963 1.3592923 0.7745353 0.5222881 0.3148945
keep <- which(lambda > 1e-8) # keep the positive eigenvalues
lambda_pos <- lambda[keep]; P_tilde_pos <- P_tilde[, keep, drop = FALSE]
length(lambda_pos)
## [1] 59
P <- sapply(seq_along(lambda_pos), function(j)
  as.vector(t(X_centered) %*% P_tilde_pos[, j] / sqrt(n * lambda_pos[j])))
dim(P)
## [1] 784 59
```

Note:

Why exactly 59 positive eigenvalues. The data are centered, so $\operatorname{rank}(\tilde{S}) = \operatorname{rank}(X_c) \leqslant n - 1 = 59$: the constraint $\mathbf{1}_n^\top X_c = 0$ forces at least one zero eigenvalue ($\tilde{S}\mathbf{1}_n = \frac{1}{n} X_c X_c^\top \mathbf{1}_n = 0$), and for generic data the other 59 are positive. The tolerance `1e-8` only discards numerical zeros. Then $\boldsymbol{p}_j = X^\top \tilde{\boldsymbol{p}}_j / \sqrt{n\tilde{\lambda}_j}$ (Suggested exercise 31) gives 59 orthonormal directions in pixel space (numerically $\max |P^\top P - I| \approx 4 \cdot 10^{-14}$).

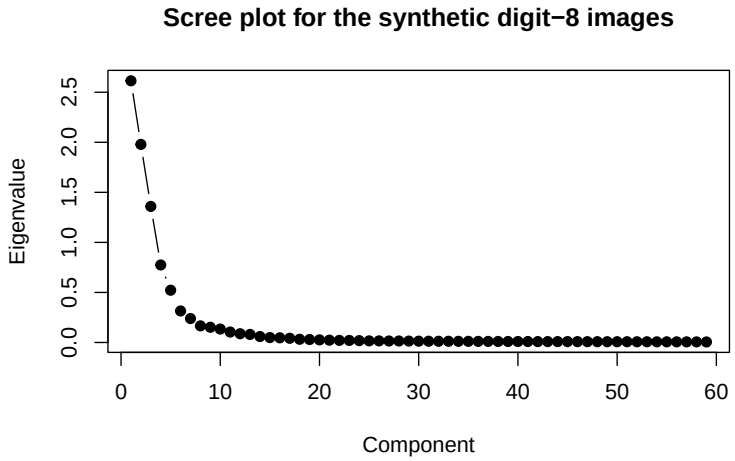


Figure 2.7: Scree plot: the first few eigenvalues dominate, because the images vary around a small number of latent shape changes.

Eigenimages

Definition 2.8.2: Eigenimage

Each principal direction $\boldsymbol{p}_j \in \mathbb{R}^p$ can be reshaped into a 28×28 array and displayed as an image: an *eigenimage*. It is not an average image: it is a **direction of variation around the mean image**.

- positive values: pixels that tend to increase together when moving along $\boldsymbol{p}_j$;
- negative values: pixels that tend to decrease;
- large absolute values: pixels most involved in that mode of variation.

When the latent score of an image on component j is large and positive, its reconstruction moves from the mean image in the direction of the eigenimage; when negative, in the opposite direction. For display, a gray scale **symmetric around zero** is used (`zlim = c(-eig_lim, eig_lim)`, here `eig_lim` = 0.159): light and dark are the two signs, middle gray means small weights.

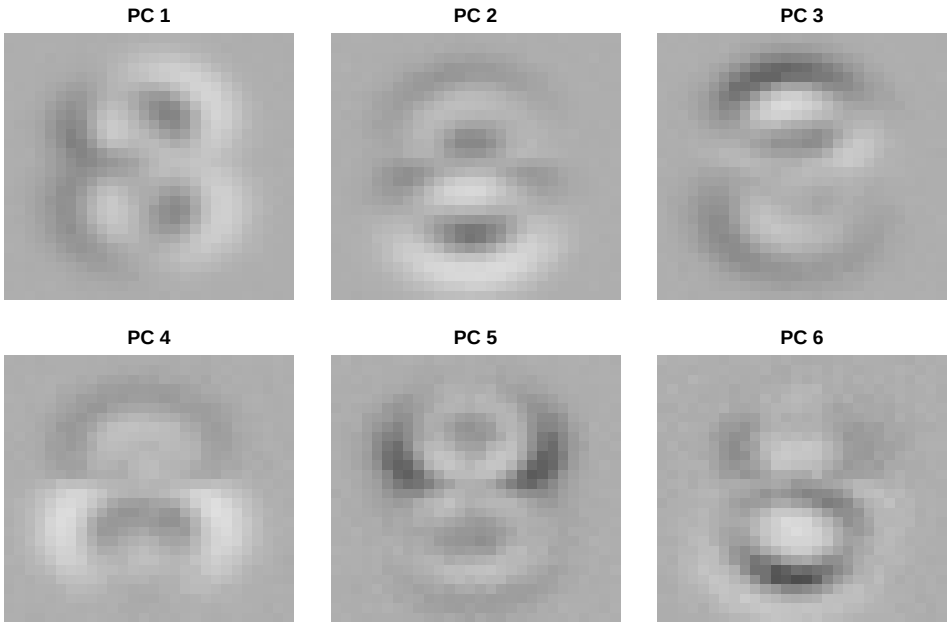


Figure 2.8: The first six eigenimages.

Interpretation of the principal directions

Eigenimages are not easy to read directly. A better way is to look at

$$\bar{x} - 2\sqrt{\lambda_j}\,\boldsymbol{p}_j, \qquad \bar{x}, \qquad \bar{x} + 2\sqrt{\lambda_j}\,\boldsymbol{p}_j,$$

because λ_j is the variance along the component, so $\sqrt{\lambda_j}$ is its standard deviation: we move by ± 2 standard deviations from the mean image.

```r
mean_img <- matrix(mu, nrow = img_h, ncol = img_w)
pj <- matrix(P[, j], nrow = img_h, ncol = img_w)
minus_img <- mean_img - 2 * sqrt(lambda_pos[j]) * pj
plus_img <- mean_img + 2 * sqrt(lambda_pos[j]) * pj
# displayed with pmin(pmax(., 0), 1) on the common range [0, 1]
```

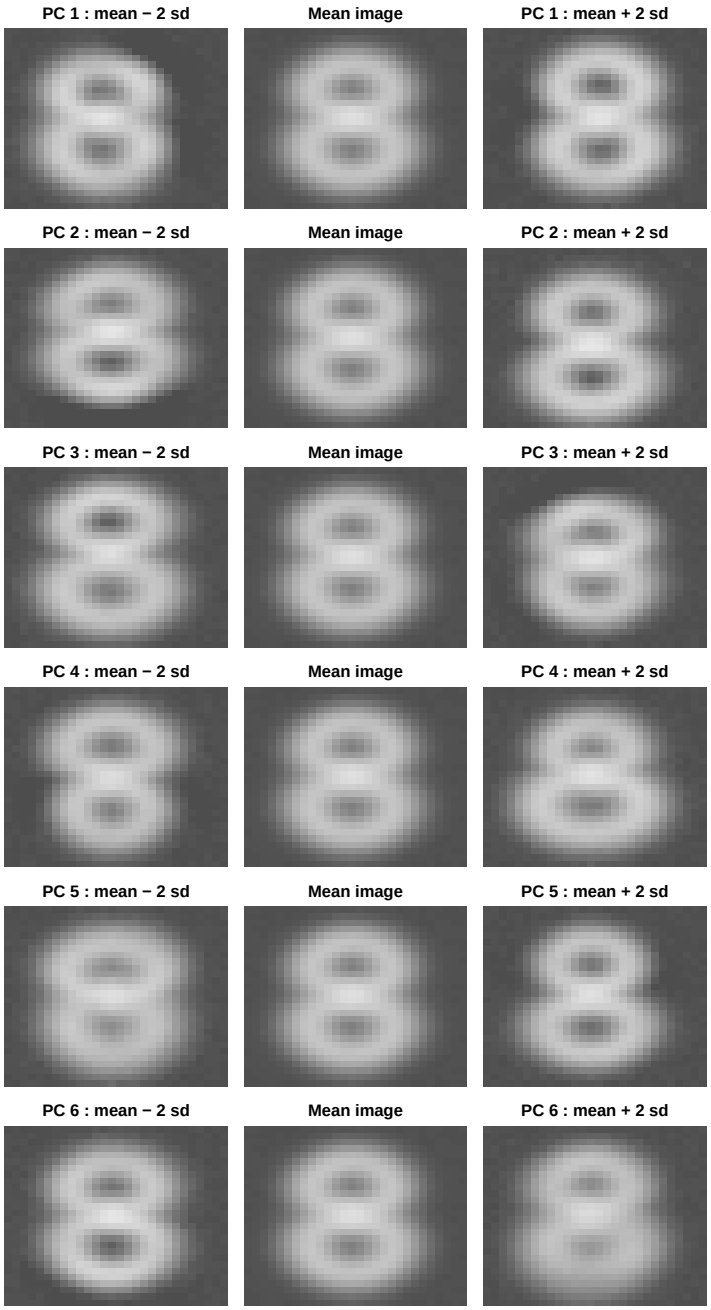


Figure 2.9: For each of the first six components: mean -2 sd, mean image, mean $+2$ sd.

Some coordinates of the theoretical endpoints lie outside $[0, 1]$ and are clipped: the endpoints are points in PCA space rather than necessarily valid pixel images. With the fixed seed, the first six directions read:

- **PC1** – **horizontal position**: a left–right translation of the whole digit;
- **PC2** – **vertical position**: an up–down displacement, stronger on the lower loop;
- **PC3** – **vertical separation of the loops**: one endpoint brings the loops closer, the other separates them;
- **PC4** – **overall width and stroke thickness**: widening/narrowing of both loops plus apparent thickness;
- **PC5** – **relative loop width**: as one loop becomes wider the other becomes narrower;
- **PC6** – **weaker mixed deformation**: localized changes in loop shape and in the central connection, not tied to a single generator parameter.

Note:

The sign of each eigenvector is arbitrary and the order of the later components may change with the simulation. These are the dominant shape variations of *this* simulated dataset, not universal features of handwritten 8s.

PCA latent scores

```
Z <- X_centered %*% P
dim(Z)
## [1] 60 59
head(Z[, 1:3])
## [,1] [,2] [,3]
## [1,] -0.3791712 0.4870712 0.6761959
## [2,] -2.5222515 0.7516631 1.5389542
## [3,] 2.1744691 1.5100075 -0.6520721
## ...
```

By the proposition on uncorrelated scores, the columns of Z have mean zero and variances λ_j (checked numerically: the mean of $Z_{\cdot j}^2$ equals $2.614, 1.979, 1.359, \dots$).

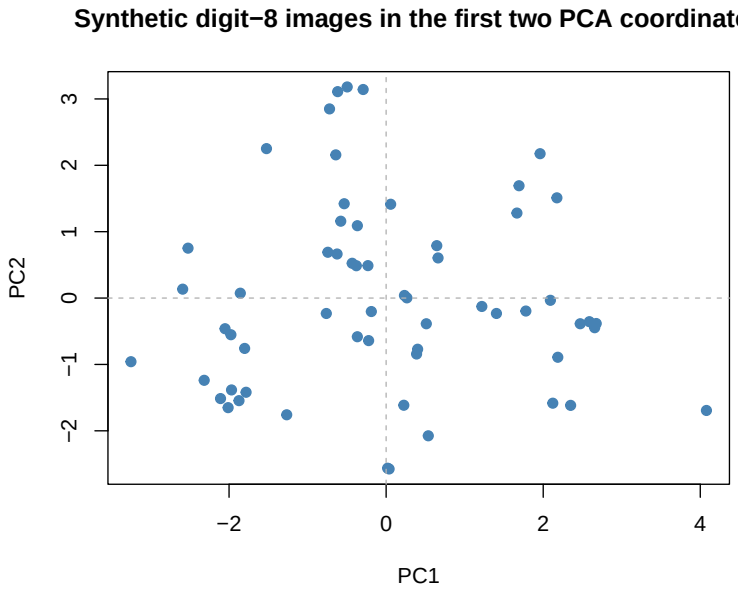


Figure 2.10: The 60 images in the first two PCA coordinates.

Note:

The scatterplot may show denser regions or an apparent grouping. This does not mean that the images come from predefined classes: each image is a combination of a few latent shape variations (horizontal and vertical position, relative loop size, thickness, pixel noise), and PCA maps images with similar values of the two dominant ones to nearby points. The exact pattern depends on the random seed.

Reconstruction with different numbers of principal components

```
reconstruct_image <- function(x, P, q, mu) {
  Pq <- P[, 1:q, drop = FALSE]
  zq <- (x - mu) %*% Pq # scores of the image
  xhat <- zq %*% t(Pq) + mu # back to pixel space
  matrix(as.vector(xhat), nrow = img_h, ncol = img_w)
}
```

i.e. $\hat{x} = \bar{x} + P_q P_q^\top (x - \bar{x})$.

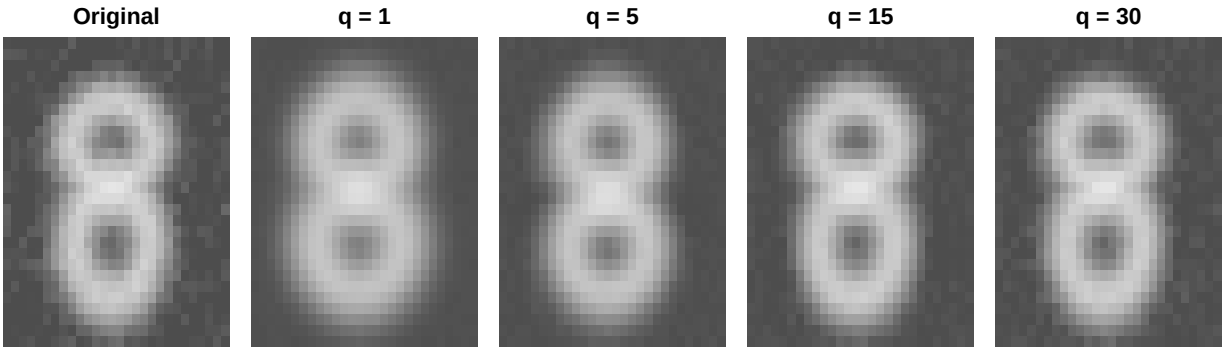


Figure 2.11: Image 1 and its reconstructions with $q = 1, 5, 15, 30$ components.

```
qs <- c(1, 2, 5, 10, 15, 20, 30, 40)
recon_error <- sapply(qs, function(q) {
  Pq <- P[, 1:q, drop = FALSE]
  Xhat <- X_centered %*% Pq %*% t(Pq)
  mean(rowSums((X_centered - Xhat)^2))
})
```

q	1	2	5	10	15	20	30	40
error	6.629	4.650	1.994	0.988	0.607	0.430	0.256	0.144

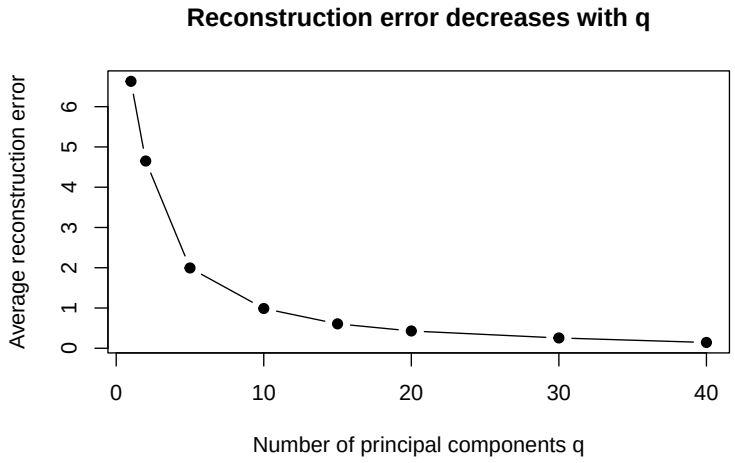


Figure 2.12: The average reconstruction error decreases with q.

Example 2.8.1

- The table is the theory in numbers: the error for q components equals exactly the lost variance $J = \sum_{j>q} \lambda_j$ (checked in R to all printed digits), and it reaches 0 at $q = 59$.
- The total variance is $J_0 = \sum_j \lambda_j \approx 6.63 + 2.61 = 9.24$, so $q = 5$ already keeps about 78% of it and $q = 15$ about 93%: with 15 numbers instead of 784 the digit is visually almost identical to the original.

- Eckart–Young check: with the SVD of the centered data and $q = 5$, `norm(X_centered - Xq, "2")` and σ_6 both equal 4.346685; and $\sigma_j^2/n = \lambda_j$ up to 10^{-14}.

Chapter 3

Introduction to Statistics

This module introduces the statistical viewpoint used in the rest of the course. The main themes are:

- **probability distributions as data generating processes**: a model is not only a formula, but a mechanism that can generate observations;
- **basic probability functions and standard distributions**, interpreted from a generative point of view;
- **population quantities, empirical distributions and empirical moments**: theoretical moments and their sample analogues;
- **estimation**: parameters are learned from data through plug-in ideas, likelihood and optimization;
- **Monte Carlo simulation**: repeated simulation to study the behaviour of statistical procedures;
- **properties of estimators and interval estimation**: uncertainty quantified through sampling distributions and confidence intervals.

The goal is to understand the chain

$$\text{probability model} \longrightarrow \text{data generated by it} \longrightarrow \text{empirical quantities} \longrightarrow \text{estimators} \longrightarrow \text{inference.}$$

Notation. Bold lowercase $\mathbf{x}$ = vectors, bold uppercase $\mathbf{X}$ = matrices; X is a random variable and x a realization; $\mathbf{X}$ may also be a random vector and $\mathbf{x}$ its observed value; $\mathbb{E}[X]$ expectation, $\mathrm{Var}(X)$ variance; $\boldsymbol{\Sigma}$ population covariance matrix, $\mathbf{S}$ empirical covariance matrix; $\overline{X}$ sample mean as a random quantity, $\overline{x}$ its observed value, $\overline{\mathbf{x}}$ an empirical mean vector.

3.1 Probability Theory for Statistics

A probability model is built from: an **outcome space**, **events**, a **probability function** and **random variables**.

3.1.1 Outcome space and events

Definition 3.1.1: Outcome space

The *outcome space* (or *sample space*) Ω is the set of all possible outcomes of a random experiment. When the experiment is performed, its outcome (*realization*) is an element $\omega \in \Omega$.

Example 3.1.1

One coin toss: $\Omega = \{\mathrm{H}, \mathrm{T}\}$; one die roll: $\Omega = \{1, \ldots, 6\}$; a nonnegative lifetime: $\Omega = [0, \infty)$.

Definition 3.1.2: Events and operations

An *event* is a subset $A \subseteq \Omega$; it *occurs* when the observed outcome belongs to it.

- **intersection**: $A \cap B = \{\omega : \omega \in A \text{ and } \omega \in B\}$ (both occur);
- **union**: $A \cup B = \{\omega : \omega \in A \text{ or } \omega \in B\}$ (at least one occurs; "or" is inclusive);
- **complement**: $A^c = \Omega \setminus A$ (A does not occur);
- A and B are **disjoint** (mutually exclusive) if $A \cap B = \varnothing$.

For finitely many events, $\bigcap_{i=1}^{m} A_i$ = outcomes in every A_i, $\bigcup_{i=1}^{m} A_i$ = outcomes in at least one A_i.

Example 3.1.2

Die roll, $A = \{2, 4, 6\}$ ("even"), $B = \{4, 5, 6\}$ ("at least 4"): $A \cap B = \{4, 6\}$, $A \cup B = \{2, 4, 5, 6\}$, $A^c = \{1, 3, 5\}$ ("odd"). "Even" and "odd" are mutually exclusive.

Venn diagrams

A Venn diagram represents Ω by a rectangle and events by regions; overlaps represent outcomes belonging to several events. Diagrams are schematic: areas are not probabilities unless stated.

Note:

Nine configurations for three events. There are three pairwise intersections $A \cap B$, $A \cap C$, $B \cap C$, each empty or not: $2^3 = 8$ patterns. But when all three pairwise intersections are nonempty, the triple intersection $A \cap B \cap C$ may itself be empty or not, so $8 + 1 = 9$ configurations must be distinguished.

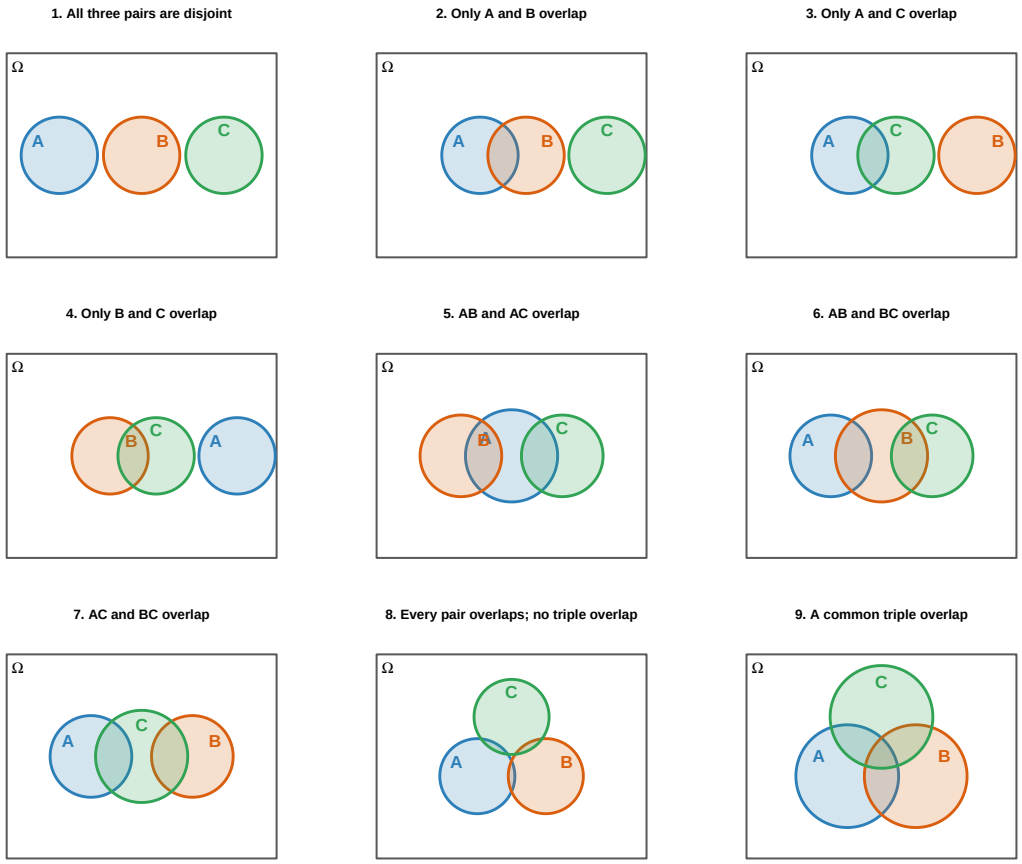


Figure 3.1: The nine configurations of disjointness and overlap for three events.

Basic set identities

Proposition 3.1.1 Set identities

For events A, B, C:

- complement laws: $A \cup A^c = \Omega$, $A \cap A^c = \varnothing$;
- commutative laws: $A \cap B = B \cap A$, $A \cup B = B \cup A$;
- distributive laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$;
- De Morgan's laws: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$.

Proof: Each identity is proved by *double inclusion*, i.e. by checking that an outcome ω belongs to the left side iff it belongs to the right side.

- Complement and commutative laws: $\omega \in A$ or $\omega \notin A$, never both; "and" and "or" are symmetric.
- Distributive: $\omega \in A \cap (B \cup C) \iff \omega \in A$ and ($\omega \in B$ or $\omega \in C$) $\iff$ ($\omega \in A$ and $\omega \in B$) or ($\omega \in A$ and $\omega \in C$) $\iff \omega \in (A \cap B) \cup (A \cap C)$. The second law is the same with "and" and "or" exchanged.
- De Morgan: $\omega \in (A \cup B)^c \iff$ not ($\omega \in A$ or $\omega \in B$) $\iff \omega \notin A$ and $\omega \notin B \iff \omega \in A^c \cap B^c$. Applying this to A^c, B^c and taking complements gives $(A \cap B)^c = A^c \cup B^c$.



Suggested exercise 32.

3.1.2 Probability function

Definition 3.1.3: Probability function

A *probability function* is a mapping $\mathbb{P} : \mathcal{A} \to [0,1]$, where $\mathcal{A}$ is a collection of events, such that

- $0 \leqslant \mathbb{P}(A) \leqslant 1$ for all $A \in \mathcal{A}$;
- $\mathbb{P}(\Omega) = 1$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$;
- for pairwise disjoint $A_1, A_2, \cdots \in \mathcal{A}$: $\mathbb{P}\big(\bigcup_{j\geqslant 1} A_j\big) = \sum_{j\geqslant 1} \mathbb{P}(A_j)$.

Proposition 3.1.2 Elementary consequences

1. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, in particular $\mathbb{P}(\varnothing) = 0$;
2. if $A \subseteq B$ then $\mathbb{P}(A) \leqslant \mathbb{P}(B)$;
3. $\mathbb{P}(A \cup B) \leqslant \mathbb{P}(A) + \mathbb{P}(B)$.

Proof: 1. A and A^c are disjoint with $A \cup A^c = \Omega$, so $1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c)$. With $A = \Omega$: $\mathbb{P}(\varnothing) = 0$.

2. $B = A \cup (B \cap A^c)$ is a disjoint union, so $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^c) \geqslant \mathbb{P}(A)$.

3. From the union formula, since $\mathbb{P}(A \cap B) \geqslant 0$.

The union formula itself follows from additivity: $A \cup B = A \cup (B \cap A^c)$ and $B = (A \cap B) \cup (B \cap A^c)$ are disjoint unions, so $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. ☺

Conditional probability

Definition 3.1.4: Conditional probability

For events A, B with $\mathbb{P}(B) > 0$, the *conditional probability* of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning restricts attention to the situations in which B has occurred.

Theorem 3.1.1

(Multiplication rule and Bayes' rule.) If $\mathbb{P}(B) > 0$, then $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\mathbb{P}(B)$. If moreover $\mathbb{P}(A) > 0$,

$$\mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) \qquad \text{and} \qquad \mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Proof: The multiplication rule is the definition multiplied by $\mathbb{P}(B)$. Writing it both ways, $\mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(A \cap B) = \mathbb{P}(B \cap A) = \mathbb{P}(B \mid A)\mathbb{P}(A)$; dividing by $\mathbb{P}(B)$ gives Bayes' rule. ☺

Suggested exercise 33.

Law of total probability

Theorem 3.1.2

(Law of total probability.) If $B_1, \dots, B_K$ form a *partition* of Ω (pairwise disjoint, $\bigcup_k B_k = \Omega$) with $\mathbb{P}(B_k) > 0$, then

$$\mathbb{P}(A) = \sum_{k=1}^{K} \mathbb{P}(A \mid B_k)\mathbb{P}(B_k).$$

Proof: By distributivity $A = A \cap \Omega = \bigcup_k (A \cap B_k)$, and the sets $A \cap B_k$ are pairwise disjoint (because the B_k are). By additivity and the multiplication rule, $\mathbb{P}(A) = \sum_k \mathbb{P}(A \cap B_k) = \sum_k \mathbb{P}(A \mid B_k)\mathbb{P}(B_k)$. ☺

Example 3.1.3

Medical screening. D = "has the condition", T = "test positive", with $\mathbb{P}(D) = 0.02$, $\mathbb{P}(T \mid D) = 0.95$ (sensitivity), $\mathbb{P}(T \mid D^c) = 0.05$ (false positive rate). Total probability with the partition $\{D, D^c\}$:

$$\mathbb{P}(T) = 0.95 \cdot 0.02 + 0.05 \cdot 0.98 = 0.068.$$

Bayes' rule:

$$\mathbb{P}(D \mid T) = \frac{\mathbb{P}(T \mid D)\mathbb{P}(D)}{\mathbb{P}(T)} = \frac{0.019}{0.068} \approx 0.279.$$

Even with an accurate test, a positive result means only a 28% chance of having the condition: $\mathbb{P}(T \mid D) = 0.95$ and $\mathbb{P}(D \mid T) \approx 0.28$ are different quantities. The reason is the low prevalence: among 1000 people, ≈ 19 true positives against ≈ 49 false positives.

```
p_D <- 0.02; p_T_given_D <- 0.95; p_T_given_not_D <- 0.05
p_T <- p_T_given_D * p_D + p_T_given_not_D * (1 - p_D) # 0.068
p_D_given_T <- p_T_given_D * p_D / p_T # 0.2794118
```

Independence of events

Definition 3.1.5: Independence

Events A and B are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Proposition 3.1.3

1. If $\mathbb{P}(B) > 0$: A, B independent $\iff$ $\mathbb{P}(A \mid B) = \mathbb{P}(A)$ (learning B does not change the probability of A).
2. If A, B are independent, so are A and B^c.
3. Two mutually exclusive events with positive probability are never independent.

Proof:

1. $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ equals $\mathbb{P}(A)$ iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
2. $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)(1 - \mathbb{P}(B)) = \mathbb{P}(A)\mathbb{P}(B^c)$.
3. $\mathbb{P}(A \cap B) = \mathbb{P}(\varnothing) = 0$, whereas $\mathbb{P}(A)\mathbb{P}(B) > 0$.

☺

Suggested exercise 34.

3.1.3 Random variables

Definition 3.1.6: Random variable

A *random variable* is a function of the outcomes $X : \Omega \to \mathbb{S}_X$; the set $\mathbb{S}_X$ of possible values is the *support*. For a set B of values, the induced probability is

$$\mathbb{P}(X \in B) = \mathbb{P}\big(\{\omega \in \Omega : X(\omega) \in B\}\big).$$

Example 3.1.4

Die roll: $X(\omega) = \omega$ (value of the die, $\mathbb{S}_X = \Omega$); $Y(\omega) = 1$ if ω is even and 0 otherwise (Y only records parity, $\mathbb{S}_Y = \{0, 1\}$). Then $\mathbb{P}(Y = 1) = \mathbb{P}(\{2, 4, 6\}) = 1/2$.

The support is usually **discrete finite** (e.g. $\{1, \dots, 6\}$), **discrete countably infinite** (e.g. $\{0, 1, 2, \dots\}$) or **continuous** (e.g. $\mathbb{R}_+$).

Definition 3.1.7: Cumulative distribution function

The *CDF* of X is $F_X(x) = \mathbb{P}(X \leqslant x)$, $x \in \mathbb{S}_X$. One writes $X \sim F_X$ ("is distributed according to"). It satisfies

- $0 \leqslant F_X(x) \leqslant 1$;
- F_X is nondecreasing;
- $\lim_{x \to \inf \mathbb{S}_X} F_X(x) = 0$ and $\lim_{x \to \sup \mathbb{S}_X} F_X(x) = 1$.

Note:

Monotonicity follows from the properties of $\mathbb{P}$: if $x \leqslant y$ then $\{X \leqslant x\} \subseteq \{X \leqslant y\}$, so $F_X(x) \leqslant F_X(y)$. Moreover $\mathbb{P}(a < X \leqslant b) = F_X(b) - F_X(a)$, since $\{X \leqslant b\}$ is the disjoint union of $\{X \leqslant a\}$ and $\{a < X \leqslant b\}$.

Definition 3.1.8: Mass function and density

- **Discrete** X: probability mass function $\mathbb{P}(X = x)$, and $F_X(x) = \sum_{t \leqslant x} \mathbb{P}(X = t)$.
- **Continuous** X: a density $f_X \geqslant 0$ such that $\mathbb{P}(a \leqslant X \leqslant b) = \int_a^b f_X(x)\,dx$, and $F_X(x) = \int_{\inf \mathbb{S}_X}^x f_X(t)\,dt$ (so $f_X = F_X'$ where differentiable).

Proposition 3.1.4 Points have probability zero

If X is (absolutely) continuous, $\mathbb{P}(X = a) = 0$ for every a, hence

$$\mathbb{P}(a < X < b) = \mathbb{P}(a \leqslant X < b) = \mathbb{P}(a < X \leqslant b) = \mathbb{P}(a \leqslant X \leqslant b).$$

Proof: $\mathbb{P}(X = a) = \int_a^a f_X(t)\,dt = 0$: a singleton has length zero. The four intervals differ only by the endpoints $\{a\}$ and $\{b\}$, which are disjoint from the rest and have probability 0, so by additivity the probabilities coincide. ☺

Suggested exercise 35.

3.1.4 Joint distributions

Definition 3.1.9: Joint density

A pair (X, Y) of continuous random variables is described by a joint density $f_{X,Y}(x, y) \geqslant 0$ with

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1$$

($f_{X,Y} = 0$ outside the supports). Probabilities of regions are integrals, e.g. $\mathbb{P}(X \leqslant a, Y \leqslant b) = \int_{-\infty}^{a} \int_{-\infty}^{b} f_{X,Y}(x, y)\, dy\, dx$.

Definition 3.1.10: Marginal and conditional densities

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx, \qquad f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)} \quad (f_X(x) > 0).$$

In the discrete case integrals become sums.

Note:

The marginal is a density: $f_X \geqslant 0$ and $\int f_X(x)\, dx = \iint f_{X,Y} = 1$. Integrating out y means "ignoring Y": $\mathbb{P}(X \leqslant a) = \mathbb{P}(X \leqslant a, Y \in \mathbb{R}) = \int_{-\infty}^{a} \big(\int f_{X,Y}(x, y)\, dy \big) dx$. Also the conditional density integrates to one in y: $\int f_{Y|X}(y \mid x)\, dy = f_X(x)/f_X(x) = 1$.

Definition 3.1.11: Independent random variables

X and Y are independent if $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ for all x, y (continuous case), or $\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$ (discrete case). Equivalently $f_{Y|X}(y \mid x) = f_Y(y)$.

Suggested exercise 36.

3.1.5 Empirical distribution function

A distribution F_X can be interpreted as a **data generating process**: repeated draws of X generate observations distributed according to F_X.

- A **random sample** $X_1, \ldots, X_n$ consists of copies of the same random variable X, usually assumed **independent and identically distributed** (i.i.d.); it is used to study sample quantities (means, estimators).
- The **observed data** $x_1, \ldots, x_n \in \mathbb{S}_X$ are realizations of them: the *sample*.

Definition 3.1.12: Empirical distribution function (EDF)

$$F^{(n)}(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(x_i \leqslant x), \qquad \mathbf{1}(A) = \begin{cases} 1 & A \text{ true} \\ 0 & A \text{ false} \end{cases}$$

It plays, for the observed sample, the role that F_X plays for the population: it is the CDF of the distribution putting mass $1/n$ on each observation (a step function with jumps of $1/n$).

Proposition 3.1.5 The EDF estimates the CDF

For an i.i.d. sample from F_X and a fixed x, $nF^{(n)}(x) \sim \mathcal{B}(n, F_X(x))$, hence

$$\mathbb{E}\big[F^{(n)}(x)\big] = F_X(x), \qquad \operatorname{Var}\big(F^{(n)}(x)\big) = \frac{F_X(x)\big(1 - F_X(x)\big)}{n} \longrightarrow 0.$$

Proof: The indicators $\mathbf{1}(X_i \leqslant x)$ are i.i.d. Bernoulli with success probability $\mathbb{P}(X_i \leqslant x) = F_X(x)$, so their sum is binomial (see the Bernoulli and binomial distributions below). Dividing the binomial mean $nF_X(x)$ and variance $nF_X(x)(1 - F_X(x))$ by n and n^2 gives the claim. ☺

Suggested exercise 37.

```
x <- rnorm(100, mean = 0, sd = 1)
Fn <- ecdf(x)
plot(Fn, main = "Empirical distribution")
curve(pnorm(x), add = TRUE, lty = 2)
```

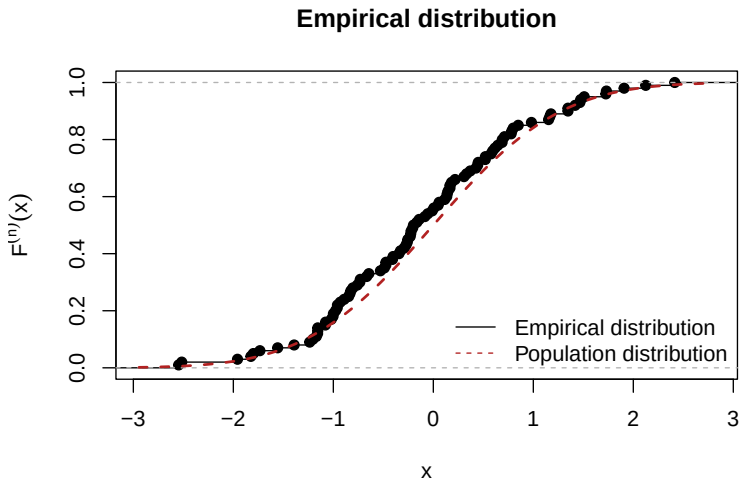


Figure 3.2: EDF of 100 draws from $\mathcal{N}(0,1)$ (steps) and the population CDF Φ (dashed).

3.1.6 Moments

A *moment* summarizes an aspect of a distribution: the mean (location), the variance (spread), higher-order moments (shape). **Population moments** are computed with F_X, **empirical moments** with $F^{(n)}$.

Definition 3.1.13: Population moment

For a suitable function g,

$$\mathbb{E}[g(X)] = \int_{S_X} g(x)\, dF_X(x) = \int_{S_X} g(x) f_X(x)\, dx,$$

since $dF_X(x) = f_X(x)\, dx$ in the continuous case.

Definition 3.1.14: Expectation and variance

- **Expectation** (first moment, population mean, $g(x) = x$): $\mu = \mathbb{E}[X] = \int x\, dF_X(x)$. It describes the centre of the distribution.
- **Second raw moment**: $\mathbb{E}[X^2] = \int x^2\, dF_X(x)$.
- **Variance** (second *central* moment):

$$\sigma^2 = \mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big] = \int (x - \mu)^2\, dF_X(x) \geqslant 0.$$

Proposition 3.1.6 Variance from the first two raw moments

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.$$

Proof: Expand the square and use linearity of the integral, with μ a constant and $\int dF_X = 1$:

$$\mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - 2\mu^2 + \mu^2 = \mathbb{E}[X^2] - \mu^2.$$

In particular $\mathbb{E}[X^2] \geqslant (\mathbb{E}[X])^2$, since the variance is nonnegative. ☺

Suggested exercises 38 and 39.

Discrete case

If X is discrete, $dF_X(x) = \mathbb{P}(X = x)$ for $x \in \mathbf{S}_X$ and 0 otherwise, so

$$\mathbb{E}[g(X)] = \sum_{x \in \mathbf{S}_X} g(x)\mathbb{P}(X = x).$$

Example 3.1.5
Fair die: $\mathbb{E}[X] = \frac{1+\cdots+6}{6} = 3.5$, $\mathbb{E}[X^2] = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$, so $\operatorname{Var}(X) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} \approx 2.92$.

Properties

Proposition 3.1.7 Properties of expectation and variance

For constants $a, b, c \in \mathbb{R}$:

1. $\mathbb{E}[c] = c$; $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$, and $\mathbb{E}\big[\sum_i a_i X_i\big] = \sum_i a_i \mathbb{E}[X_i]$; $\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$;
2. if X, Y are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ (the converse is false in general);
3. $\operatorname{Var}(c) = 0$, $\operatorname{Var}(X + b) = \operatorname{Var}(X)$, $\operatorname{Var}(aX) = a^2 \operatorname{Var}(X)$, $\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X)$;
4. $\operatorname{Var}(aX + bY) = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab \operatorname{Cov}(X, Y)$, in particular $\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y)$;
5. if X, Y are independent, $\operatorname{Cov}(X, Y) = 0$ and $\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$; more generally, for independent $X_1, \ldots, X_n$, $\operatorname{Var}\big(\sum_i a_i X_i\big) = \sum_i a_i^2 \operatorname{Var}(X_i)$.

Proof:

1. Linearity of the integral (sum): $\mathbb{E}[aX + bY] = \iint (ax + by) f_{X,Y}\, dx\, dy = a \int x f_X\, dx + b \int y f_Y\, dy$, using the marginals. $\mathbb{E}[c] = c \int dF = c$. The finite-sum version follows by induction.
2. By independence $f_{X,Y} = f_X f_Y$, so $\mathbb{E}[XY] = \iint xy\, f_X(x) f_Y(y)\, dx\, dy = \int x f_X(x)\, dx \int y f_Y(y)\, dy$.
3. With $\mu = \mathbb{E}[X]$: $\mathbb{E}[aX + b] = a\mu + b$, hence $\operatorname{Var}(aX + b) = \mathbb{E}\big[(aX + b - a\mu - b)^2\big] = a^2 \mathbb{E}[(X - \mu)^2] = a^2 \operatorname{Var}(X)$. The other three are special cases ($a = 0$; $a = 1$; $b = 0$).
4. With μ_X, μ_Y the means, $\operatorname{Var}(aX + bY) = \mathbb{E}\big[(a(X - \mu_X) + b(Y - \mu_Y))^2\big] = a^2 \operatorname{Var}(X) + b^2 \operatorname{Var}(Y) + 2ab\, \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]$.
5. $\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$ by point 2. For n variables, $\operatorname{Var}\big(\sum_i a_i X_i\big) = \sum_i a_i^2 \operatorname{Var}(X_i) + \sum_{i \neq j} a_i a_j \operatorname{Cov}(X_i, X_j)$ by the same expansion, and all covariances vanish.

☺

Example 3.1.6

Uncorrelated but dependent. Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Then $\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}[X]\mathbb{E}[X^2] = 0$ (odd moments of a symmetric distribution vanish), but Y is a function of X, so they are clearly not independent. This is why "$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$" does not imply independence.

Suggested exercise 40.

Covariance

Definition 3.1.15: Covariance

For jointly distributed X, Y with finite second moments,

$$\operatorname{Cov}(X,Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Positive: large values of one tend to occur with large values of the other; negative: large with small; near zero: little *linear* association (it does not imply independence).

Why the two expressions agree: Expanding, $\mathbb{E}[(X-\mu_X)(Y-\mu_Y)] = \mathbb{E}[XY] - \mu_X\mathbb{E}[Y] - \mu_Y\mathbb{E}[X] + \mu_X\mu_Y = \mathbb{E}[XY] - \mu_X\mu_Y$.

If X, Y are independent (finite second moments) then $\operatorname{Cov}(X,Y) = 0$; the converse is generally false. The covariance enters the variance of a sum: $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y)$.
Suggested exercise 41.

Correlation

Definition 3.1.16: Correlation

$$\rho_{X,Y} = \operatorname{cor}(X,Y) = \frac{\operatorname{Cov}(X,Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}.$$

Theorem 3.1.3

1. $-1 \leqslant \rho_{X,Y} \leqslant 1$, with $|\rho_{X,Y}| = 1$ iff $Y = aX + b$ almost surely for some $a \neq 0$;
2. correlation is invariant to positive changes of scale: $\operatorname{cor}(aX + c, bY + d) = \operatorname{cor}(X,Y)$ for $ab > 0$ (and it changes sign if $ab < 0$).

Proof: **1.** Let $U = (X - \mu_X)/\sigma_X$ and $V = (Y - \mu_Y)/\sigma_Y$, so $\mathbb{E}[U^2] = \mathbb{E}[V^2] = 1$ and $\mathbb{E}[UV] = \rho$. Then

$$0 \leqslant \mathbb{E}\big[(U-V)^2\big] = 2 - 2\rho, \qquad 0 \leqslant \mathbb{E}\big[(U+V)^2\big] = 2 + 2\rho,$$

hence $-1 \leqslant \rho \leqslant 1$. Equality $\rho = 1$ means $\mathbb{E}[(U-V)^2] = 0$, i.e. $U = V$ almost surely, a linear relation between X and Y with positive slope (similarly $\rho = -1$ with negative slope).
2. Constants do not change covariances or variances, $\operatorname{Cov}(aX+c, bY+d) = ab\operatorname{Cov}(X,Y)$ and $\sqrt{\operatorname{Var}(aX+c)\operatorname{Var}(bY+d)} = |a||b|\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}$, so $\operatorname{cor}(aX+c, bY+d) = \frac{ab}{|ab|}\operatorname{cor}(X,Y)$.

Suggested exercise 42.

Empirical moments

For a sample $x_1, \dots, x_n$ the EDF puts mass $1/n$ on each observation ($dF^{(n)}(x_i) = 1/n$), so the plug-in moments are

$$\overline{x} = \int x \, dF^{(n)}(x) = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad s^2 = \int (x-\overline{x})^2 \, dF^{(n)}(x) = \frac{1}{n}\sum_{i=1}^{n}(x_i-\overline{x})^2$$

(*sample* or *empirical* mean and variance), and for paired data (x_i, y_i)

$$\widehat{\mathrm{Cov}}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i-\overline{x})(y_i-\overline{y}).$$

Note:

Empirical moments are close to, but not equal to, the population values: they depend on the random sample and fluctuate from sample to sample even when all samples come from the same F_X. The usual *unbiased* sample covariance divides by $n-1$; R's `cov()` and `var()` use this convention. The **correlation is unaffected** as long as the same denominator is used in the covariance and in the standard deviations (the factor cancels).

Proposition 3.1.8 Computational formulas

$$\frac{1}{n}\sum_i (x_i-\overline{x})^2 = \frac{1}{n}\sum_i x_i^2 - \overline{x}^2, \qquad \frac{1}{n}\sum_i (x_i-\overline{x})(y_i-\overline{y}) = \frac{1}{n}\sum_i x_i y_i - \overline{x}\,\overline{y}, \qquad \sum_i (x_i-\overline{x}) = 0.$$

Proof: These are $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ and $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$ applied to the empirical distribution $F^{(n)}$. Directly: $\sum_i (x_i-\overline{x}) = n\overline{x} - n\overline{x} = 0$, and $\sum_i (x_i-\overline{x})(y_i-\overline{y}) = \sum_i x_i y_i - \overline{y}\sum_i x_i - \overline{x}\sum_i y_i + n\overline{x}\,\overline{y} = \sum_i x_i y_i - n\overline{x}\,\overline{y}$.

Suggested exercises 43 and 44.

```
set.seed(123)
n <- 200
x <- rnorm(n)
y <- 0.8 * x + rnorm(n, sd = 0.6)
c(covariance = cov(x, y), correlation = cor(x, y))
## covariance correlation
## 0.6960265 0.7772878
```

Example 3.1.7

Population values: $\mathrm{Cov}(X,Y) = \mathrm{Cov}(X, 0.8X + \varepsilon) = 0.8\,\mathrm{Var}(X) = 0.8$, $\mathrm{Var}(Y) = 0.64 + 0.36 = 1$, so $\rho = 0.8/\sqrt{1\cdot 1} = 0.8$. The sample gives 0.696 and 0.777: close, but not equal.

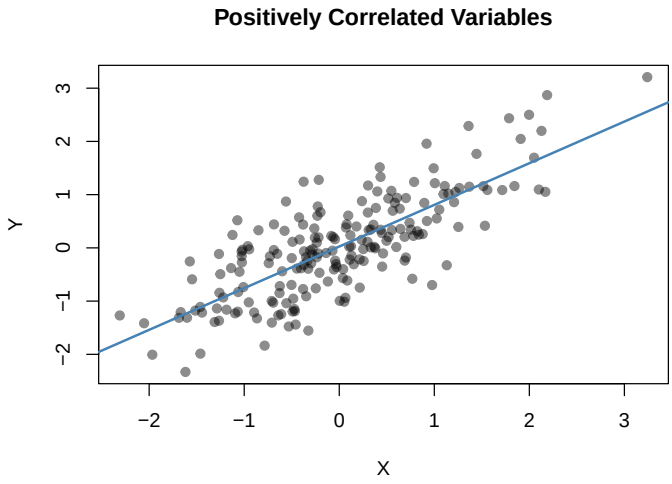


Figure 3.3: Simulated positively correlated variables with the least-squares line.

3.1.7 Quantiles

Definition 3.1.17: Quantile

For $p \in (0,1)$, a *p-quantile* is a value q_p with $F_X(q_p) \geqslant p$. The standard definition is the *generalized inverse* of the CDF:

$$q_p = \inf\{x \in \mathbb{S}_X : F_X(x) \geqslant p\}.$$

If F_X is continuous and strictly increasing, $q_p = F_X^{-1}(p)$, i.e. $F_X(q_p) = p$.

Special quantiles: the **median** $m = q_{0.5}$; the **quartiles** $q_{0.25}$, $q_{0.5}$, $q_{0.75}$; the $100p$**-th percentile** is q_p (e.g. the 90-th percentile is $q_{0.90}$).
Suggested exercise 45.

```
qnorm(0.95) ## 1.644854
qbinom(0.95, size = 10, prob = 0.3) ## 5
qunif(0.25, min = -1, max = 3) ## 0
qexp(0.5, rate = 2) ## 0.3465736
set.seed(123); x <- rnorm(100)
quantile(x, probs = c(0.25, 0.5, 0.75)) # quantiles of the EDF
## 25% 50% 75%
## -0.49385424 0.06175631 0.69181917
```

Example 3.1.8

Checks by hand: $\mathcal{U}(-1,3)$ has $F(x) = (x+1)/4$, so $q_{0.25}$ solves $(x+1)/4 = 0.25$, $x = 0$. Exp(2) has $F(x) = 1 - e^{-2x}$, so the median solves $e^{-2x} = 0.5$, $x = \log 2/2 \approx 0.3466$. For the discrete binomial the generalized inverse is needed: 5 is the smallest x with $F(x) \geqslant 0.95$.

3.1.8 Some well-known distributions

Bernoulli distribution

Definition 3.1.18: Bernoulli

$X \sim \mathcal{B}(1,\pi)$ models a single binary outcome, $\mathbb{S}_X = \{0,1\}$: $\mathbb{P}(X=1) = \pi$ (success), $\mathbb{P}(X=0) = 1-\pi$ (failure); compactly

$$\mathbb{P}(X = x \mid \pi) = \pi^x (1-\pi)^{1-x}, \qquad x \in \{0,1\}.$$

Examples: a coin flip, whether a customer clicks on an ad, whether a classifier's prediction is correct.

Proposition 3.1.9 Moments and quantiles of the Bernoulli

$\mathbb{E}[X] = \pi$, $\mathrm{Var}(X) = \pi(1-\pi)$. The p-quantile is 0 if $p \leqslant 1-\pi$ and 1 if $p > 1-\pi$.

Proof: $\mathbb{E}[X] = 0\cdot(1-\pi) + 1\cdot\pi = \pi$. Since $X^2 = X$ for $X \in \{0,1\}$, $\mathbb{E}[X^2] = \pi$ and $\mathrm{Var}(X) = \pi - \pi^2 = \pi(1-\pi)$. For the quantiles, $F_X(x) = 0$ for $x < 0$, $= 1-\pi$ on $[0,1)$, $= 1$ for $x \geqslant 1$: the smallest x with $F_X(x) \geqslant p$ is 0 when $p \leqslant 1-\pi$, and 1 otherwise. ☺

So several probability levels correspond to the same quantile. In R: `rbinom(n, size = 1, prob = p)`.
Suggested exercise 46.

Binomial distribution

Definition 3.1.19: Binomial

X counts the successes in N independent Bernoulli trials with success probability π: $\mathbb{S}_X = \{0,\dots,N\}$, $X \sim \mathcal{B}(N,\pi)$ and

$$\mathbb{P}(X = x \mid N,\pi) = \binom{N}{x}\pi^x(1-\pi)^{N-x}, \qquad \binom{N}{x} = \frac{N!}{x!\,(N-x)!}.$$

Proposition 3.1.10

$\mathbb{E}[X] = N\pi$, $\quad \mathrm{Var}(X) = N\pi(1-\pi)$.

Proof: **Mass function.** A specific sequence with x successes and $N-x$ failures has probability $\pi^x(1-\pi)^{N-x}$ by independence, and there are $\binom{N}{x}$ such sequences (choice of the positions of the successes).

Moments. Write $X = \sum_{k=1}^{N} B_k$ with $B_k \overset{\text{i.i.d.}}{\sim} \mathcal{B}(1,\pi)$. By linearity $\mathbb{E}[X] = N\pi$; by independence $\mathrm{Var}(X) = \sum_k \mathrm{Var}(B_k) = N\pi(1-\pi)$. ☺

In R: `rbinom(n, size = N, prob = p)`.

Uniform distribution

Definition 3.1.20: Uniform

$X \sim \mathcal{U}(a,b)$ if all values of $[a,b]$ are equally plausible: $\mathbb{S}_X = [a,b]$,

$$f_X(x) = \frac{1}{b-a} \quad (a \leqslant x \leqslant b), \qquad F_X(x) = \frac{x-a}{b-a} \quad (a \leqslant x \leqslant b).$$

Proposition 3.1.11

$\mathbb{E}[X] = \dfrac{a+b}{2}$, $\quad \mathrm{Var}(X) = \dfrac{(b-a)^2}{12}$, $\quad$ median $q_{0.5} = \dfrac{a+b}{2}$.

Proof:

$$\mathbb{E}[X] = \int_a^b \frac{x}{b-a}\,dx = \frac{b^2-a^2}{2(b-a)} = \frac{a+b}{2}, \qquad \mathbb{E}[X^2] = \int_a^b \frac{x^2}{b-a}\,dx = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}.$$

Hence $\mathrm{Var}(X) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4} = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12} = \frac{(b-a)^2}{12}$. The median solves $\frac{x-a}{b-a} = \frac{1}{2}$, i.e. $x = \frac{a+b}{2}$. ☺

In R: `runif(10, min = -1, max = 1)`.
Suggested exercise 47.

Theorem 3.1.4

(Probability integral transform and inverse transform sampling.) Let $X \sim F_X$ with F_X continuous and strictly increasing. Then

$$U = F_X(X) \sim \mathcal{U}(0,1).$$

Conversely, if $U \sim \mathcal{U}(0,1)$, then $X = F_X^{-1}(U)$ has CDF F_X.

Proof: For $u \in [0,1]$, since F_X is strictly increasing, $F_X(X) \leqslant u \iff X \leqslant F_X^{-1}(u)$, so

$$\mathbb{P}(U \leqslant u) = \mathbb{P}(F_X(X) \leqslant u) = \mathbb{P}\big(X \leqslant F_X^{-1}(u)\big) = F_X\big(F_X^{-1}(u)\big) = u,$$

which is the CDF of $\mathcal{U}(0,1)$. Conversely, $\mathbb{P}(F_X^{-1}(U) \leqslant x) = \mathbb{P}(U \leqslant F_X(x)) = F_X(x)$, because the uniform CDF is the identity on $[0,1]$. ☺

Note:

This is why the uniform distribution is central in simulation: pseudo-random generators produce $\mathcal{U}(0,1)$ numbers, which are then transformed by F_X^{-1}. Example: for $\mathrm{Exp}(\lambda)$, $F^{-1}(u) = -\log(1-u)/\lambda$, so $-\log(1-U)/\lambda \sim \mathrm{Exp}(\lambda)$.

Suggested exercise 48.

Gaussian distribution

Definition 3.1.21: Gaussian (normal)

$X \sim \mathcal{N}(\mu, \sigma^2)$ has density

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big), \qquad \mathbb{E}[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2.$$

It is the most important continuous distribution in statistics. Its CDF is denoted Φ when $\mu = 0$, $\sigma = 1$.

Theorem 3.1.5

(Affine transformations.) If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = aX + b$ with $a \neq 0$, then $Y \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$. Consequently the *standardized* variable

$$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1).$$

Proof: Take $a > 0$ (the case $a < 0$ is analogous, with a sign flip). Then $F_Y(y) = \mathbb{P}(aX + b \leqslant y) = F_X\big(\frac{y-b}{a}\big)$, and differentiating,

$$f_Y(y) = \frac{1}{a} f_X\Big(\frac{y-b}{a}\Big) = \frac{1}{\sqrt{2\pi a^2\sigma^2}} \exp\Big(-\frac{\big(\frac{y-b}{a} - \mu\big)^2}{2\sigma^2}\Big) = \frac{1}{\sqrt{2\pi a^2\sigma^2}} \exp\Big(-\frac{(y-a\mu-b)^2}{2a^2\sigma^2}\Big),$$

the $\mathcal{N}(a\mu + b, a^2\sigma^2)$ density. For Z take $a = 1/\sigma$, $b = -\mu/\sigma$. ☺

Suggested exercise 49.

Exponential distribution

Definition 3.1.22: Exponential

$X \sim \text{Exp}(\lambda)$, $\lambda > 0$, has density $f(x) = \lambda e^{-\lambda x}$ for $x \geqslant 0$. It is the standard model for a positive waiting time (inter-arrival times).

Proposition 3.1.12

$$F(x) = 1 - e^{-\lambda x}, \quad \mathbb{E}[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}, \quad \text{median} = \frac{\log 2}{\lambda}.$$

Proof: $F(x) = \int_0^x \lambda e^{-\lambda t}\, dt = 1 - e^{-\lambda x}$. By parts,

$$\mathbb{E}[X] = \int_0^\infty x\lambda e^{-\lambda x} dx = \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x} dx = \frac{1}{\lambda}, \qquad \mathbb{E}[X^2] = \left[-x^2 e^{-\lambda x}\right]_0^\infty + \int_0^\infty 2xe^{-\lambda x} dx = \frac{2}{\lambda}\mathbb{E}[X] = \frac{2}{\lambda^2},$$

so $\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$. The median solves $e^{-\lambda x} = \frac{1}{2}$.

In R: `rexp(10, rate = 2)`.

Combinations of Gaussian random variables

A sum of *jointly Gaussian* random variables is Gaussian; its mean is the sum of the means, and its variance includes covariance terms unless the variables are independent. In the independent case, if $X_j \sim \mathcal{N}(\mu_j, \sigma_j^2)$, $j = 1, \ldots, p$,

$$\sum_{j=1}^p X_j \sim \mathcal{N}\Big(\sum_{j=1}^p \mu_j,\ \sum_{j=1}^p \sigma_j^2\Big).$$

Note:

The *parameters* follow from the properties of moments alone (linearity of $\mathbb{E}$, additivity of Var under independence); what is special to the Gaussian is that the *shape* stays Gaussian. Combined with the affine property, the sample mean of i.i.d. $\mathcal{N}(\mu, \sigma^2)$ variables is exactly $\overline{X} \sim \mathcal{N}(\mu, \sigma^2/n)$. Generatively, a Gaussian produces observations concentrated around μ with spread governed by σ^2.

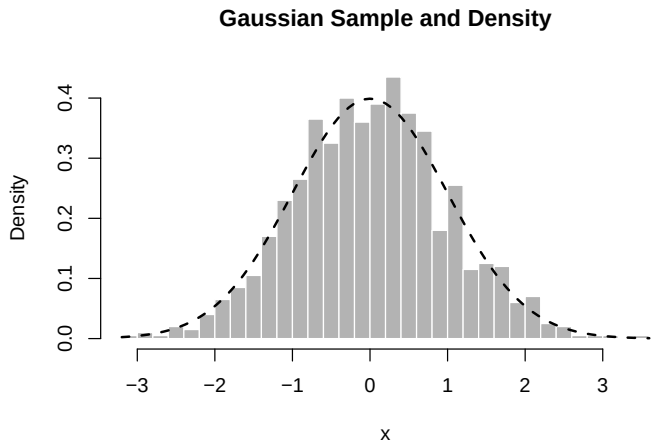


Figure 3.4: Histogram of 1000 draws from $\mathcal{N}(0, 1)$ and the density.

Multivariate Gaussian distribution

Definition 3.1.23: Multivariate Gaussian

A random vector $\mathbf{X} \in \mathbb{R}^p$ is $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with mean $\boldsymbol{\mu} \in \mathbb{R}^p$ and covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$ (symmetric positive definite), if its density is

$$f(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = (2\pi)^{-p/2} |\boldsymbol{\Sigma}|^{-1/2} \exp\Big(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \Big).$$

Note:

- For $p = 1$, $\boldsymbol{\Sigma} = \sigma^2$ and the formula reduces to the univariate density.
- If $\boldsymbol{\Sigma} = \operatorname{diag}(\sigma_1^2, \ldots, \sigma_p^2)$, the quadratic form is $\sum_j (x_j - \mu_j)^2 / \sigma_j^2$ and the density factorizes into the product of the marginals: uncorrelated jointly Gaussian components are **independent** (a property specific to the Gaussian).
- The population covariance matrix $\boldsymbol{\Sigma} = \mathbb{E}\big[(\mathbf{X} - \mathbb{E}[\mathbf{X}])(\mathbf{X} - \mathbb{E}[\mathbf{X}])^\top\big]$ is exactly the object approximated by the empirical covariance matrix $\mathbf{S} = \frac{1}{n} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ of the PCA chapter: the multivariate Gaussian is the natural population model behind covariance matrices and PCA.

```r
set.seed(123)
mu <- c(0, 1)
Sigma <- matrix(c(1, 0.7,
                  0.7, 2), nrow = 2, byrow = TRUE)
X <- MASS::mvrnorm(n = 500, mu = mu, Sigma = Sigma)
```

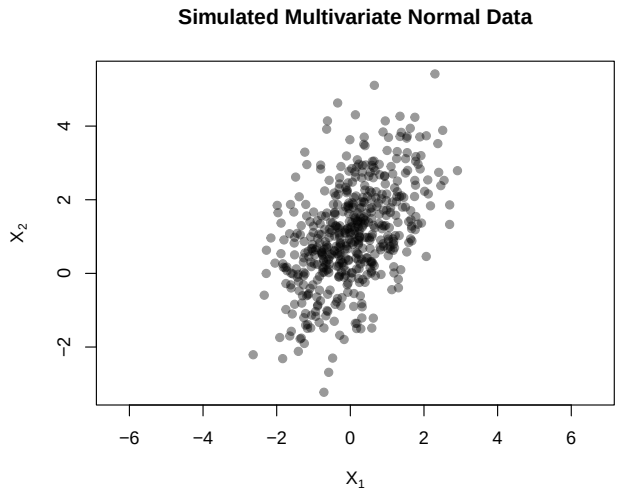


Figure 3.5: 500 draws from $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$; the correlation is $0.7/\sqrt{2} \approx 0.49$.

3.2 Estimation

A distribution (*statistical model*) structures our thinking about data; the observed data and their EDF contain information on the underlying model. If a sample looks approximately normal, we may model it as $\mathcal{N}(\mu, \sigma^2)$ with *unknown* parameters. **Estimation** uses the data to learn the parameters of the data generating process.

Definition 3.2.1: Point estimator

A *point estimator* is a function of the data $X_1, \dots, X_n$. Examples: the sample mean $\overline{x}$ for the population mean μ; the empirical covariance matrix $\mathbf{S}$ for $\mathbf{\Sigma}$.

3.2.1 Plug-in estimators

Definition 3.2.2: Plug-in estimator

If the target is a *functional* of the distribution, $\theta = T(F_X)$, the plug-in estimator replaces F_X by the EDF:

$$\widehat{\theta} = T\big(F^{(n)}\big).$$

Example 3.2.1

$$\mu = \int x\, dF_X(x) \implies \widehat{\mu} = \int x\, dF^{(n)}(x) = \overline{x}, \qquad \sigma^2 = \int (x-\mu)^2\, dF_X(x) \implies \widehat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \overline{x})^2.$$

Likewise the median $q_{0.5}(F_X)$ is estimated by the sample median, and $\mathbf{\Sigma}$ by $\mathbf{S}$ (multivariate case).

3.2.2 Maximum likelihood estimation

Definition 3.2.3: Likelihood and MLE

Let the data be modelled by $\mathbb{P}(X = x \mid \theta) = p(x \mid \theta)$ (a mass function or a density), with unknown parameter θ. For i.i.d. observations $x_1, \dots, x_n$, the *likelihood* is the joint probability of the sample as a function of θ,

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta),$$

and the *maximum likelihood estimator* is $\widehat{\theta} = \arg\max_\theta L(\theta)$. One usually maximizes the *log-likelihood*

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta),$$

which has the same maximizer because log is increasing, and turns products into sums.

Proposition 3.2.1 MLE of a Bernoulli

If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{B}(1, \pi)$, $\pi \in [0,1]$, then $\widehat{\pi} = \frac{1}{n}\sum_i x_i = \overline{x}$.

Proof: With $p(x_i) = \pi^{x_i}(1-\pi)^{1-x_i}$ and $k = \sum_i x_i$ (number of successes),

$$\ell(\pi) = \sum_{i=1}^{n} \big[x_i \log \pi + (1 - x_i)\log(1-\pi)\big] = k \log \pi + (n-k)\log(1-\pi).$$

For $0 < k < n$:

$$\ell'(\pi) = \frac{k}{\pi} - \frac{n-k}{1-\pi} = 0 \iff k(1-\pi) = (n-k)\pi \iff \pi = \frac{k}{n},$$

and $\ell''(\pi) = -\frac{k}{\pi^2} - \frac{n-k}{(1-\pi)^2} < 0$, so it is a maximum. If $k = 0$ (resp. $k = n$), $\ell(\pi) = n\log(1-\pi)$ (resp. $n \log \pi$) is maximized at $\pi = 0$ (resp. 1), again k/n. ∎

So familiar empirical quantities are not only intuitive summaries: they also arise from a probabilistic estimation principle.
Suggested exercise 50.

Proposition 3.2.2 MLE of a Gaussian

If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then

$$\widehat{\mu}_{\text{ML}} = \overline{x}, \qquad \widehat{\sigma}^2_{\text{ML}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2.$$

Proof:

$$\ell(\mu, \sigma^2) = \sum_{i=1}^{n} \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x_i-\mu)^2/(2\sigma^2)}\right] = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.$$

In μ: for any σ^2, maximizing ℓ means minimizing $\sum_i (x_i-\mu)^2$; $\frac{\partial}{\partial\mu}\sum_i(x_i-\mu)^2 = -2\sum_i(x_i-\mu) = 0 \iff \mu = \overline{x}$ (a minimum, the function being a convex parabola). Equivalently, $\sum_i(x_i-\mu)^2 = \sum_i(x_i-\overline{x})^2 + n(\overline{x}-\mu)^2$.
In σ^2: set $v = \sigma^2$ and $Q = \sum_i(x_i-\overline{x})^2$. Then $\frac{\partial \ell}{\partial v} = -\frac{n}{2v} + \frac{Q}{2v^2} = 0 \iff v = Q/n$; the derivative is positive for $v < Q/n$ and negative for $v > Q/n$, so this is the maximum.

Note:

For the Gaussian, the MLEs coincide with the plug-in estimators. Note that $\widehat{\sigma}^2_{\text{ML}}$ divides by n: it is *biased* (see below).

Suggested exercise 51.

3.2.3 Properties of estimators

- An **estimator** is a function of the random sample $X_1, \dots, X_n$: it is itself a *random variable*, with a *sampling distribution* (often unknown, though some of its moments can be derived).
- An **estimate** is the value of the estimator on an observed sample.

The moments of the sampling distribution are used to study the properties of estimators.

Bias

Definition 3.2.4: Bias

$\text{Bias}(\widehat{\theta}) = \mathbb{E}[\widehat{\theta}] - \theta$. The estimator is *unbiased* if $\mathbb{E}[\widehat{\theta}] = \theta$: on average over repeated samples it hits the true parameter.

Proposition 3.2.3 The sample mean is unbiased

If $X_1, \dots, X_n$ are i.i.d. with mean μ, then $\mathbb{E}[\overline{X}] = \mu$.

Proof: By linearity, $\mathbb{E}[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \frac{1}{n}\cdot n\mu = \mu$.

Variance of an estimator

Definition 3.2.5: Variance and efficiency

$\text{Var}(\widehat{\theta}) = \mathbb{E}\big[(\widehat{\theta} - \mathbb{E}[\widehat{\theta}])^2\big]$; a good estimator has small variance. Among unbiased estimators of the same parameter, the one with smaller variance is more *efficient*. The *standard error* is $\sqrt{\text{Var}(\widehat{\theta})}$.

Proposition 3.2.4 Variance of the sample mean

If $X_1, \dots, X_n$ are i.i.d. with variance σ^2, then

$$\operatorname{Var}(\overline{X}) = \frac{\sigma^2}{n}, \qquad \operatorname{se}(\overline{X}) = \frac{\sigma}{\sqrt{n}}.$$

Proof: By independence, $\operatorname{Var}\left(\frac{1}{n}\sum_i X_i\right) = \frac{1}{n^2}\sum_i \operatorname{Var}(X_i) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}$. ✆

So increasing the sample size reduces the variability of the estimator. For Gaussian data, $\overline{X}_n \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly.

```
set.seed(123)
B <- 5000; sample_sizes <- c(5, 10, 30, 100, 300); mu <- 2; sigma <- 3
# for each n: B sample means of Gaussian samples
xbar <- replicate(B, mean(rnorm(n, mean = mu, sd = sigma)))
```

n	mean of $\overline{X}$	bias	empirical var.	σ^2/n	$\sigma/\sqrt{n}$
5	1.9805	−0.0195	1.8050	1.80	1.3416
10	2.0151	0.0151	0.8786	0.90	0.9487
30	2.0156	0.0156	0.3006	0.30	0.5477
100	1.9954	−0.0046	0.0903	0.09	0.3000
300	1.9969	−0.0031	0.0305	0.03	0.1732

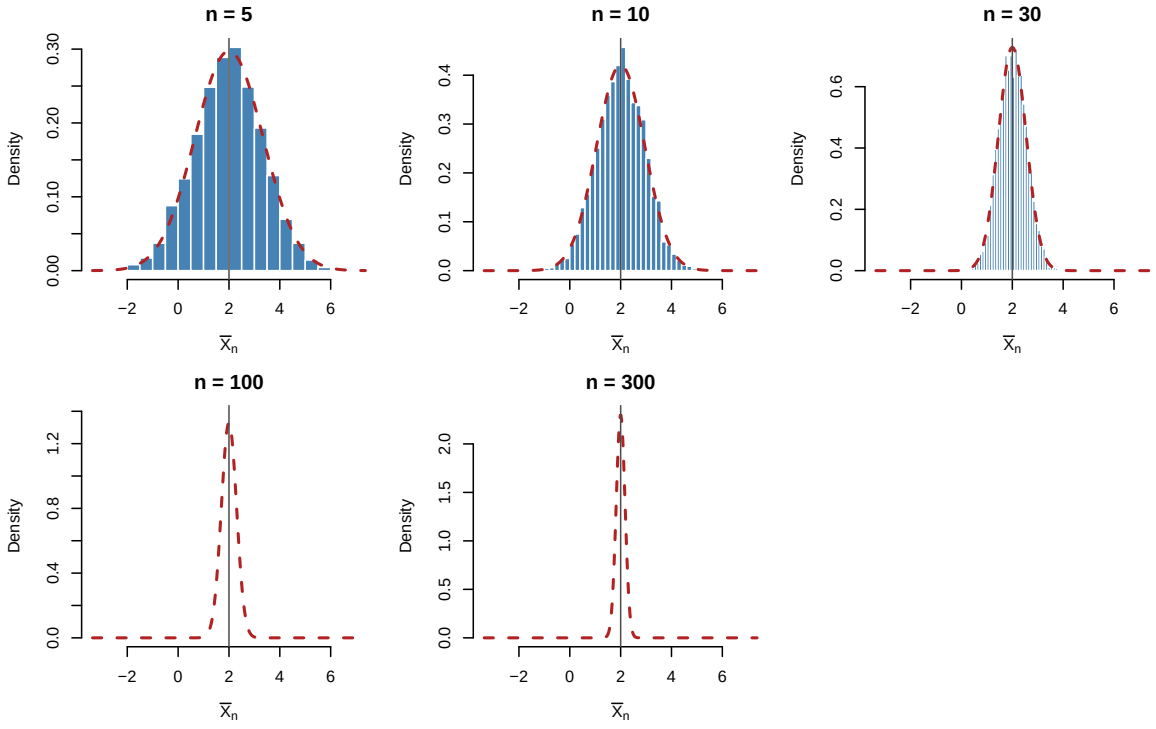


Figure 3.6: Sampling distribution of $\overline{X}_n$ ($B = 5000$ replications) with the exact density $\mathcal{N}(\mu, \sigma^2/n)$ (dashed).

The simulation shows that the estimator is centred near μ (unbiasedness), has nonzero variance, and becomes more concentrated as n grows, in agreement with σ^2/n.

Mean squared error

Definition 3.2.6: Mean squared error

$\mathrm{MSE}(\widehat{\theta}) = \mathbb{E}\big[(\widehat{\theta} - \theta)^2\big]$.

Theorem 3.2.1

(Bias–variance decomposition.)

$$\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}) + \mathrm{Bias}(\widehat{\theta})^2.$$

Proof: Add and subtract $\mathbb{E}[\widehat{\theta}]$:

$$\mathbb{E}\big[(\widehat{\theta} - \theta)^2\big] = \mathbb{E}\big[(\widehat{\theta} - \mathbb{E}[\widehat{\theta}])^2\big] + 2\big(\mathbb{E}[\widehat{\theta}] - \theta\big)\underbrace{\mathbb{E}\big[\widehat{\theta} - \mathbb{E}[\widehat{\theta}]\big]}_{=0} + \big(\mathbb{E}[\widehat{\theta}] - \theta\big)^2,$$

where $\mathbb{E}[\widehat{\theta}] - \theta$ is a constant and can be taken out of the expectation.

This shows that an estimator with a small bias but a much smaller variance can be preferable to an unbiased one with high variance.
Suggested exercise 52.

Proposition 3.2.5 Bias of the empirical variance

If $X_1, \dots, X_n$ are i.i.d. with variance σ^2, then for $\widehat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \overline{X})^2$

$$\mathbb{E}\big[\widehat{\sigma}^2\big] = \frac{n-1}{n}\sigma^2, \qquad \mathrm{Bias}(\widehat{\sigma}^2) = -\frac{\sigma^2}{n},$$

while $s^2 = \frac{1}{n-1}\sum_i (X_i - \overline{X})^2$ is unbiased.

Proof: By the computational formula, $\sum_i (X_i - \overline{X})^2 = \sum_i X_i^2 - n\overline{X}^2$. Using $\mathbb{E}[Y^2] = \mathrm{Var}(Y) + \mathbb{E}[Y]^2$,

$$\mathbb{E}[X_i^2] = \sigma^2 + \mu^2, \qquad \mathbb{E}[\overline{X}^2] = \frac{\sigma^2}{n} + \mu^2.$$

Hence $\mathbb{E}\big[\sum_i (X_i - \overline{X})^2\big] = n(\sigma^2 + \mu^2) - n\big(\frac{\sigma^2}{n} + \mu^2\big) = (n-1)\sigma^2$. Dividing by n or by $n-1$ gives the two statements. Intuitively, deviations are measured from $\overline{X}$, which is closer to the data than μ: one "degree of freedom" is used to estimate the mean.

Consistency

Bias, variance and MSE are *finite-sample* properties. When they cannot be computed explicitly, one studies the behaviour as $n \to \infty$: *asymptotic* properties.

Theorem 3.2.2

(Weak law of large numbers.) Let $X_1, X_2, \dots$ be i.i.d. from any F_X with finite mean μ. Then the sample mean converges in probability to μ:

$$\overline{X} \xrightarrow{\mathbb{P}} \mu, \qquad \text{i.e.} \qquad \mathbb{P}\big(|\overline{X} - \mu| > \varepsilon\big) \to 0 \quad \forall \varepsilon > 0.$$

Proof (under the extra assumption of a finite variance σ^2): **Chebyshev's inequality**: for any Y with finite variance and $\varepsilon > 0$,

$$\mathbb{P}\big(|Y - \mathbb{E}[Y]| > \varepsilon\big) \leqslant \frac{\mathrm{Var}(Y)}{\varepsilon^2},$$

because $\mathrm{Var}(Y) = \mathbb{E}[(Y-\mathbb{E}Y)^2] \geqslant \mathbb{E}\big[(Y-\mathbb{E}Y)^2\,\mathbf{1}(|Y-\mathbb{E}Y|>\varepsilon)\big] \geqslant \varepsilon^2\,\mathbb{P}(|Y-\mathbb{E}Y|>\varepsilon)$. Applied to $Y=\overline{X}$, with $\mathbb{E}[\overline{X}]=\mu$ and $\mathrm{Var}(\overline{X})=\sigma^2/n$:

$$\mathbb{P}\big(|\overline{X}-\mu|>\varepsilon\big) \leqslant \frac{\sigma^2}{n\varepsilon^2} \longrightarrow 0.$$

(The general case with only a finite mean needs more advanced tools.)

Definition 3.2.7: Consistency

An estimator is *consistent* if $\widehat{\theta} \xrightarrow{\mathbb{P}} \theta$ as $n \to \infty$. The LLN says that $\overline{X}$ is consistent for μ.

Note:

The same Chebyshev argument shows that **any estimator whose MSE tends to 0 is consistent**: $\mathbb{P}(|\widehat{\theta}-\theta|>\varepsilon) \leqslant \mathbb{E}[(\widehat{\theta}-\theta)^2]/\varepsilon^2 = \mathrm{MSE}/\varepsilon^2$ (Markov's inequality applied to $(\widehat{\theta}-\theta)^2$). Hence a *biased* estimator can be consistent, provided bias and variance both vanish: this is the case of $\widehat{\sigma}^2$ (bias $-\sigma^2/n$). Proving consistency in general requires fundamental theorems of statistics, "left to statisticians".

Asymptotic normality

Theorem 3.2.3

(Central limit theorem.) Let $X_1,\dots,X_n$ be i.i.d. from F_X with mean μ and variance $\sigma^2<\infty$, and $S_n = X_1+\cdots+X_n$, so that $\mathbb{E}[S_n]=n\mu$ and $\mathrm{Var}(S_n)=n\sigma^2$. Then the standardized sum

$$Z_n = \frac{S_n - n\mu}{\sqrt{n}\,\sigma} = \sqrt{n}\,\frac{\overline{X}_n-\mu}{\sigma}$$

converges in distribution to $\mathcal{N}(0,1)$. Informally, for moderately large n, $S_n \approx \mathcal{N}(n\mu, n\sigma^2)$ and $\overline{X}_n \approx \mathcal{N}(\mu,\sigma^2/n)$, *whatever* the shape of F_X.

Note:

The moments of S_n are exact and elementary (linearity and independence); the deep content of the CLT is that the *shape* becomes Gaussian. The two expressions of Z_n agree because $S_n - n\mu = n(\overline{X}_n-\mu)$ and $n/\sqrt{n}=\sqrt{n}$.

Example 3.2.2

Sum of Gamma variables. $X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Gamma}(\alpha,\theta)$ (shape α, scale θ) has $\mathbb{E}[X_i]=\alpha\theta$, $\mathrm{Var}(X_i)=\alpha\theta^2$, so $\mathbb{E}[S_n]=n\alpha\theta$, $\mathrm{Var}(S_n)=n\alpha\theta^2$. With $n=30$, $\alpha=2$, $\theta=1.5$: $\mathbb{E}[S_n]=90$, $\mathrm{sd}(S_n)=\sqrt{30\cdot 2\cdot 2.25}=11.619$. Although each X_i is skewed, the histogram of 5000 simulated sums is close to $\mathcal{N}(90, 11.619^2)$ (Monte Carlo mean 90.15, sd 11.51). Here $Z_n = \sqrt{n}\,(\overline{X}_n-\alpha\theta)/\sqrt{\alpha\theta^2}$.

```r
set.seed(123)
sum_gamma <- replicate(5000, sum(rgamma(30, shape = 2, scale = 1.5)))
```

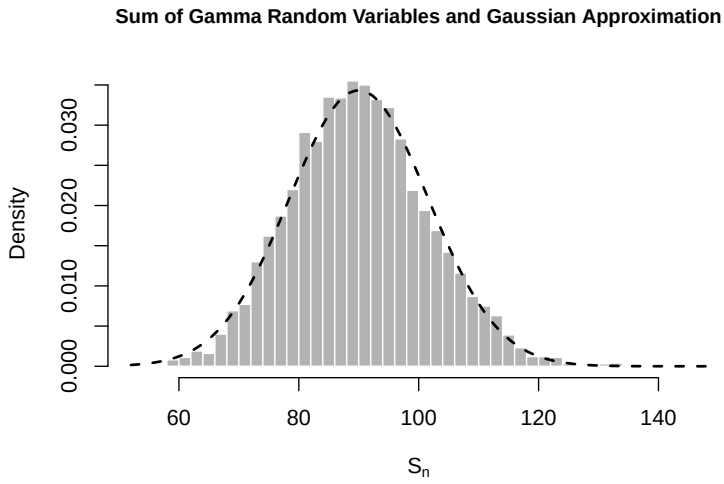


Figure 3.7: Simulated sums of 30 Gamma variables and the Gaussian density with the same mean and variance.

Monte Carlo study

Definition 3.2.8: Monte Carlo simulation

Repeatedly generate data from a statistical model and compute quantities of interest:

1. choose a probability model and true parameter values;
2. generate a sample from the model;
3. compute a statistic or estimator;
4. repeat many times (B replications);
5. study the distribution (EDF, histograms, boxplots) and moments of the outputs.

It lets us study how estimators behave under controlled conditions, approximating their sampling distribution.

Example 3.2.3

Bias and consistency of $\widehat{\sigma}^2$. Gaussian samples with $\sigma^2 = 4$, $B = 5000$ replications for each n:

n	MC mean of $\widehat{\sigma}^2$	MC bias	MC sd	theory $\frac{n-1}{n} \cdot 4$
3	2.6587	−1.3413	2.6701	2.667
5	3.2063	−0.7937	2.2687	3.200
10	3.6169	−0.3831	1.7180	3.600
20	3.8204	−0.1796	1.2352	3.800
50	3.9155	−0.0845	0.7969	3.920
100	3.9560	−0.0440	0.5419	3.960
500	3.9931	−0.0069	0.2515	3.992

The Monte Carlo means follow the theoretical value $\frac{n-1}{n}\sigma^2$ of the bias proposition: for $n = 3$ the estimator is clearly below 4; as n grows the bias and the spread vanish (consistency).

```
s2_hat <- replicate(B, {
  x <- rnorm(n, mean = 0, sd = 2)
  mean((x - mean(x))^2) # denominator n
```

```
})
```

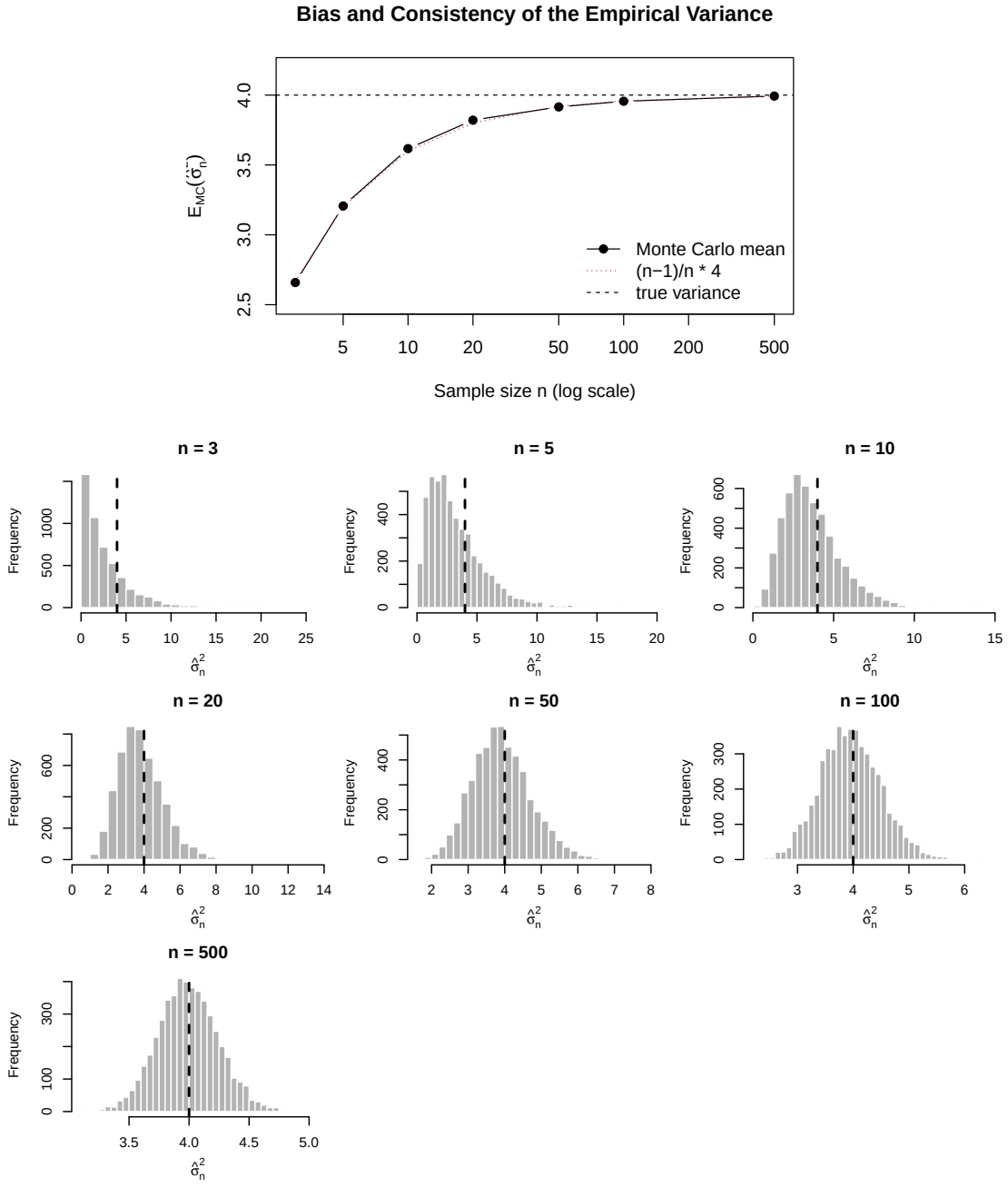


Figure 3.8: Top: Monte Carlo mean of $\widetilde{\sigma}_n^2$ against n (dashed: true value 4). Bottom: sampling distributions; for small n they are very spread out and centred below the true value (dashed line).

3.3 Interval Estimation

Point estimators give one number but do not describe uncertainty; **confidence intervals** do.

Definition 3.3.1: Confidence interval

A confidence interval (CI) for θ is a *random* interval $[C_l(X_1,\dots,X_n),\ C_u(X_1,\dots,X_n)]$ constructed so that, under repeated sampling, it contains the true θ with a prescribed probability $1-\alpha$ (the *confidence level*, usually 95%). The bounds are functions of the random sample, hence random.

3.3.1 Construction of confidence intervals

A CI is built from the CDF $F_{\widehat{\theta}}$ of an unbiased or consistent estimator, derived theoretically or by simulation. It is **exact** (valid for every n) or **asymptotic** (valid as $n \to \infty$; its accuracy at finite n should then be checked, e.g. by Monte Carlo). The recipe:

1. find T_l, T_u with $\mathbb{P}(T_l \leqslant \widehat{\theta} \leqslant T_u) = \int_{T_l}^{T_u} dF_{\widehat{\theta}} = 1-\alpha$ (a common choice is the central interval);
2. since $F_{\widehat{\theta}}$ depends on θ, **invert** the inequality into $\mathbb{P}\big(g(T_l) \leqslant \theta \leqslant g(T_u)\big) = 1-\alpha$, with bounds not depending on θ;
3. $[g(T_l), g(T_u)]$ is a $(1-\alpha)$ CI, exact or asymptotic according to $F_{\widehat{\theta}}$.

Example with an asymptotic CDF

Asymptotically most consistent estimators are Gaussian: typically, via the CLT,

$$Z = \sqrt{n}\,\frac{\widehat{\theta}-\theta}{\sigma_{\widehat{\theta}}} \ \sim\ \mathcal{N}(0,1) \quad \text{as } n \to \infty.$$

Theorem 3.3.1

(Asymptotic Gaussian CI.) Let $z_{1-\alpha/2} = \Phi^{-1}(1-\alpha/2)$. Then, asymptotically,

$$\mathbb{P}\Big(\widehat{\theta} - z_{1-\alpha/2}\frac{\sigma_{\widehat{\theta}}}{\sqrt{n}} \leqslant \theta \leqslant \widehat{\theta} + z_{1-\alpha/2}\frac{\sigma_{\widehat{\theta}}}{\sqrt{n}}\Big) = 1-\alpha.$$

Proof: Let $z_{\alpha/2} = \Phi^{-1}(\alpha/2)$. By symmetry of $\mathcal{N}(0,1)$, $z_{\alpha/2} = -z_{1-\alpha/2}$, and $\mathbb{P}(z_{\alpha/2} < Z < z_{1-\alpha/2}) = (1-\alpha/2) - \alpha/2 = 1-\alpha$. Now invert:

$$-z_{1-\alpha/2} < \sqrt{n}\,\frac{\widehat{\theta}-\theta}{\sigma_{\widehat{\theta}}} < z_{1-\alpha/2} \iff \widehat{\theta} - z_{1-\alpha/2}\frac{\sigma_{\widehat{\theta}}}{\sqrt{n}} < \theta < \widehat{\theta} + z_{1-\alpha/2}\frac{\sigma_{\widehat{\theta}}}{\sqrt{n}},$$

multiplying by $\sigma_{\widehat{\theta}}/\sqrt{n} > 0$ and rearranging (the inequalities flip when isolating $-\theta$). For a continuous distribution strict and non-strict inequalities give the same probability.

Note:

The central interval is the **shortest** interval with probability $1-\alpha$ for the $\mathcal{N}(0,1)$, because the density is symmetric and unimodal. For $\alpha = 0.05$, $z_{0.975} = 1.96$. In practice the unknown $\sigma_{\widehat{\theta}}$ is replaced by a consistent estimate, e.g. for the mean $\overline{X} \pm 1.96\, s/\sqrt{n}$.

Suggested exercise 53.

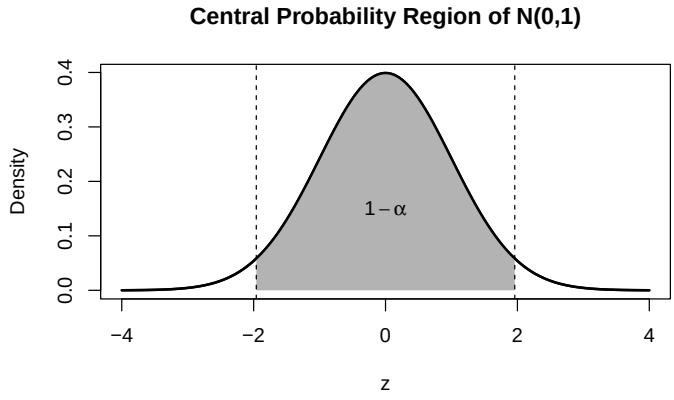


Figure 3.9: The shaded region $z_{\alpha/2} < Z < z_{1-\alpha/2}$ has probability $1-\alpha$.

Example with an exact CDF

Theorem 3.3.2

If $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and $s^2 = \frac{1}{n-1}\sum_i (X_i - \overline{X})^2$, then

$$\sqrt{n}\,\frac{\overline{X} - \mu}{s} \sim St_{n-1},$$

the *Student* distribution with $n-1$ degrees of freedom (stated without proof). Hence the **exact** CI

$$\mu \in \left[\overline{X} - t_{n-1,\,1-\alpha/2}\frac{s}{\sqrt{n}},\ \overline{X} + t_{n-1,\,1-\alpha/2}\frac{s}{\sqrt{n}}\right].$$

Derivation of the interval: The Student distribution is symmetric around 0, so $\mathbb{P}\big(-t_{n-1,1-\alpha/2} \leqslant T \leqslant t_{n-1,1-\alpha/2}\big) = 1-\alpha$ for $T = \sqrt{n}(\overline{X}-\mu)/s$, and the inequalities are inverted exactly as in the Gaussian case.

Note:

The Student distribution has heavier tails than the Gaussian, so $t_{n-1,1-\alpha/2} > z_{1-\alpha/2}$ (e.g. $t_{9,0.975} = 2.262$ against 1.96): the interval is wider to account for the estimation of σ by s. As $n \to \infty$, $St_{n-1} \to \mathcal{N}(0,1)$ and the two intervals coincide. In R: `qt(0.975, df = n - 1)`.

Suggested exercise 54.

3.3.2 Bootstrap confidence intervals

When the distribution of an estimator is hard to derive (even asymptotically), or its standard error is cumbersome, the **bootstrap** gives a computational alternative.

Definition 3.3.2: Nonparametric bootstrap

Given the sample $x_1, \ldots, x_n$ and a consistent estimator $\widehat{\theta}$:

1. resample n observations **with replacement** from the data, B times;
2. compute the estimator on each resample: $\widehat{\theta}^{*1}, \ldots, \widehat{\theta}^{*B}$ (bootstrap replicates);
3. use the EDF of the replicates to approximate the sampling distribution of $\widehat{\theta}$.

The *percentile* interval is $[q^*_{\alpha/2},\ q^*_{1-\alpha/2}]$, with q^* the empirical quantiles of the replicates.

Note:

Resampling from the data means sampling from the EDF $F^{(n)}$: the bootstrap is the plug-in principle applied to the *whole* sampling distribution, i.e. "the unknown F_X is replaced by $F^{(n)}$".

Example 3.3.1

Median of an exponential sample. The sample median has a much less convenient sampling distribution than the mean. We draw $n = 50$ observations from Exp(2), whose population median is $\log 2/2 \approx 0.3466$, and take $B = 10\,000$ bootstrap samples.

- observed sample median $\widehat{\theta} = 0.4223$;
- bootstrap replicates: mean 0.4273, variance 0.00635, standard error 0.0797;
- 95% percentile CI: $[q^*_{0.025}, q^*_{0.975}] = [0.2919,\ 0.5896]$; with `boot::boot.ci` (different resamples): $[0.2919,\ 0.5960]$.

The two intervals differ only by Monte Carlo variation, which decreases with B. Here the interval contains the true median 0.3466.

```r
set.seed(123)
x <- rexp(50, rate = 2)
theta_hat <- median(x)
set.seed(456)
theta_boot <- replicate(10000, median(sample(x, size = 50, replace = TRUE)))
quantile(theta_boot, probs = c(0.025, 0.975)) # percentile CI

# with the recommended package boot: statistic(data, indices)
median_statistic <- function(data, indices) median(data[indices])
boot_fit <- boot::boot(data = x, statistic = median_statistic, R = 10000)
boot::boot.ci(boot_fit, conf = 0.95, type = "perc") # endpoints in $percent[4:5]
```

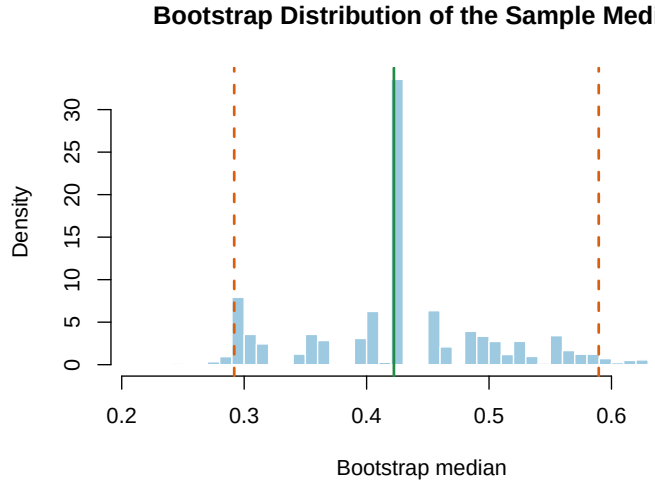


Figure 3.10: Bootstrap distribution of the sample median: percentile endpoints (orange, dashed) and observed median (green).

3.3.3 Coverage

Definition 3.3.3: Coverage probability

For an interval procedure $[C_l(X_1,\dots,X_n), C_u(X_1,\dots,X_n)]$, the *coverage* is

$$\mathbb{P}\big(C_l(X_1,\dots,X_n) \leqslant \theta \leqslant C_u(X_1,\dots,X_n)\big).$$

A method advertised as a 95% CI should have coverage close to 0.95. Coverage is a property of the **procedure**, not of one realized interval: over many repetitions, about 95% of the intervals should contain θ.

Note:

Once the data are observed, a given interval either contains θ or not: the "95%" refers to the random interval *before* sampling.

Monte Carlo estimate of coverage: choose a model with known θ and a sample size n; generate many samples; compute the CI on each; record whether it contains θ; average. With B replications, the estimate is the proportion of covering intervals (a Bernoulli mean, with Monte Carlo standard error $\sqrt{0.95 \cdot 0.05/B} \approx 0.005$ for $B = 2000$).

Example 3.3.2

$X_1,\dots,X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$ (skewed), $\theta = \mathbb{E}[X] = 1$. Three 95% CIs for the mean:

- **asymptotic Gaussian**: $\overline{X} \pm z_{0.975}\, s/\sqrt{n}$;
- **bootstrap normal**: $\overline{X} \pm z_{0.975}\, \widehat{\text{se}}$, with $\widehat{\text{se}}$ the standard deviation of the bootstrap means $\overline{X}^*_1,\dots,\overline{X}^*_B$;
- **bootstrap percentile**: empirical quantiles of the bootstrap means.

$B_{\text{mc}} = 2000$ Monte Carlo samples, $B_{\text{boot}} = 1000$ bootstrap replicates each. For $n = 30$:

Method	Coverage	Average length
Asymptotic normal	0.9060	0.6959
Bootstrap normal	0.9020	0.6840
Bootstrap percentile	0.9075	0.6781

As a function of n:

n	10	20	30	50	100
Asymptotic normal	0.8665	0.8975	0.9210	0.9450	0.9395
Bootstrap normal	0.8535	0.8885	0.9180	0.9435	0.9380
Bootstrap percentile	0.8620	0.8930	0.9190	0.9410	0.9375

A good method has coverage close to the nominal 0.95 and intervals not unnecessarily wide. For small n all three **undercover** (around 0.86 at $n = 10$), because the skewness of the exponential makes the Gaussian approximation poor; the coverage approaches 0.95 as n grows. The two runs at $n = 30$ (0.906 vs 0.921) differ by Monte Carlo variation.

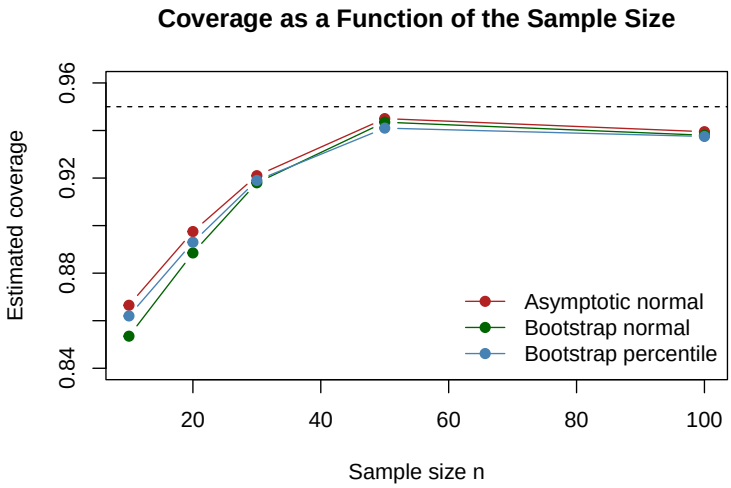


Figure 3.11: Estimated coverage of the three 95% CIs against n (dashed: nominal level).